ALM7 &
5th iLS Conference
Perth, Western Australia | October 18-21, 2026

OFFICIAL CONFERENCE PUBLICATION

BOOK OF ABSTRACTS

18–21 OCTOBER 2026

Perth, Western Australia

Joint conference between the Australian Lipid Society and the International Lipidomics Society

NAVIGATE THE PUBLICATION

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WELCOME TO SUNNY

Perth 2026

Blue skies, sparkling water, amazing food and a little quokka charm — Perth is ready to welcome you. Between sessions, enjoy river views, sunsets, beaches and the relaxed Western Australian vibe while connecting with the global lipidomics community.

ABOUT US

About the Conference

We are delighted to announce the first joint conference of the Australian Lipid Meeting 7 (ALS) and the 5th International Lipidomics Society (iLS) Conference, taking place October 18–21, 2026 at the Novotel Perth Langley, in Perth, Western Australia.

This special event unites two major gatherings in lipid research, bringing together researchers, scientists, clinicians and industry leaders from around the world to share discoveries, promote collaborations and shape the future of lipidomics.

The program will feature plenary and keynote speakers, oral and poster presentations and networking sessions; all designed to encourage the exchange of ideas and promote interdisciplinary collaboration.

- Microbial & Plant Lipids in Health and Biotechnology
- Lipid Imaging & Single-Cell Lipidomics
- AI, Computational Tools & Data Standards for Lipidomics
- Lipids in Health and Disease
- Advancements in Lipidomic Technologies
- Translational and Precision Lipidomics
- Clinical Lipidomics: From Pre-analytics to Cohort Studies
- Lipid Particles and Delivery Systems

Whether you are a biochemist, clinician, analytical chemist, nutritionist or cell biologist, this conference will provide a forum to engage with the global lipid research community.

We look forward to welcoming you to sunny Perth in 2026 for this exciting joint celebration of lipidomics!

PEOPLE BEHIND THE MEETING

Committee Members

LOCAL ORGANISING COMMITTEE

Peter Meikle
Baker Institute

Nicola Gray
Murdoch University

Luke Whiley
Curtin University

Monique Ryan
Murdoch University

Dana Hicks
Murdoch University

Jayden Lee Roberts
University of Notre Dame

Corey Giles
Baker Institute

Kevin Huynh
Baker Institute

AUSTRALIAN LIPID MEETING COMMITTEE

Peter Meikle
Baker Institute

Nicola Gray
Murdoch University

Luke Whiley
Curtin University

Monique Ryan
Murdoch University

Dana Hicks
Murdoch University

Jayden Lee Roberts
University of Notre Dame

Corey Giles
Baker Institute

Anthony Don
The University of Sydney

Gavin Reid
University of Melbourne

Maria Fuller
The University of Adelaide

Stephen Blanksby
Queensland University of Technology

Todd Mitchell
University of Wollongong

Kevin Huynh
Baker Institute

INTERNATIONAL LIPIDOMICS SOCIETY REPRESENTATIVES

Kim Ekroos
Lipidomics Consulting

Peter Meikle
Baker Institute

Robert Ahrends
University of Vienna

Xue Li Guan
University of Copenhagen

Shane Ellis
University of Wollongong

Óscar Pastor
Hospital Universitario Ramón y Cajal

Gerhard Liebisch
University of Regensburg

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EVENT SPONSOR

BUSINESS EVENTS PERTH

FEATURED SPEAKERS

Plenary & Keynote Speakers

Invited speakers are highlighted with a dedicated editorial profile before the contributed abstracts.

FEATURED SPEAKER

Prof. Kerry-Anne Rye

Head of the Cardiometabolic Disease Research Group, University of New South Wales, Australia & Editor-in-Chief of Journal of Lipid Research

Perth 2026

PLENARY SPEAKER

Lipidomics and Lipid Function—Measuring Lipids is no Longer Enough

ABSTRACT

Lipid dysregulation is a major feature of cardiovascular disease (CVD). The diagnosis and management of CVD has traditionally focussed on measuring and reducing circulating cholesterol and triglyceride levels. Although this decreases cardiovascular risk, it provides limited information about the biological properties and functions of individual lipid species that contribute to disease development and progression. Over the past two decades lipidomics has generated detailed insights into changes in individual lipid species in CVD that cannot be captured by conventional lipid measurements. However, the biological significance of most of these changes remains uncertain. This presentation explores the transition from measuring lipids to understanding their biology and function in CVD. It is based on the biological effects of individual lipid species being dependent on their abundance, as well as their cellular distribution, biological activity and ability to interact with proteins and lipoproteins. High density lipoproteins (HDLs) provide an important example of this complexity. Although HDL-cholesterol is inversely associated with CVD risk, interventions that increase HDL-C do not always decrease cardiovascular risk or events. This highlights that lipid concentration does not necessarily reflect lipid function and that both may be profoundly altered in disease states. The integration of lipidomics with single-cell and spatial transcriptomics, proteomics, metabolomics, imaging and functional studies provides an opportunity to determine how individual lipid species affect disease mechanisms. This integrated approach could facilitate clinically relevant biomarker discovery, improve our understanding of CVD pathogenesis and identify new therapeutic targets. In conclusion, the transition from lipid identification and quantification to lipid biology has the potential to transform CVD treatment by enabling the development of targeted approaches that reduce morbidity and mortality.

FEATURED SPEAKER

Prof. Hiroshi Tsugawa

Head of Systems Omics Laboratory, Kyoto University, Japan & Lead Developer of MS-DIAL

Perth 2026

KEYNOTE SPEAKER

Toward agent-assisted lipidomics: extending the MS-DIAL ecosystem from laboratory-scale

ABSTRACT

Advances in mass spectrometry and data processing are rapidly expanding the structural information accessible by untargeted lipidomics. As lipid annotations become more detailed, however, it becomes increasingly important to preserve not only the reported lipid identities but also the analytical evidence and processing steps that support them. I will first introduce our recent developments in untargeted lipidomics. ZT Scan DIA combines scanning data-independent acquisition with computational deconvolution along the quadrupole and retention-time dimensions, enabling reconstruction of compound-specific MS/MS spectra from highly multiplexed data. In lipidomics, deconvolution improved annotation precision by removing contaminant product ions and enabled separate quantification of co-eluting isomers using MS2 chromatograms, although common diagnostic ions could also be erroneously removed. Nevertheless, optimization of analysis parameters minimized this negative effect. In parallel, we developed a sodium-adduct electron-activated dissociation (EAD) MS/MS workflow for phospholipids that provides diagnostic information on acyl chain positions on the glycerol backbone. Application of this approach to Entamoeba histolytica revealed lipid class-dependent positional distributions of very long chain fatty acids that were difficult to resolve by conventional MS/MS. Improvements in lipidomics society also increase the importance of reproducible and traceable data processing. We are therefore extending the MS-DIAL ecosystem beyond individual desktop analyses. In addition to MS-DIAL 5 and the pathway-centered lipidomics environment MSLipidMapper, we are developing a cross-platform local application that executes MS-DIAL Console workflows and exports standardized results including mzTab-M. We are also organizing public mass spectrometry studies into technically coherent analysis units and developing provenance-aware catalogues that connect repository files with processing conditions, deconvoluted spectra, alignment features, consensus spectra, and the evidence supporting annotation claims. These recent activities will be introduced.

FEATURED SPEAKER

A/Prof. Hyungwon Choi

Director of Singapore Lipidomics Incubator (SLING), National University of Singapore, Singapore

Perth 2026

KEYNOTE SPEAKER

Scalable workflows for population-scale lipidomics in cardiometabolic diseases

ABSTRACT

Population-scale lipidomics studies require seamless workflows integrating automated sample preparation, fast and robust analytical methods, and accurate and scalable data processing. In this talk, we present a combination of robotics-based sample preparation and a highly efficient data processing workflow, MRMhub, which together substantially reduce analysis time and manual effort across the entire experimental process. We demonstrate this workflow in a plasma lipidomics study of multiple Singapore cohorts of individuals with type 2 diabetes at risk of diabetic kidney disease (n > 8,000), in which we predict rapid renal function decline using a network-based machine learning approach.

We also present an alternative high-throughput MS workflow using the RapidFire-MS system, which replaces chromatographic separation with rapid solid-phase extraction and enables targeted analysis of a focused panel of lipids. We demonstrate this capability in a study of plasma ceramides as biomarkers of cardiovascular outcomes in two Singapore cohorts, where MS analysis of thousands of samples can be completed within approximately 24 hours. Together with our automated data processing tool, RFkit, this shifts the major time requirement of the workflow from MS acquisition and data processing to sample preparation.

These complementary workflows demonstrate the power of automation and underscore the importance of scalability in population-scale lipidomics. Our software ecosystem also enables flexible workflow customization while digitalizing the data processing pipeline, improving transparency and reproducibility and facilitating knowledge transfer between laboratories.

FEATURED SPEAKER

Prof. Gerald Watts

Winthrop Professor University of Western Australia Medical School, Australia

Perth 2026

KEYNOTE SPEAKER

Go with the flow: unravelling lipoprotein metabolism with stable isotopes

ABSTRACT

Use of stable isotope tracers and compartmental modelling have provided deeper knowledge of the mechanisms underlying lipid disorders in patients at high risk of atherosclerotic cardiovascular disease. These studies underpin our understanding of the origins of atherogenic dyslipidaemias and provide insight into the mode of action of established therapies and support the discovery of new molecular treatments. The principles of such studies with be reviewed and data presented that have 1) challenged the dysfunction of the LDL receptor as the exclusive paradigm for the pathogenesis of familial hypercholesterolaemia, 2) unravelled the complexities of Lp(a) metabolism and its therapeutic regulation, 3) characterised the dyslipidaemia and therapies in the cardiometabolic syndrome, 4) investigated the metabolic effects of specific genetic variants that have led to development of new molecular therapies targeting the mRNA of the APOC3 and ANGPTL3 genes.

FEATURED SPEAKER

Dr. Laura Bindila

Head of Clinical Lipidomics Unit, Institute of Physiological Chemistry, University Medical Center Mainz, Germany

Perth 2026

KEYNOTE SPEAKER

Multi-modal 4D- and targeted lipidomic strategies for neuro- and immuno-metabolism

ABSTRACT

In-depth investigation of shifts in lipid metabolism that are specific to health and different diseases require development and integration of multi-modal analytical pipelines to capture membrane lipids, energy lipids, lipid mediators and/or neuro-immune epitopes carrying lipids. Combinatorial strategies of high resolution untargeted lipidomics using 4_dimensional trapped ion mobility mass spectrometry, 4D-TIMS, with parallel reaction monitoring PRM- PASEF and multiple reaction monitoring, MRM, using single sample extracts are highlighted here to expedite the clinical research by maximizing lipidome coverage and fingerprint readout for health and diseases. Expanding the portfolio of reproducible (semi)-quantitative profiling in the discovery of lipid markers of metabolic dysregulation up to large cohorts enables new mechanistic hypotheses for disease differentiation and phenotyping. Application of such pipelines for plasma/serum, tissues and cells analysis in neurological and cardiovascular diseases, as well as cancer will be discussed.

FEATURED SPEAKER

A/Prof. Shane Ellis

Group Leader of Mass Spectrometry Imaging (MSI) Laboratory, University of Wollongong, Australia

Perth 2026

KEYNOTE SPEAKER

Next-Generation Approaches for Enhancing Spatial Lipidomics

ABSTRACT

Mass spectrometry imaging (MSI) is enabling new research opportunities in lipidomics. A key focus of our group is improving the analytical capabilities of MSI technologies in terms of spatial resolution, sensitivity and coverage, and their application across many diseases, in particular neurodegeneration. Firstly, to address the lack of quantitative data obtained by MSI (Q-MSI), we have developed multiplexed Q-MSI workflows whereby multiple internal standards are applied homogeneously to the tissue. The most recent implementation has focussed on gangliosides and has been applied to a cohort of 90 human tissue samples with and without Parkinson's Disease (PD). This has revealed novel ganglioside changes associated with PD, including significant differences between male and female PD cases. Conventional ionisation methods can suffer from poor ionisation efficiencies. To overcome this, we are developing and applying MALDI-2 and plasma ionisation to improve lipid coverage and sensitivity. The resulting increase in sensitivity has also enabled gains in spatial resolution. Using two prototype instruments equipped with transmission mode optics and either plasma ionisation or MALDI-2, MSI can now be obtained at low-micron and sub- micron pixel sizes for tissue and cell imaging. Combined with novel sample preparation workflows, rich spectra of lipids and nucleotides can be obtained at pixel sizes of 1 micron and below. This can be applied to study heterogeneity at cellular and sub-cellular scales, and allows in-situ single cell lipidomics, whereby lipid profiles of individual cells can be acquired directly from complex tissue microenvironments. Finally, in the context of PD and environmental pollutants, we are developing high throughput single cell lipidomics workflows to elucidate heterogeneity within cell cultures and cellular responses to pollutants. Our workflows allow automatic microscopy registration, segmentation and analysis of many thousands of cells. For example, distinct lipid profiles are observed in iPSC-derived cells upon nanoplastic exposure and in the presence of mutations in PD-relevant genes such as SNCA.

FEATURED SPEAKER

Prof. Aleksandra Filipovska

Louis Landau Chair in Child Health Research, University of Western Australia, Australia

Perth 2026

PLENARY SPEAKER

Mitochondrial Architecture in Health and Disease: from Membrane Complexes to Organelle

ABSTRACT

Mitochondrial function depends on far more than the activity of individual enzymes or pathways. It emerges from the organisation of mitochondrial membranes, the assembly of macromolecular complexes, and dynamic physical and metabolic interactions with other organelles. Disruption of these relationships is increasingly recognised as a feature of human disease, yet how defects at different levels of cellular organisation are connected remains poorly understood. Our work combines quantitative imaging, transcriptomics, proteomics, lipidomics, metabolomics and synthetic cell biology to investigate mitochondrial organisation from molecular interactions within membranes to communication across the organelle network. Using disease models and systematic perturbation of organelle biogenesis, we find that defects in the endoplasmic reticulum, Golgi and peroxisomes can propagate throughout the cell, altering mitochondrial morphology and function together with lipid, metabolic and glycoprotein homeostasis. At the molecular level, we have developed approaches to map native protein–lipid interactions, revealing how structural lipids such as cardiolipin contribute to the assembly, stability and organisation of mitochondrial respiratory complexes and how these relationships are altered in disease. We are also moving from defining these interactions to manipulating them. Synthetic biology approaches that modify mitochondria have shown us that mitochondrial location can reshape cellular proteomes and metabolism and, in cells with disrupted organelle architecture, restore connectivity between compartments, improve organelle morphology and rescue trafficking defects. These findings suggest that some consequences of cellular dysfunction reflect not only loss of individual proteins or pathways, but disruption of the spatial and molecular organisation of the cell. I will discuss how mitochondrial membrane organisation and inter-organelle communication are integrated across scales, how these networks are remodelled in disease, and how targeted manipulation of cellular architecture can be used to identify mechanisms of pathology and explore new strategies for restoring cellular function.

CONTRIBUTED ABSTRACTS

Oral Presentations

Oral abstracts are presented in presentation-number order with compact clickable entries in the contents.

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SPONSOR TALK

SP-01

Large panel of lipid identification and quantitation using ZT Scan DIA analysis

PRESENTING AUTHOR

Paul Baker

Thusitha Rupasinghe[1], Jason Cousen[2], Rebekah Sayers[3], Paul Baker[4]

[1] SCIEX, Australia; [2] SCIEX, Canada; ^3 SCIEX, UK; [4] SCIEX, US

PRESENTATION

SP-01

Monday 19

15:50

Riverside Ballroom

SCIEX

BACKGROUND

Data-independent acquisition (DIA) enables broad MS/MS coverage and retrospective quantitation in untargeted lipidomics. However, conventional DIA workflows often generate chimeric spectra because multiple precursor ions are co-isolated and fragmented simultaneously. This spectral complexity can reduce confidence in lipid annotation and compromise quantitative specificity. To address these limitations, we evaluated Zeno Trap-enabled scanning DIA (ZT Scan DIA), a continuously scanning Q1 acquisition strategy that improves precursor-fragment association through post-acquisition Q1 and retention time deconvolution.

METHODS

NIST SRM 1950 human plasma was extracted using a protein precipitation workflow and analyzed on a SCIEX ZenoTOF 8600 system coupled to an Exion LC platform. Lipid separation was performed using an ACQUITY UPLC CSH C18 column with a 15-minute water/acetonitrile/isopropanol gradient containing 0.1% acetic acid and 10 mM ammonium acetate. Data were acquired using both data-dependent acquisition (DDA) and ZT Scan DIA workflows, with Zeno trap pulsing applied to enhance MS/MS sensitivity. ZT Scan DIA datasets were processed in MS-DIAL 5.6 to evaluate lipid coverage, spectral quality, precursor specificity, and MS/MS-level quantitative performance. Selected lipid species were further processed in SCIEX OS 5.0 using Analytics for semi-quantitative analysis based on internal standards.

RESULTS

and Discussion

This study demonstrates how ZT Scan DIA helps bridge the gap between untargeted discovery and quantitative lipidomics. By improving precursor selectivity and reducing spectral interference, ZT Scan DIA enables more accurate MS/MS-level quantitation directly from untargeted datasets. The ability to perform retrospective quantitation unlocks additional value from discovery experiments, reducing the need for follow-up targeted analyses while increasing confidence in biological interpretation. Through practical examples, we show how ZT Scan DIA transforms discovery datasets into long-term quantitative assets that can support both current and future investigations.

CONCLUSION

s

These results demonstrate that continuously scanning Q1 DIA provides a practical strategy for simultaneous identification and quantitation in untargeted lipidomics. By combining DIA-level coverage with improved precursor selectivity, ZT Scan DIA reduces spectral ambiguity, increases confidence in the characterization of co-eluting and isobaric lipid species, and creates a durable digital record that can be reanalyzed for future qualitative and quantitative studies. This workflow delivers a robust balance of coverage, selectivity, and quantitative reliability for complex biological samples.

ORAL PRESENTATION

O-01

LipiRich: An Open-Source, Browser-Based Application for Normalisation, Statistical Analysis, and Pathway Enrichment using Untargeted Lipidomics Data

PRESENTING AUTHOR

Sarah Hancock

Sarah Hancock[1], Nigel Turner[1]

[1] Cellular Bioenergetics Laboratory, Victor Chang Cardiac Research Institute, Sydney NSW, Australia

PRESENTATION

O-01

Monday 19

09:30

Riverside Ballroom

KEYWORDS

lipidomics

untargeted analysis

pathway enrichment

R Shiny

open source

Untargeted lipidomics generates increasingly large, complex datasets requiring robust normalisation, statistical testing, and pathway-level interpretation. Downstream analysis is frequently handled through ad hoc, non-reproducible scripts requiring substantial programming expertise, limiting reproducibility and standardisation across laboratories without dedicated bioinformatics support.

We present LipiRich, an open-source R Shiny application providing an interactive, browser-based pipeline for analysing untargeted lipidomics data exported from MS-DIAL 5, a widely used tool for lipid feature detection and annotation. LipiRich guides users through a structured workflow encompassing data import, internal standard handling, background subtraction and normalisation, exploratory visualisation, principal component analysis, and pairwise and multi-group statistical testing (via rstatix). Pathway-level interpretation is supported through three complementary approaches: lipid set enrichment (ORA or FGSEA) across class, fatty acid identity, saturation, chain length, and ether-subclass sets; Pearson correlation network analysis of significant lipid species; and enzyme activity proxy scoring based on lipid class ratios, visualised as a z-scored heatmap with group comparison statistics. Additional outputs include significance heatmaps, grouped class bar plots, and standardised, publication-ready plot export across all analysis tabs. No local installation or coding expertise is required, as the application runs entirely in-browser (www.lipirich.sarahehancock.com) via a self-hosted server (source code available at github.com/sarahehancock/LipiRich).

By consolidating lipidomics analysis and pathway interpretation into a single, transparent, freely available tool, LipiRich lowers the technical barrier to rigorous lipidomics analysis and addresses common friction points in comparable software, such as inconsistent lipid nomenclature handling and rigid input data formatting requirements.

ORAL PRESENTATION

O-02

An atlas of imputed plasma lipidomic biomarkers for health and disease in 197,840 individuals from the UK Biobank

PRESENTING AUTHOR

Changyu Yi

Changyu Yi[1], Yu Xu[2], Kevin Huynh[1], Tingting Wang[1], Habtamu B Beyene[1], Aleksandar Dakic[1], Jingqin Wu[1], Michelle Cinel[1], Natalie A Mellett[1], Thy Duong[1], Gerald F Watts[3], Joseph Hung[3], Jennie Hui[4], John Beilby[4], John Blangero[5], Eric K Moses[6], Rima Kaddurah-Daouk[7], Minna K Karjalainen[8], Mika Ala-Korpela[8], Johannes Kettunen[8], Terho Lehtimäki[9], Olli T Raitakari[10], Paul Lacaze[11], Chenglong Yu[11], Allison M Hodge[12], Agus Salim[13], Michael Inouye[2], Corey Giles[1], Peter J Meikle[1]

[1] Baker Heart and Diabetes Institute, Melbourne, Australia; [2] Victor Phillip Dahdaleh Heart and Lung Research Institute, University of Cambridge, Cambridge, UK; [3] School of Medicine, University of Western Australia, Perth, Australia; [4] School of Biomedical Sciences, University of Western Australia, Perth, Australia; [5] South Texas Diabetes and Obesity Institute, The University of Texas Rio Grande Valley, Brownsville, Texas, USA; [6] Menzies Institute for Medical Research, University of

PRESENTATION

O-02

Monday 19

09:45

Riverside Ballroom

KEYWORDS

- Lipidomics
- UK Biobank
- polygenic score
- cardiovascular disease
- asthma

Lipids are essential components of cellular membranes and signalling pathways, playing critical roles in many biological processes. Although dysregulated lipid metabolism is implicated in cardiovascular disease (CVD) and type 2 diabetes, its contribution to many other diseases remains poorly understood. We comprehensively investigated associations between lipid metabolism and disease using data from the UK Biobank. As lipidomic measurements are unavailable in the UK Biobank, we developed lipid prediction models in the Young Finns Study (YFS; n = 2,076) by integrating genetic and nuclear magnetic resonance (NMR)-based metabolomic data. Model performance was externally validated in the Alzheimer's Disease Neuroimaging Initiative (ADNI; n = 1,393), achieving a median correlation of 0.55 between predicted and measured concentrations across 519 lipid species and classes. The lipid profiles were then imputed in the UK Biobank and tested for associations with 1,162 clinical outcomes using Cox proportional hazards models. We identified 54,884 significant associations (false discovery rate q < 0.05), involving 497 clinical outcomes. The most associated disease categories were circulatory, digestive, and endocrine/metabolic disorders. For ischaemic heart disease, phosphatidylcholines (41 species), ceramides (37 species), and sphingomyelins (34 species) showed the greatest number of associations. We also identified 180 lipid species associated with asthma, with phosphatidylethanolamine, alkylphosphatidylethanolamine, alkenylphosphatidylethanolamine, phosphatidylinositol, and lysophosphatidylinositol classes showing predominantly positive associations, whereas most lysophosphatidylcholine, lysoalkylphosphatidylcholine, and lysoalkenylphosphatidylcholine species were inversely associated. These findings provide a comprehensive resource linking lipid metabolism to diverse diseases and identify candidate biomarkers for future mechanistic and translational studies.

ORAL PRESENTATION

O-03

A pilot study of AI assisted LC-MS lipid structure identification for under represented lipid categories

PRESENTING AUTHOR

Zhixu Ni

Zhixu Ni[1], Mingjie Huang[1], Haiyue Hou[2], Weishen Zhou[2], Yu Xia[2]

[1] Institute of Data and Information, Tsinghua Shenzhen International Graduate School, Lishui Road 2279, Shenzhen, 518055;
[2] Department of Chemistry, Tsinghua University, Hetian Building Tsinghua University, Beijing, 100084

PRESENTATION

O-03

Monday 19

10:00

Riverside Ballroom

KEYWORDS

Structural Lipidomics

Glycosphingolipids

Lipid identification

AI for science

Lipids are essential small biomolecules with diverse chemical structures, physicochemical properties, and biological functions. As lipidomics moves from class-level profiling toward high-resolution structural characterization and discovery of previously under represented lipid categories, current coventional data analysis workflows generaly facing challengies in identification accuracy and coverage.

Many lipid classes, including glycolipids and oxidized lipids (epilipids), remain less represented in mainstream structural and spectral databases, and lack of commercial standards makes it difficult to construct reference library. Thus, databases dependent identification algorithms have challengies to identify such lipid classes.

To address this gap, several tools have been developed to identify lipids without using databases. For example. LPPtiger enables prediction and identification of epilipid structures, while LipidCXSMILES, provides dynamic serialization of lipid structures for computational analysis. Building on these resources, we propose a next-generation AI assisted lipid structure identification algorithm and corresponding workflow including hybrid wet-lab and dry-lab improvements.

We validated this hybrid strategy using glycosphingolipids and gangliosides. The workflow successfully identified lipid structures absent from current databases and substantially improved analysis efficiency compared with manual spectral interpretation, while preserving high identification accuracy. This work advances lipidomics toward comprehensive, accurate, and extensible structural identification beyond the constraints of existing structural and spectral reference libraries.

ORAL PRESENTATION

O-04

A high-throughput LC-MS and machine learning framework for population-scale plasma lipidomics

PRESENTING AUTHOR

Amy W. Liang

Amy W. Liang[1], Thy Duong[1], Matilda Van Burren-Milne[1], Natalie A. Mellet[1], Michelle Cinel[1], Habtamu B. Beyene[1], Pukjera J. Rungreang[1], Mark Sartain[2], Dianna J. Magliano[1], Jonathan E. Shaw[1], Alysha De Livera[3], Peter J. Meikle[1], Corey Giles[1], Kevin Huynh[1]

[1] Baker Heart and Diabetes Institute, Melbourne, Australia; [2] Agilent Technologies, Santa Clara, CA 95051, USA; [3] Mathematics and Statistics, School of Computing, Engineering and Mathematical Sciences, La Trobe University, Melbourne, Australia

PRESENTATION

O-04

Monday 19

10:15

Riverside Ballroom

KEYWORDS

Ultra-high-throughput

Comprehensive lipidomics

Machine learning

Population lipidomics

Comprehensive plasma lipidomics provides valuable insights into disease biology but population lipidomics remains challenging due to the trade-off between throughput and depth of coverage. We have developed a new approach by coupling high-throughput untargeted lipidomics with machine learning models trained using our comprehensive targeted lipidomics platform to increase analytical throughput.

We applied a 15-minute targeted LC-MS/MS platform (989 lipids across 52 classes, run time ~4months) on a subset of the AusDiab 2005 collection (n = 5270). These samples were subsequently rerun on a 1-minute LC-MS platform (Agilent Revident Q-TOF, run time of ~10days). Least absolute shrinkage and selection operator (LASSO) model was trained using 6065 features extracted from the 1-minute method to estimate the lipids measured in the 15-minute method within a 5-fold cross-validation framework.

We were able to infer the lipids measured in the 15-minute method (mean/median R^2 of 0.78/0.84, >575 lipids with $R^2 > 0.80$). Triglycerides and phosphatidylcholine showed best performance. Inferred lipids retained biological associations with age, sex and BMI, with regression coefficients showing near-identical concordance between measured and inferred lipids (R2 >0.98, 989 species). Associations with complex clinical outcomes highlighted similar performance (type-II diabetes onset at 5-years, R2 = 0.92, 989 species), with good concordance between significant species such as LPC O-20:1 (measured, β=0 .87 [0.82-0.94], FDR p = 2.09 x 10-2, inferred, β=0.87 [0.82-0.93], FDR p = 6.56 x 10-3).

Our new ultra-high-throughput platform enables accurate prediction of the plasma lipidome, providing opportunities to scale lipidomics into large population studies.

ORAL PRESENTATION

O-05

Melanoma cells remodel their lipidome yet maintain membrane biophysical properties during drug adaptation

PRESENTING AUTHOR

Aurelie Benfield

Aurelie Benfield[1], Reuben Young[2], Grégoire Philippe[3], Nicholas Condon[4], Nicolau Barbosa Palma Filho[1], Helmut Schaider[5], Stephen Blanksby[6], Sónia Troeira Henriques[3]

[1] School of Biomedical Sciences, Faculty of Health, Queensland University of Technology, Translational Research Institute, Australian Research Council Centre of Excellence for Innovations in Peptide and Protein Science; [2] Central Analytical Research Facility, Queensland University of Technology, Molecular Horizons and School of Science, University of Wollongong; [3] School of Biomedical Sciences, Faculty of Health, Queensland University of Technology, Translational

PRESENTATION

O-05

Monday 19

11:15

Riverside Ballroom

KEYWORDS

drug resistance

lipid metabolism

phospholipase digestion

leaflet asymmetry

Acquired drug resistance continues to undermine the long-term success of targeted cancer therapies, yet genetic explanations alone cannot account for the full complexity of resistant phenotypes. We investigated whether the cell membrane, its lipid composition, and biophysical states act as a non-genetic driver of resistance, and whether these adaptations expose new therapeutic vulnerabilities.

Using an integrated multiomics strategy, we profiled how membrane lipids and proteins involved in lipid metabolic pathway evolve across drug-naïve, drug-tolerant, and fully resistant BRAF-mutant melanoma cells treated with dabrafenib. This included quantitative lipidomics, outer-leaflet phospholipid mapping, and live-cell biophysical measurements providing side-by-side comparison of total and surface-resolved membrane composition across these states.

The targeted therapy drug dabrafenib, triggered extensive lipidome remodelling, most strikingly in drug-tolerant persister cells. We observed shifts in fatty-acid unsaturation, cholesterol and sphingomyelin levels, and increased exposure of negatively charged phospholipids at the cell surface. Yet despite these dramatic compositional changes, membrane fluidity and tension remained remarkably stable. Proteomics results showed drug-resistant cells rewired lipid metabolic pathways and preserved core membrane biophysical properties under therapeutic pressure.

These findings uncover a conserved membrane “biophysical homeostasis” that cancer cells maintain during drug adaptation, and highlight lipid-metabolic proteins as potential therapeutic targets in melanoma resistance.

ORAL PRESENTATION

O-06

ER translocon gating governs metabolic allocation through selective lipid routing

PRESENTING AUTHOR

Carl Mousley

Carl Mousley[1], Lee Thomasson[1], Emma Hadfield[1], Rob Steuart[1], Danielle Dye[1], Robert Trengove[1]

[1] Curtin University

PRESENTATION

O-06

Monday 19

11:27

Riverside Ballroom

KEYWORDS

- Endoplasmic reticulum
- Sec61 complex
- Lipid metabolism
- Metabolic allocation

The ER is the principal site of lipid synthesis, yet whether intrinsic properties of the ER membrane directly organise lipid metabolism remains unclear. Here, we investigated the consequences of defective ER translocon gating on lipid homeostasis using quantitative lipidomics, transcriptomics and genetic suppression in Saccharomyces cerevisiae. The translocon-gating mutant sss1-7 displayed a distinct lipidomic state that was partially restored by the channel-closing suppressor SEC61N302L, indicating that lipid remodelling is tightly coupled to translocon gating.

Defective gating promoted extensive neutral-lipid remodelling, characterised by expansion of TAG pools and selective enrichment of mixed-unsaturation and very-long-chain fatty acid (VLCFA)-containing species. Similar increases in C26-containing ceramides demonstrated coordinated redistribution of VLCFAs across glycerolipid and sphingolipid pathways. Analysis of diacylglycerol precursors revealed that this programme originates from a structurally biased precursor pool that is subsequently amplified during TAG synthesis.

Membrane lipid metabolism was remodelled in parallel. Broad depletion of monomethylphosphatidylethanolamine and reciprocal accumulation of phosphatidylcholine indicated increased flux through the phosphatidylethanolamine methylation pathway, while elevated lysophosphatidylcholine levels revealed enhanced membrane turnover and acyl editing. Despite substantial lipidomic changes, transcriptomic analysis identified minimal induction of lipid metabolic genes, indicating that remodelling occurs primarily through redistribution of existing lipid intermediates rather than transcriptional rewiring.

Importantly, constitutive ER stress failed to reproduce these features, demonstrating that the response is distinct from canonical unfolded protein response signalling. These findings identify ER translocon gating as a regulator of metabolic allocation and reveal a translocon-dependent programme of lipid compositional filtering that selectively routes structurally distinct lipid species through competing metabolic pathways.

ORAL PRESENTATION

O-07

Co-fractionation multiomics for mapping the composition of native membrane lipid-protein complex interactomes

PRESENTING AUTHOR

Liuyu Peng

Liuyu Peng[1], Richard Lee[2], Laetitia Hughes[2], Nikki Van Den Berg[2], Oliver Rackham[3], Gavin Reid[4], Aleksandra Filipovska[2]

[1] School of Chemistry, The University of Melbourne, The Kids Research Institute Australia; [2] The Kids Research Institute Australia, ARC Centre of Excellence in Synthetic Biology, University of Western Australia; [3] The Kids Research Institute Australia, ARC Centre of Excellence in Synthetic Biology, University of Western Australia, Curtin Medical School, Curtin University; [4] School of Chemistry, The University of Melbourne, Bio21 Molecular Science and Biotechnology Institute, The

PRESENTATION

O-07

Monday 19

11:39

Riverside Ballroom

KEYWORDS

- Lipidomics
- Multi-omics
- Cofractionation-MS
- lipid-protien interactions
- mitochondrial dysfunction

Structural lipids are essential components of cellular membranes, where they regulate cell and organelle morphology, dynamics and communication by stabilizing and modulating membrane proteins and protein complexes. Despite advances in mass spectrometry (MS)-based profiling of proteins, lipids and metabolites, scalable methods for systematically mapping lipid–protein interactomes remain limited. Here, we developed a co-fractionation MS strategy that integrates size-exclusion chromatography with multi-omics workflows to globally profile lipid–protein associations under native conditions. We applied this multi-omics analysis of protein–lipid engagement (MAPLE) approach to healthy controls and three mitochondrial disease models involving cardiolipin depletion or impaired assembly of large energy-converting protein complexes. MAPLE identified Complex I modules that depend on cardiolipin for biogenesis and Complex III subunits required for respirasome assembly. Loss of mitochondrial translation further revealed a requirement for cardiolipin in the terminal maturation and dimerization of ATP synthase. MAPLE provides a scalable platform for investigating native lipid–protein associations and offers new insights into membrane biology, mitochondrial organization and inter-organelle communication.

ORAL PRESENTATION

O-08

Effects of Reactive Oxygen Species on Liposomal Biomimetics of the Mitochondrial Membranes: A Biophysical Approach

PRESENTING AUTHOR

Susana Mesen-Porras

Susana Mesen-Porras[1], Jana Eisermann[1]

[1] University of Stuttgart, Institute of Physical Chemistry, 70569 Stuttgart, Germany

PRESENTATION

O-08

Monday 19

11:51

Riverside Ballroom

KEYWORDS

- Reactive oxygen species
- Cardiolipin
- Cellular ageing
- Biomimetic approach
- Oxidative stress

Reactive oxygen species (ROS), generated as by-products of mitochondrial oxidative phosphorylation, play dual roles in cellular physiology by mediating redox signaling but contributing to oxidative damage associated with metabolic, cardiovascular, and inflammatory diseases. The inner mitochondrial membrane (IMM) is particularly vulnerable to ROS-induced alterations that can disrupt lipid–protein interactions and compromise membrane function. To investigate these processes under controlled conditions, we study simplified biomimetic lipid vesicles as models of the IMM via scattering and spectroscopic techniques.

Phosphatidylcholine (PC)-based vesicles containing varying amounts of cardiolipin (CL), phosphatidylethanolamine (PE), or phosphatidylglycerol (PG) were prepared to examine how individual phospholipids influence membrane organization and oxidative reactivity. Our findings indicate that CL affects membrane properties differently from PE and PG. Above a threshold of approximately 5 wt% CL, hydroxyl radical penetration into the lipid bilayer appears to be restricted, whereas increasing PE or PG content correlates with enhanced hydroxyl radical levels. These results identify CL as an active modulator of redox processes and suggest a potential role in mechanisms linked to cellular ageing.

This biomimetic approach provides mechanistic insight into how oxidative modifications shape membrane organization and signaling, while the transition toward more native-like systems – together with matching protein systems (e.g. cytochrome c) – will further improve the physiological relevance of these findings.

ORAL PRESENTATION

O-09

Characterization of a lipid droplet-associated microalgal protein as a novel phosphatidic acid biosensor

PRESENTING AUTHOR

Bilge Ercan

Bilge Ercan[1], Nian Kee Tan[1], Julia Lin[2], Jingyao You[1], Stephen M. Butler[1], Kate A. Jolliffe[1]

[1] The University of Sydney & ARC Centre of Excellence for Innovations in Peptide and Protein Science (CIPPS); [2] The University of Sydney

PRESENTATION

O-09

Monday 19

12:03

Riverside Ballroom

KEYWORDS

phosphatidic acid

lipid droplets

amphipathic helix

lipid packing defects

fluorescent biosensor

Phosphatidic acid (PA) is an essential signaling lipid in eukaryotic cells. Despite its importance, biosensors recognizing PA-rich membranes are in short supply, partly because there is no known consensus motif for detecting PA. Here, we demonstrate that a lipid droplet-associated protein domain from Chlamydomonas reinhardtii can be repurposed as a biosensor for PA-rich lipid membranes. We found that the lipid binding domain of Delayed in TAG Hydrolysis 1 (DTH1LBD) of Chlamydomonas reinhardtii binds to artificial lipid droplets (aLDs) containing Phosphatidylethanolamine (PE) or PA. In contrast to the PE-binding observed on aLDs, DTH1LBD does not recognize PE in bilayer membranes in vitro. Cysteine mutagenesis coupled with covalent incorporation of various environment-sensitive fluorophores and fluorescent confocal microscopy imaging of PA-containing giant unilamellar vesicles showed that the N-terminal amphipathic helices of DTH1LBD are sufficient to recognize PA-containing membranes. Our findings demonstrate that the N-terminal amphipathic helices of DTH1LBD recognize PA-rich membranes through lipid headgroup and membrane packing defects imparted by the unique structure of PA.

ORAL PRESENTATION

O-10

Comprehensive, multi-ancestry lipidomic GWAS reveals molecular pathways linking the human lipidome to cardiometabolic, neurodegenerative, and cancer risk

PRESENTING AUTHOR

Corey Giles

Corey Giles[1], Minna Karjalainen[2], Tingting Wang[1], Kevin Huynh[1], Changyu Yi[1], Jingqin Wu[1], Aleksandar Dakic[1], Agus Salim[1], Kari North[3], Terho Lehtimäki[4], Olli Raitakari[5], Mika Kähönen[6], Eric Moses[7], Rima Kaddurah-Daouk[8], John Blangero[9], Joanne Curran[9], Mika Ala-Korpela[2], Johannes Kettunen[2], Peter Meikle[1]

[1] Baker Heart and Diabetes Institute, Melbourne, Victoria, Australia; [2] Systems Epidemiology, Research Unit of Population Health, Faculty of Medicine, University of Oulu and Biocenter Oulu, Oulu, Finland; [3] Department of Epidemiology, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA; [4] Department of Clinical Chemistry, Fimlab Laboratories, and Finnish Cardiovascular Research Center Tampere, Faculty of Medicine and Health Technology, Tampere University,

PRESENTATION

O-10

Monday 19

14:30

Riverside Ballroom

KEYWORDS

- Genomics
- Lipidomics
- Cancer
- Alzheimer's disease
- Genes

Dysregulation of lipid metabolism is intimately linked to the aetiology and pathogenesis of human disease, yet the genetic architecture regulating lipid metabolism remains incompletely defined. To address this, we performed a comprehensive, multi-ancestry lipidomic genome-wide association study of 881 species across 50 classes in up to 16,814 participants from seven non-Finnish European, Finnish, and admixed Mexican American cohorts.

Meta-analysis identified 264,239 significant SNP-lipid associations (P<3.73x10-10), comprising 24,400 variants across 162 genomic regions. Nearly all lipid species (808; 92%) had at least one genome-wide significant association, with hexosylceramide and cholesteryl ester species being the most polygenic. Colocalisation with NMR-derived lipoprotein traits revealed that the majority of these associations (101; 62%) were lipoprotein-independent.

Missense variants provided near end-to-end pathway mapping of sphingolipid metabolism from de novo synthesis (SPTLC1, SPTLC3) and ceramide synthesis (CERS2, CERS4, CERS5) to desaturation (DEGS1), glycosphingolipid metabolism (UGT8, GAL3ST1, A4GALT, ST3GAL5), and sphingosine-1-phosphate synthesis and transport (SPHK2, MFSD2B). Associations between deoxyceramides and PHGDH, PSAT1, PSPH, and CPS1 linked serine availability to atypical sphingolipid production. Several associations were ancestry-restricted, including missense MFSD2B and DEGS1 variants observed only in admixed Mexican American participants.

Colocalisation with cancer GWAS identified shared signals between CERS2-associated hexosylceramide and basal cell carcinoma and prostate cancer, and between very-long-chain lysophosphatidylcholine and prostate cancer at VPS53. Mendelian randomisation provided consistent evidence that dihexosylceramide species causally influence Alzheimer's disease risk.

Together, these findings establish a comprehensive, multi-ancestry map of lipid metabolism and identify molecular pathways linking lipid biology to human disease.

ORAL PRESENTATION

O-11

Lipoprotein and lipid perturbations induced by acute and chronic inflammation

PRESENTING AUTHOR

Samantha Lodge

Samantha Lodge[1], Fiona Wood[2]

[1] Murdoch University; [2] Burn Injury Research Unit, School of Biomedical Sciences, University of Western Australia, Perth, Australia, Fiona Wood Foundation, Perth, Australia, Burns service of Western Australia, WA Department of Health, Murdoch, Australia.

PRESENTATION

O-11

Monday 19

14:45

Riverside Ballroom

KEYWORDS

Lipoproteins

Burn Injury

Nuclear Magnetic Resonance

Paediatric burn trauma is a global health concern associated with long-term adverse health outcomes, including increased morbidity and mortality. However, the underlying mechanisms and pathological processes that predispose children with burn injuries to these long-term risks remain poorly understood. In this study, 1H Nuclear Magnetic Resonance (NMR) spectroscopy was used to comprehensively characterise longitudinal lipoproteomic changes in blood samples collected from 142 paediatric burn patients during both the acute injury phase and long-term recovery.

During the acute phase of burn injury, 55% of lipoprotein main fractions and subfractions were significantly altered compared with non-burn controls. The burn cohort exhibited increased low-density lipoprotein (LDL) triglyceride concentrations, an elevated apolipoprotein B100/apolipoprotein A1 (ApoB100/ApoA1) ratio, and reduced high-density lipoprotein (HDL) levels. At one-year post-burn injury, 53% of lipoprotein fractions and subfractions remained significantly altered relative to non-burn controls. This persistent lipoproteomic profile was indicative of increased cardiovascular risk, characterised by a reduction in larger LDL subfractions and an increase in smaller, more atherogenic LDL subfractions. Significant changes in the lipid composition of very low-density lipoprotein (VLDL), intermediate-density lipoprotein (IDL), LDL, and HDL were also observed at one year post-burn, with the greatest compositional shifts occurring in IDL compared with non-burn controls.

These findings demonstrate that paediatric burn injury induces substantial lipoproteomic alterations during the acute phase that persist well into long-term recovery. The sustained metabolic disturbances identified in this study may contribute to an increased risk of future cardiovascular disease following paediatric burn injury.

ORAL PRESENTATION

O-12

A lipidomic risk score improves cardiovascular risk assessment in statin treatment across large cohorts

PRESENTING AUTHOR

Jingqin Wu

Jingqin Wu[1], Aleksandar Dakic[1], Tingting Wang[1], Thomas Meikle[1], Kevin Huynh[1], Changyu Yi[1], Habtamu Beyene[1], Agus Salim[2], Michelle Cinel[1], Nat Mellett[1], Thy Duong[1], Anjali Bhagwat[1], Jonathan Shaw[1], Dianna Magliano[1], John Simes[3], John McNeil[4], Paul Lacaze[5], Chenglong Yu[5], Rory Wolfe[5], Jean Yang[6], Stuart Grieve[7], Clara Chow[8], Stephen Vernon[9], Michael Gray[9], Gemma Figtree[10], Jedidiah Morton[1], Bernie Pope[11], Melinda Carrington[1], Corey Giles[1], Peter Meikle[1]

[1] Baker Heart and Diabetes Institute, Melbourne, Australia; [2] Baker Department of Cardiometabolic Health, The University of Melbourne, Melbourne, Australia; [3] National Health and Medical Research Council Clinical Trials Centre, University of Sydney, Sydney, NSW, Australia; [4] School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia.; [5] School of Public Health and Preventive Medicine, Monash University, Melbourne, Australia; [6] School of

PRESENTATION

O-12

Monday 19
15:00
Riverside Ballroom

KEYWORDS

- lipidomics
- cardiovascular risk prediction
- statin therapy
- residual cardiovascular risk
- biomarkers

Despite substantial reductions in low-density lipoprotein cholesterol (LDL-C) with statin therapy, residual cardiovascular disease (CVD) risk persists. Whether lipidomic risk captures treatment-responsive biology while retaining prognostic value during statin therapy remains unclear. We evaluated whether a lipidomics-derived cardiovascular risk score (L.CVDRisk) reflects statin-associated biological response and provides prognostic information across primary and secondary prevention settings. Associations between statin use, lipid measures, and L.CVDRisk were examined using multivariable linear regression in the LIPID trial (n=4,811), with validation in ASPREE (n=2,945) and AusDiab (n=8,082) cohorts. Cox proportional hazards models were used to assess associations of L.CVDRisk with subsequent CVD events in LIPID and incident CVD in the validation cohorts, including analyses stratified by statin use and baseline LDL-C. Residual risk detection was evaluated among individuals achieving LDL-C<3.0 mmol/L after statin therapy. Mediation analyses quantified the contribution of L.CVDRisk to statin-associated reductions in CVD risk. Statin therapy was associated with significant reductions in L.CVDRisk (β=-4.05%; 95% CI:-4.42–-3.77). L.CVDRisk independently of LDL-C predicted CVD events (HR=2.08; 95% CI:1.59-2.72) and in joint models with LDL-C, L.CVDRisk remained predictive (HR=1.28; 95% CI: 1.16-1.41). Among statin-treated individuals achieving LDL-C<3.0 mmol/L(n=1,368), L.CVDRisk identified subgroups accounting for 44% of subsequent events with an association between L.CVDRisk reclassification and future CVD (χ^2 test, p=0.0002). Adding L.CVDRisk to LDL-C improved risk classification (NRI=14.9%; p=0.0004). Mediation analyses suggested that L.CVDRisk mediated 45% of the effect of statin therapy on CVD risk (p<0.001). L.CVDRisk is a treatment-responsive and prognostically informative biomarker that complements LDL-C and enables refined detection of residual CVD risk.

ORAL PRESENTATION

O-13

When recovery differs: Sex-based lipidomic changes after childhood burn injury

PRESENTING AUTHOR

Monique J Ryan

Monique J Ryan[1], Eva Kierath[2], Samantha Lodge[3], Reika Masuda[1], Jacqueline Davis[4], Nina D'Vaz[4], Lucy W Barrett[5], Nicola Gray[6], Elaine Holmes[3], Fiona M Wood[7], Luke Whiley[8]

[1] Australian National Phenome Centre, Murdoch University, Centre for Computational Systems Medicine, Murdoch University; [2] Centre for Computational Systems Medicine, Murdoch University, Fiona Wood Foundation, Fiona Stanley Hospital, Burn Injury Research Unit, University of Western Australia; [3] Centre for Computational Systems Medicine, Murdoch University; [4] The Kids Research Institute Australia, Perth Children's Hospital; [5] Fiona Wood Foundation, Fiona Stanley Hospital, Burn

PRESENTATION

O-13

Monday 19

16:22

Riverside Ballroom

KEYWORDS

- Lipidomics
- Liquid chromatography-mass spectrometry
- Sex differences
- Nuclear magnetic resonance
- Paediatric burn injury

Childhood burn injuries represent a significant global health issue with potential long-term implications for physical and psychological well-being, even when the burn is non-severe (<15% total body surface area). Promoting a good quality of life for all children requires a deeper understanding of the factors that may shape recovery. One such factor, biological sex, remains overlooked in burn research and clinical care despite emerging evidence of worsened short- and long-term outcomes for females compared to males post-burn. This study investigates sex-specific responses to non-severe paediatric burn injury with a focus on lipid metabolism and systemic inflammation, as these interlinked pathways can influence recovery and long-term health post-burn.

Children under five years old were recruited at hospital admission through the Childhood Burn Injury Biobank (n=118) and followed longitudinally for 2 years post-injury, with matched controls from ORIGINS (n=299). Plasma lipid profiles and systemic inflammatory markers (GlycA/GlycB) were assessed using LC-MS/MS and NMR, respectively.

Both male and female children exhibited early post-burn disruptions in fatty acids and lysophospholipids (adj. p-values < 0.001). However, only females demonstrated persistent changes beyond one year post-injury, including elevated arachidonic acid (FA20:4), reduced monoacylglycerols, and a larger spike in acute GlycA/GlycB. Males, when 'recovered' (2 years post-injury), appeared similar to non-burn controls except for some persistent elevations in fatty acid species.

These findings reveal a distinct, sex-specific biological response to burn injury that may place young females at increased risk of sustained metabolic dysregulation. Recognising and addressing such differences is vital to promoting equitable recovery.

ORAL PRESENTATION

O-14

Distinct plasma lipidomic patterns across cardiovascular outcomes with contrasting profiles for stroke and heart failure

PRESENTING AUTHOR

Jaya Ajay Singh

Jaya Ajay Singh[1,3], Luke Whiley[1,2], Monique Ryan[3], Reika Masuda[1,3], Leon Flicker[4], Graeme J. Hankey[5], Bu Yeap[6], Jeremy Nicholson[7], Elaine Holmes[1,7], Nicola Gray[1,8]

[1] Centre for Computational and Systems Medicine, Health Futures Institute, Murdoch University, Murdoch, Western Australia, Australia; [2] School of Diagnostic and Therapeutic Sciences and Curtin Medical Research Institute, Curtin University, Perth, Western Australia, Australia; [3] Australian National Phenome Centre, Murdoch University, Murdoch, Western Australia, Australia; [4] Medical School and WA Centre for Health and Ageing, University of Western Australia, Perth, Western Australia,

PRESENTATION

O-14

Monday 19

16:34

Riverside Ballroom

KEYWORDS

Stroke

Cardiovascular Health

Lipidomics

Fatty acids

Mass Spectrometry

BACKGROUND

Cardiovascular disease (CVD) comprises clinically distinct outcomes that may have different underlying lipid profiles. Combining these outcomes into a composite endpoint may obscure associations specific to individual conditions.

AIM

We examined plasma lipidomic associations with incident stroke, coronary heart disease (CHD), heart failure, and composite CVD in older men.

METHODS

Plasma lipid quantification of 695 lipids was performed using liquid chromatography-tandem mass spectrometry (LC-MS/MS) in 689 men aged 79–97 years from the community-based Health in Men Study (HIMS) who were free of CVD at baseline. Participants were followed for approximately 8 years for first incident stroke (n=55), CHD (n=83), heart failure (n=77), and composite CVD (n=189) events. Associations were tested using Cox proportional hazards models adjusted for age, body mass index, diabetes, hypertension, current smoking, and lipid-lowering medication. Lipid associations were further examined by lipid family, lipid-set enrichment, carbon-chain length, and degree of unsaturation.

RESULTS

Global lipidome analyses showed different association patterns across outcomes, involving triacylglycerols (TAGs), diacylglycerols (DAGs), phosphatidylethanolamines (PEs), PE plasmalogens [PE(P)s], and sphingomyelins (SMs). Stroke showed higher-hazard estimates, whereas heart failure showed predominantly lower-hazard patterns. At the lipid family level, contrasting estimates for stroke and heart failure were observed for TAGs (HR 1.21 vs 0.82), DAGs (HR 1.22 vs 0.81), and SMs (HR 1.14 vs 0.82); PE estimates were 1.24 and 0.98, respectively. These patterns were attenuated towards the null for composite CVD. When lipid associations were examined according to carbon number and unsaturation TAG estimates for stroke shifted towards higher hazard with increasing carbon number and unsaturation whereas for heart failure, higher-carbon TAGs were predominantly associated with lower hazard, with the strongest lower-hazard estimates among more polyunsaturated species.

CONCLUSION

Plasma lipidomic patterns differed across cardiovascular outcomes, with contrasting profiles for stroke and heart failure. Combining heterogeneous cardiovascular outcomes may attenuate lipid patterns apparent when individual phenotypes are examined separately.

ORAL PRESENTATION

O-15

Identification of novel lipidomic biomarkers in first trimester maternal serum for the screening of aneuploidy specifically trisomy 21

PRESENTING AUTHOR

Khoa Lam

Khoa Lam[1], Tomas Rozek[1], Jennie Louise[1]

[1] SA Pathology and Adelaide University

PRESENTATION

O-15

Monday 19

16:46

Riverside Ballroom

KEYWORDS

lipidomics

screening

machine learning

BACKGROUND

First trimester screening for trisomy 21 (T21) integrates maternal age-related risk with biochemical markers (free βhCG and PAPPA) and ultrasound derived nuchal translucency (NT) measurements using Bayesian modelling. Although this combined test achieves approximately 90% detection at a 5% false-positive rate, its performance is constrained by biological variability and dependence of NT assessment. Lipidomics offers a direct approach to identify novel biomarkers reflective of placental and fetal pathophysiology.

AIM

To identify novel first-trimester maternal serum lipid biomarkers associated with T21 and develop a lipid-based screening algorithm independent of NT measurement.

METHODS

A two-stage discovery-validation study design was implemented. Stage 1, untargeted lipidomic profiling was performed on 802 archived first trimester maternal serum samples using high-resolution quadrupole time-of-flight mass spectrometry (AB Sciex TripleTOF 5600+). Supervised machine learning statistical analysis were applied to identify discriminatory lipid species. Stage 2 comprised targeted validation of shortlisted lipids using an independent cohort of more than 5000 samples using high-throughput triple quadrupole (AB Sciex Triple Quad 6500+) mass spectrometer to evaluate algorithm performance.

RESULTS

Untargeted lipidomic analysis detected 2,229 lipid features per sample. Machine learning based feature selection reduced this dataset to 50 highly discriminatory lipid features. These candidate lipids were transferred to the targeted validation cohort for assessment to predictive performance with a lipid-based screening algorithm.

Expected outcome: This large-scale lipidomic study demonstrates the feasibility of translating untargeted discovery into high throughput targeted validation for prenatal screening. A lipid-based algorithm may provide a robust, ultrasound independent approach for improved first trimester T21 risk assessment.

ORAL PRESENTATION

O-16

Defining the potential of alkylglycerol supplementation for treating diverse metabolic disorders

PRESENTING AUTHOR

Richard G. Lee

Richard G. Lee[1], Oliver Rackham[1], Aleksandra Filipovska[2]

[1] Curtin University, The Kids Research Institute Australia, ARC Centre of Excellence in Synthetic Biology; [2] The Kids Research Institute Australia, ARC Centre of Excellence in Synthetic Biology

PRESENTATION

O-16

Tuesday 20

09:30

Riverside Ballroom

KEYWORDS

- Ether lipids
- Multi-omics
- organelle communication
- metabolic diseases

Mutations in genes regulating lipid metabolism and organelle biogenesis have been associated with diverse multi-system disorders that have varied clinical presentations, many of which have no known cure. Our previous work has showed that disrupted ether lipid metabolism and mitochondrial dysfunction lies at the centre of many of these disorders, and that supplementation with ether lipid precursors called alkylglycerols can reverse mitochondrial phenotypes associates with inter-organelle biogenesis breakdown. To investigate the mechanisms underlying this rescue, we performed whole-cell lipidomics, proteomics, transcriptomics, and metabolomics across nine model cell lines of organelle biogenesis disorders that show rescue of mitochondrial dysfunction upon alkylglycerol supplementation. This analysis showed that alkylglycerols restored defects in disrupted pathways including those related to organelle communication and lipid metabolism. Live cell and immunofluorescence imaging with organelle markers further showed diverse changes in organelle morphologies underpinning these metabolic rescues. Finally, we used CRISPR-Cas9 gene editing to ablate key genes highlighted from these analyses to validate their essential role in alkylglycerol-mediated rescue of inter-organelle biogenesis, confirming key metabolic regulators that are central to the rescue response. This work has characterised the diverse role of ether lipids in supporting inter-organelle biogenesis and demonstrated the broad clinical potential of alkylglycerol supplementation therapies in the treatment of metabolic disorders.

ORAL PRESENTATION

O-17

Non-Canonical Sphingolipid Remodeling in Adipose Tissue Tracks Insulin Resistance and Type 2 Diabetes Remission

PRESENTING AUTHOR

Elin Chorell

Tova Eurén[1], Timotej Strmen[1], Helena Edlund[1], Pär Steneberg[1], Julia Otten[1], Elin Chorell[1]

[1] Umeå University

PRESENTATION

O-17

Tuesday 20

09:45

Riverside Ballroom

KEYWORDS

- type 2 diabetes
- sphingolipidomics
- adipose tissue
- mass spectrometry
- mRNA sequencing

Adipose tissue insulin resistance (IR) is a central determinant of type 2 diabetes (T2D). We asked whether non-canonical sphingolipid metabolism in subcutaneous white adipose tissue (WAT), particularly 1-deoxyceramide formation, distinguishes metabolically unhealthy adipose expansion from restored insulin sensitivity during T2D remission.

We integrated targeted sphingolipidomics with mRNA-seq in human adipose tissue from individuals with T2D sampled before and after a diet-induced remission intervention, alongside BMI-matched normoglycaemic controls. Matched plasma and adipose tissue secretome samples were analysed to assess whether adipose-derived sphingolipids reflect systemic lipid stress. Parallel analyses were performed in high fat diet (HFD)-induced IR mice and in HFD mice treated with ATX-304, a mitochondrial uncoupler that preserves insulin sensitivity despite maintained weight gain, which allows us to disentangle IR from adipose tissue expansion. WAT, plasma and adipose-conditioned media were profiled to link tissue lipid remodeling with endocrine sphingolipid release.

Across species, IR adipose tissue showed enrichment of saturated and monounsaturated 1-deoxyceramides compared with metabolically healthy controls. The same lipid class was reduced in humans achieving diet-induced T2D remission and depleted in high-fat-fed ATX-304-treated mice, despite persistent obesity. These changes contrasted with canonical ceramides and were accompanied by transcriptional remodeling of pathways related to amino-acid handling, de novo sphingolipid synthesis, elongation and mitochondrial stress. Adipose secretome analyses further suggested that 1-deoxyceramides may be exported from dysfunctional WAT and contribute to circulating sphingolipid signatures.

These findings identify adipose 1-deoxyceramide metabolism as a conserved lipid mechanism linking impaired WAT insulin sensitivity with T2D and as a reversible feature of diabetes remission.

ORAL PRESENTATION

O-18

Targeting Endothelial Plasma Membrane Cholesterol to Mitigate Atherogenesis

PRESENTING AUTHOR

Jacob (Yanfei) Qi

Jacob (Yanfei) Qi[1]

[1] Heart Research Institute, University of Sydney

PRESENTATION

O-18

Tuesday 20

10:00

Riverside Ballroom

KEYWORDS

- Cholesterol transport
- Atherosclerosis
- Endothelial cells
- Inflammation

Despite the success of lipid-lowering therapies, atherosclerosis remains a leading cause of cardiovascular morbidity and mortality. Beyond reducing circulating risk factors, enhancing the resilience of the vascular wall to a pro-atherogenic environment may represent a new therapeutic strategy. Circulating cholesterol has long been recognised as a major pro-atherogenic factor; however, how cholesterol is handled by vascular cells, particularly after uptake by endothelial cells, the innermost lining of the blood vessels, and how this process shapes vascular responsiveness to pro-atherogenic stimuli, remain poorly understood.

In this study, we identify ORP2 as a unique intracellular cholesterol transporter that specifically delivers both biosynthesised and endocytosed cholesterol to the plasma membrane. Knockdown of ORP2 blunted cholesterol transport to the endothelial plasma membrane, suppressed endothelial inflammatory activation, and reduced monocyte recruitment to the endothelial monolayer. To further define the physiological role of ORP2, we generated the first endothelial cell-specific ORP2 knockout mouse model. On an Apoe-knockout background and under a high-fat, high-cholesterol diet, endothelial ORP2 deficiency reduced atherosclerotic plaque burden by more than 50%.

These findings reveal endothelial plasma membrane cholesterol as a previously underappreciated driver of vascular inflammation and atherogenesis. This study opens a new avenue for cholesterol-targeted therapy: beyond suppressing cholesterol biosynthesis or promoting cholesterol clearance, targeting intracellular cholesterol mobilisation and subcellular distribution can confer anti-atherosclerotic benefits.

ORAL PRESENTATION

O-19

Integrative lipidomic analysis across six cohorts reveals biological pathways underlying cardiovascular disease risk

PRESENTING AUTHOR

Tingting Wang

Tingting Wang[1], Kevin Huynh[1], Changyu Yi[1], Habtamu Beyene[1], Thy Duong[1], Natalie A Mellett[1], Thomas G Meikle[1], Jingqin Wu[1], Aleksandar Dakic[1], Michelle Cinel[1], Anjali Bhagwat[1], Eric K Moses[2], John J McNeil[3], Paul Lacaze[3], Roger L Milne[4], John Simes[5], John Chalmers[6], Bernard Pope[7], Gemma A Figtree AM[8], Jean Yang[8], Beilei Bian[8], Jonathan E Shaw[1], Dianna J Magliano[1], ACDC Investigators[1], Corey Giles[1], Peter J Meikle[1]

[1] Baker Heart and Diabetes Institute; [2] Menzies Institute for Medical Research, University of Tasmania; [3] School of Public Health and Preventive Medicine, Monash University; [4] Cancer Epidemiology Division, Cancer Council Victoria; [5] National Health and Medical Research Council of Australia (NHMRC) Clinical Trials Centre, University of Sydney; [6] The George Institute for Global Health, The University of New South Wales; [7] Australian BioCommons, Faculty of Medicine, Dentistry and

PRESENTATION

O-19

Tuesday 20

10:15

Riverside Ballroom

KEYWORDS

CVD

Lipidome

Diabetes

BACKGROUND

Metabolic dysfunction is intimately linked to cardiovascular disease (CVD), yet the specific lipid pathways mediating this relationship remain poorly characterised. While lipidomics enables comprehensive assessment of lipid metabolism, studies to date have been restricted to single cohorts or cross-sectional associations, limiting generalisability and mechanistic insights.

METHODS

Using the Australian Cardiovascular disease Data Commons (ACDC; https://commons.heartdata.baker.edu.au/login), we performed an integrative lipidomic analysis across six prospective cohorts comprising approximately 30,000 participants (AusDiab, BHS, MCCS, ASPREE, LIPID, and ADVANCE). Incident CVD outcomes were harmonised across cohorts, and associations between circulating lipid species, established cardiometabolic risk factors, and future CVD events were evaluated using cohort-specific survival analyses followed by meta-analysis. Enrichment and mediation analyses were performed to identify dysregulated pathways and quantify lipid-mediated effects linking diabetes and CVD.

RESULTS

Across cohorts, 28% of 657 measured lipid species were associated with diabetes and 32% with smoking, and 30% predicted incident CVD. Enrichment analyses identified ceramide, ether lipid, and omega-3/omega-6 pathways as key contributors to CVD risk. Mediation analyses revealed 45 lipid species (6.8%) that mediated the association between diabetes and CVD, explaining 3–15% of the total effect. Variance decomposition showed that 25.9% (95% CI 9.7–41.4%) of lipidomic variance associated with CVD was shared with diabetes, indicating substantial overlap between diabetes- and CVD-related lipidomic signatures.

CONCLUSION

s: This multi-cohort analysis demonstrates that lipidomic alterations link metabolic dysfunction to CVD. These findings provide mechanistic insights and identify potential lipid biomarkers and pathways for improved CVD risk prediction, prevention, and therapeutic intervention.

ORAL PRESENTATION

O-20

Free fatty acid consumption, metabolism and biosynthesis in honeybees

PRESENTING AUTHOR

Sam Hodgkinson

Sam Hodgkinson[1], Daniel Stabler[1], Phil Stevenson[2]

[1] University of Southampton; [2] Royal Botanical Gardens Kew

PRESENTATION

O-20

Tuesday 20

11:15

Riverside Ballroom

KEYWORDS

- Honeybee
- Apis mellifera
- biosynthesis
- fatty acid
- LC-MS

Honeybees (Apis mellifera) are obligate palynivorous and nectivorous pollinators, obtaining all dietary nutrients from pollen and nectar. Pollen provides honeybees with their main source of dietery protein and lipid, but the pollen of insect-pollinated plant species vary greatly in their quality and quantity of these nutrients. Therefore, honeybees must balance intake of different pollen species to construct a nutritional profile suited to their requirements. Of particular research interest is the dietary ratio of omega-3 to omega-6 polyunsaturated fatty acids (PUFAs), and the downstream impacts on physiology.

Using chemically defined, holidic diets alongside deuterated water, we tested how bees respond behaviourally and physiologically to diets that were defined by their fatty acid composition. Diets were formulated using a representative range of concentrations of reagent-grade fatty acids, to measure dose-dependent physiological responses. Using newly-emerged honeybees, we confined bees to these diets for five days, before dissecting to obtain pooled samples of key tissues. Lipid fractions were extracted using a dichloromethane/methanol mixture, and using LC-MS/MS, the limitations of their de-novo fatty acid synthesis as well as the differential metabolism of fatty acids in distinct tissues throughout the body was measured

We found that honeybees exhibit extreme nutrient balancing behaviour to mitigate overconsumption of free PUFAs in the diet, despite high relative absorption and generation of triglyceride stores. Validated using NMR, we also found that bees were unable to synthesise ‘essential’ PUFAs, but can synthesise long-chain saturated fatty acids, along with monounsaturated omega-9 acids.

ORAL PRESENTATION

O-21

Untargeted lipidomics and proteomics to expand molecular coverage of human milk

PRESENTING AUTHOR

Alexandra George

Alexandra George[1], Thy Duong[1], Peter Meikle[1], Kevin Huynh[1]

[1] Baker Heart and Diabetes Institute

PRESENTATION

O-21

Tuesday 20

11:27

Riverside Ballroom

KEYWORDS

untargeted

human milk

proteomics

lipidomics

nutrition

Compared with other areas of ‘omics, human milk remains comparatively under-characterised, despite its importance for infant nutrition and development. Our targeted lipidomics platform currently quantifies ~800 lipid species in human milk, but targeted approaches are inherently limited to predefined analytes. The complementary human milk proteome also remains incompletely characterised. In this exploratory study, we applied complementary untargeted lipidomics and proteomics to expand molecular coverage of human milk. Matched colostrum and mature human milk samples were extracted for lipids using butanol–methanol (1:1) and aqueous butanol–methanol (1:2:2) protocols, to maximise recovery of chemically diverse lipid species, and analysed using 60-minute reverse-phase liquid chromatography–quadrupole time-of-flight mass spectrometry (LC-QTOF-MS). Remaining protein pellets underwent tryptic digestion followed by LC-QTOF-MS proteomic analysis. MS-DIAL and MSFragger were used for feature detection. To date, >1000 lipid species (of ~10,000 features) have been curated with consistent accurate mass, retention time and fragmentation characteristics. Distinct lipid profiles were observed between colostrum and mature milk and between extraction methods, demonstrating complementary recovery of the milk lipidome. This analysis expanded coverage to include new immune-associated lipid classes, including gangliosides and lysophospholipids, revealing marked remodelling across lactation. Proteomic profiling identified >240 proteins and similarly demonstrated substantial differences between colostrum and mature milk, consistent with the transition from an immune-enriched colostrum proteome to a distinct mature milk proteome. Together, these findings expand the molecular landscape of human milk and provide a resource for future targeted assay development and investigation of milk bioactives in infant growth, immune development and long-term health.

ORAL PRESENTATION

O-22

High marbled Wagyu beef vs. low marbled Water buffalo beef – are there differences in the lipidome of muscle between both breeds?

PRESENTING AUTHOR

Dirk Dannenberger

Dirk Dannenberger[1], Beate Fuchs[1], Christina Galuska[1], Tobias Becker[2], Almut Scheler[3]

[1] Research Institute for Farm Animal Biology (FBN Dummerstorf), Wilhelm-Stahl-Allee 2, 18196 Dummerstorf, Germany; [2] Marblelution GmbH, Friedrichswerther Str. 4A, Nessetal OT Haina,Germany; [3] Institute of Livestock Farming, State Research Center of Agriculture and Fisheries Mecklenburg-Vorpommern, Dummerstorf, Germany

PRESENTATION

O-22

Tuesday 20

11:39

Riverside Ballroom

KEYWORDS

Wagyu

Water buffalo

muscle

PUFA

Intramuscular fat (IMF) deposition in beef muscle is one of the most prominent traits determining beef quality and palatability attributes. In this study longissimus muscles from male Wagyu cattle (n=15) and male Water buffalo (n=15) bred and raised under extensive conditions in Germany were included. The IMF contents differed between 21.2% (Wagyu) and 2.4% (Water buffalo). The distribution of lipid classes (TLC) revealed significant differences, with the largest variations for FFA (2.8 vs. 13%) and TG (85 vs. 60%), and low variation of PLs in muscle of Wagyu and Water buffalo, respectively. TG species were dominated by saturated fatty acids (Wagyu) and LC-PUFA (Water buffalo), with highest deposition of TG(16:0/16:0/18:1) and TG(16:0/18:1/18:1), and TG(16:0/18:2/22:6) and TG(14:/16:1 /18:1) in Wagyu and Water buffalo, respectively. PC and PE were the most abundant PL classes in muscle of both breeds and represent more than 80% of total PLs. PC and PE species containing LC-PUFA was elevated in muscle of Water buffalo with highest peak area ratios for PE(O-16:2/20:5), PC(18:2/20:5) or PE(16:1/20:5) compared to Wagyu muscle. Based on this, higher levels of lysophospholipids of LPE and LPC, were detected in muscle of Water buffalo compared with Wagyus. In accordance to TG species distribution, higher peak area ratio of PC and PE containing saturated fatty acids, PE(16:0/18:1, PC16:0/16:0), were detected in Wagyu compared to Water buffalo.

In conclusion, the lipidomic approach revealed differences in muscle of high marbled Wagyu beef and low marbled Water buffalo beef and can improve the understanding of muscle lipid metabolism/physiology.

ORAL PRESENTATION

O-23

Whole-grain rye feeding drives tissue-specific host-metabolic and microbial responses beyond conventional clinical endpoints

PRESENTING AUTHOR

Ambrin Farizah Babu

Ambrin Farizah Babu[1], Marjo Tuomainen[2], Artur Sannikov[1], Tomi Tuomainen[3], Ville Koistinen[1], Topi Meuronen[1], Marko Lehtonen[4], Olli Kärkkainen[4], Iman Zarei[2], Retu Haikonen[2], Anna Kårlund[1], Pasi Tavi[3], Kati Hanhineva[1]

[1] Food Sciences Unit, Department of Life Technologies, University of Turku, Finland; [2] School of Medicine, Institute of Public Health and Clinical Nutrition, University of Eastern Finland, Finland; [3] A. I. Virtanen Institute for Molecular Sciences, University of Eastern Finland, Finland; [4] School of Pharmacy, Faculty of Health Science, University of Eastern Finland, Finland

PRESENTATION

O-23

Tuesday 20
11:51
Riverside Ballroom

KEYWORDS

whole-grain rye

fermentation

odd-chain lysophospholipids

acylcarnitines

metabolic diseases

Clinical evaluation of metabolic diseases relies on circulating biomarkers that provide limited insight into tissue-specific metabolism. Whole-grain rye, a cereal linked to metabolic health, is commonly consumed as sourdough-fermented bread, where fermentation structurally and biochemically transforms this complex food matrix. Consequently, how these transformations influence organ-specific metabolic responses remains unknown.

To investigate this, mice received low-fat (LF) diets supplemented with fermented (FRB) or unfermented rye bread (URB), alongside LF and high-fat (HF) controls. Clinical phenotyping and [18F]FDG-PET imaging were integrated with fecal metagenomics and untargeted metabolomics across plasma, heart, muscle, intestinal tissues, and feces.

Despite indistinguishable body weight, circulating clinical markers, and tissue glucose uptake, both rye diets enriched fibre-fermenting Lachnospiraceae and reduced Adlercreutzia mucosicola, accompanied by increased proximal intestinal phosphatidylcholines and depletion of distal odd-chain lysophospholipids (LPC15:0, LPC15:1, and LPE17:0). FRB uniquely increased muscle medium-chain acylcarnitines, consistent with altered fatty acid oxidation, and elevated intestinal glycerophospholipids, sphingolipids, and monoacylglycerols. Conversely, URB reduced p-cresol sulphate (uremic toxin) and increased antioxidant ergothioneine along the plasma–gut axis. Cardiac metabolites remained largely unchanged, suggesting that dietary adaptation was primarily confined to tissues directly exposed to dietary components and involved in nutrient processing. Unlike HF-induced phospholipid depletion and long-chain acylcarnitine accumulation across plasma, intestine, and heart, rye-mediated adaptations under LF conditions were localized and predominantly observed in muscle and gut.

These findings demonstrate that dietary interventions producing comparable conventional clinical outcomes can nevertheless reshape tissue metabolism through distinct host–microbiome interactions, providing a mechanistic framework for evaluating nutritional interventions for metabolic disease beyond circulating biomarkers.

ORAL PRESENTATION

O-24

Interactive effects of lactation stage, dietary supplementation, and maternal metabolic state on human milk lipidome

PRESENTING AUTHOR

Baoru Yang

Baoru Yang[1], Timo Seitz[2], Mikael Fabritius[3], Dattatray Mongad[4], Noora Houttu[5], Lotta Saros[5], Maaria Kortesniemi[3], Marika Kalpio[3], Leo Lahti[4], Kirsi Laitinen[6]

[1] Food Sciences, Department of Life Technologies, University of Turku, Finland; [2] Food Sciences, Department of Life Technologies, Nutrition and Food Research Center, Faculty of Medicine, University of Turku, Finland; [3] Food Sciences, Department of Life Technologies, Faculty of Technology, University of Turku, Finland; [4] Department of Computing, Faculty of Technology, University of Turku, Finland; [5] Nutrition and Food Research Center, Faculty of Medicine, University of Turku,

PRESENTATION

O-24

Tuesday 20

12:03

Riverside Ballroom

KEYWORDS

Human milk lipidomics

Lactation stage

Dietary supplementation

Fish oil

Metabolic state

Milk lipids play a crucial role in the growth, development, and health of infants. Despite the high importance, there is lack of in-depth knowledge on changes in breast milk lipidome across lactation stages at the molecular level. Furthermore, it is vital to improve the understanding on how milk lipidome respond to maternal dietary supplementation and how interactive effects of lactation stage, dietary intervention, and maternal metabolic state shape the milk lipidome. To bridge these knowledge gaps, 281 mothers of overwight or with obesity were recruited and divided into four groups receiving dietary supplmentation with fish oil, probiotics, fish oil plus probiotics, and placebo, respectively. The supplementation started during early pregancy and lasted through the lactation stage. Nontargeted lipidomics analysis was performed of colostrum and mature milk (3-months postpartum), using ultra-high performance liquid chromatography-quadrupole time-of-fly mass spectrometry. Compared to mature milk, colostrum is more abundant in lipids containing long-chain polyunsaturated fatty acids, highlighting the importance of colostrum for the cell membrane structure and function of organs and tissues, as well as early programming of physiology in infants. While fish oil supplementation increased the level of docosahexaenoic acid (DHA)-containing lipid species across the milk samples, the impact was stronger in mature milk than in colostrum, counteracting the decline in these lipid species during transition from colostrum to mature milk. Furthermore, maternal body mass index and gestational diabetes were associated with lower lipid content in colostrum, suggesting a regulating role of maternal metabolic state in lipid biosynthesis and remodelling in mammary glands.

ORAL PRESENTATION

O-25

Multi-Region Brain Lipidomics in Alzheimer's Disease Neuropathology

PRESENTING AUTHOR

Kevin Huynh

Joe Yi[1], Matilda van Buuren-Milne[1], Thy Duong[1], Nat Mellett[1], Corey Giles[1], Amy Liang[1], David Bennett[2], Matthias Arnold[3], Xianlin Han[4], Rima Kaddurah-Daouk[5], Peter Meikle[1], Kevin Huynh[1]

[1] Baker Heart and Diabetes Institute; [2] Department of Neurological Sciences, Rush University Medical Center, Chicago, IL, USA; [3] Institute of Computational Biology, Helmholtz Zentrum München, Germany; [4] Barshop Institute for Longevity and Aging Studies, University of Texas Health Science Center at San Antonio, San Antonio; [5] Duke Institute of Brain Sciences, Duke University, Durham, NC, USA

PRESENTATION

O-25

Tuesday 20

14:30

Riverside Ballroom

KEYWORDS

Lipidomics

Alzheimer's disease

Brain

Ferroptosis

Alzheimer's disease (AD) pathology is accompanied by extensive lipid remodelling and lipid related pathologies. Brain lipidomic studies are complicated by strong regional heterogeneity, variable myelin content, post-mortem effects and incomplete cross-cohort harmonisation.

We developed comprehensive lipidomic approaches spanning >1,300 lipid species across 54 classes, measuring lipids in >800 brain slices across the ROSMAP and Mayo brain cohorts. These included dorsolateral prefrontal cortex, cerebellum, temporal cortex and superior temporal gyrus. Models evaluated lipid associations with AD neuropathology, particularly Braak stage, while adjusting for age at death, sex, post-mortem interval, study covariates and lipid-derived myelin content.

Across cohorts and regions, brain lipids were more strongly associated with tau-related pathology than amyloid measures. Multi-region trajectories identified a consistent Braak-linked oxidative lipid signature, including increasing oxidised phosphatidylcholine and phosphatidylethanolamine species. The pattern converged with in vitro GPX4-inhibition experiments in HT22 hippocampal neuronal cells, which reproduced key ferroptosis-like features, including the accumulation of oxidised lipid products, altered alkenyl-/alkyl-phosphatidylethanolamine ratios and increased triglycerides. In parallel, glycosphingolipid ratios (HexCer, Hex2Cer, Hex3Cer, gangliosides) and related lysosomal lipid markers, were associated with Braak stage, implicating lysosomal and lipid-droplet biology in disease progression.

These findings show that targeted multi-region brain lipidomics, coupled with myelin adjustment and harmonised analytical workflows, can resolve coherent biochemical signatures of AD pathology. The data support oxidative phospholipid metabolism, ether-lipid remodelling and glycosphingolipid/lysosomal dysregulation as linked features of Alzheimer's neurodegeneration and provide a framework for further genetic, cell-type and experimental validation.

ORAL PRESENTATION

O-26

Analysis of Total Fatty Acids by SFC/MS: Separation of Omega-3 and Omega-6, Branched and Hydroxy Fatty Acids

PRESENTING AUTHOR

Michaela Chocholouskova

Michaela Chocholouskova[1], Bernice Huimin Wong[2], David L. Silver[2], Federico Torta[1]

[1] Signature Research Program in Cardiovascular and Metabolic Disorders, Duke- NUS Medical School, Singapore, Singapore Lipidomics Incubator (SLING), National University of Singapore, Singapore; [2] Signature Research Program in Cardiovascular and Metabolic Disorders, Duke- NUS Medical School, Singapore

PRESENTATION

O-26

Tuesday 20

14:45

Riverside Ballroom

KEYWORDS

Supercritical Fluid Chromatography

Mass Spectrometry

Total Fatty Acids

omega-3 fatty acids

omega-6 fatty acids

Fatty acids (FA) exist either in free form or as components of complex lipids. They have structural functions as constituents of phospholipids, the building blocks of cell membranes. They also have storage functions in neutral lipids and act as precursors of oxylipins, which regulate immune responses, inflammation, cell growth and apoptosis. The standard method for the analysis of total FA is gas chromatography–mass spectrometry (GC/MS), which requires derivatization steps and involves long analysis run times (30 min and more). Here, we present a new approach for the analysis of total FA using Supercritical Fluid Chromatography coupled with Mass Spectrometry (SFC/MS) with a short run time of 7.5 min. The method, using a C18 column, was optimized for the separation of 62 FA including omega-3 and omega-6 FA (such as FA 18:3 n-3 and n-6, FA 20:3 n-3 and n-6, FA 22:5 n-3 and n-6), branched FA (iso/anteiso) from their unbranched counterparts and hydroxy-FA. The SFC system was coupled to a high-resolution time-of-flight mass spectrometer (QTOF) to obtain highly accurate mass measurements, as FA show limited fragmentation and identification. The type of modifier, temperature and back-pressure regulator settings were optimised to achieve the best and fastest separation of individual FA and their isomers. After full validation, we measured FA in the NIST Standard Reference Material (SRM) 1950 human plasma, demonstrating the high precision of our SFC/MS method based on quantitative data. The method was also applied to the study of other biological samples, such as cells and tissues.

ORAL PRESENTATION

O-27

In vivo stable isotope labelling and mass spectrometry reveals that the APOE4 is a loss of function allele for brain lipid turnover

PRESENTING AUTHOR

Yushan Cai

Yushan Cai[1], Jun Yup Lee[1], Georgia Watt[1], Catherine Zhang[1], Lyndsey Swift[2], Bryce Vissel[2], Anthony Don[1]

[1] School of Medical Sciences, Faculty of Medicine and Health, The University of Sydney, Camperdown, NSW, 2006, Australia Charles Perkins Centre, The University of Sydney, Camperdown, NSW, 2006, Australia; [2] St. Vincent's Hospital Centre for Applied Medical Research, St Vincent's Hospital, Australia, School of Clinical Medicine, UNSW Medicine & Health, St Vincent's Healthcare Clinical Campus, Faculty of Medicine and Health, UNSW Sydney, Australia

PRESENTATION

O-27

Tuesday 20
15:00
Riverside Ballroom

KEYWORDS

- Alzheimer's Disease
- Brain lipid turnover
- Brain aging
- Heavy isotope labelling

Aging and APOE genotype are by far the greatest risk factors of Alzheimer's disease. APOE and other genetic risk factors strongly implicate altered brain lipid trafficking and catabolism as key contributors to the etiology of the disease. We hypothesise that risk genes including APOE4 promote neurodegeneration through impaired brain lipid turnover.

This study introduces a new method to track brain lipid turnover by administering deuterium oxide in the drinking water of mice for 8 weeks, commencing at 3 and 12 months of age, resulting in deuteration of newly synthesised brain lipids. High-resolution liquid chromatography-tandem mass spectrometry was employed to quantify deuterium incorporation into lipids (~250 lipids from 17 structural classes) from corpus callosum, hippocampus, biochemically-isolated myelin, cortex and brain stem.

This approach revealed tremendous heterogeneity in turnover rates for different brain lipids. The rate of Myelin-specific glycosphingolipids (hexosylceramides and sulfatides) and cholesterol replacement declined significantly between 3 and 12 months of age, while deletion of APOE selectively impaired cholesterol turnover specifically. Accumulation of cholesterol and myelin sphingolipids, particularly sulfatide, were seen in corpus callosum, hippocampus and isolated myelin of APOE knockout and APOE4 knock-in compared to WT and APOE3 knock-in mice. This difference between genotype was not seen in the cortex.

Our data supports a model in which APOE4 is a loss of function allele for brain lipid turnover, and ageing synergises with APOE4 genotype to impair myelin cholesterol and sphingolipid turnover. This is predicted to result in myelin degeneration and neuroinflammation, which is strongly correlated with Alzheimers's disease pathology.

ORAL PRESENTATION

O-28

Rapid, High-Throughput Lipid Class Separation Using Supercritical Fluid Chromatography-Mass Spectrometry (SFC-MS)

PRESENTING AUTHOR

Rangika Perera

Rangika Perera[1], Caryn Hepburn[2], Aurelie Benfield[3], Therese Fulloon[4], Marten F. Snel[2], Sónia Troeira Henriques[3], Stephen J. Blanksby[1]

[1] Central Analytical Research Facility, Queensland University of Technology, Brisbane, Queensland, Australia; [2] Waters Australia, Rydalmere 2116, Australia; [3] School of Biomedical Sciences, Faculty of Health, Queensland University of Technology, Princess Alexandra Hospital, Translational Research Institute, Australia; [4] School of Chemistry and Physics, Queensland University of Technology, Brisbane, Queensland, Australia

PRESENTATION

O-28

Tuesday 20

16:00

Riverside Ballroom

KEYWORDS

- Supercritical fluid chromatography
- lipidomics
- high-throughput screening
- class-based separation
- tandem mass spectrometry (MS/MS)

The growing demand for high-throughput lipidomic screening has exposed significant bottlenecks in conventional LC-MS workflows due to the long run times required for adequate resolution of apolar lipids. Supercritical fluid chromatography (SFC) offers a promising pathway to overcome these hurdles, yet its full potential in lipid analysis remains underexplored. In this study, we establish an optimized SFC-MS workflow that achieves rapid, class-based resolution of a broad spectrum of lipid categories. The experiments were carried out on a Waters UPC² system integrated with a Viridis™ BEH column (3 × 100 mm, 1.7 µm), coupled to a Synapt G2S quadrupole time-of-flight mass spectrometer. Using this stationary phase, we reliably separated 15 distinct lipid classes spanning glycerolipids, glycerophospholipids, sphingolipids, and sterols within an 8-minute runtime. Method performance was assessed through replicate injections of quality control samples, demonstrating reproducibility with retention time RSDs ≤ 0.60% and peak area RSDs < 4%.

The importance of class-based separation lies in mitigating ion suppression and matrix effects by segregating lipids according to headgroups, enabling co-elution of class-specific internal standards for more accurate quantitation. Post-column addition of sodium acetate in the electrospray makeup solvent enhanced ionization of several lipid classes and provided complementary MS/MS spectra via consistent formation of $[M+Na]^+$ adduct ions. Compared to conventional LC-MS, SFC-MS also enables manipulation of ion type and collision energy for enhanced characterization, exemplified by differentiation of bis(monoacylglycero)phosphate (BMP) from phosphatidylglycerol (PG) through diagnostic $[M+Na]^+$ fragment ions in positive ion mode. This integrated workflow represents a robust, high-throughput platform for quantitative lipidome profiling.

ORAL PRESENTATION

O-29

A novel targeted reversed-phase LC-QqQ-MS/MS methodology reveals that ER translocon gating governs storage and membrane lipid partitioning in Saccharomyces cerevisiae

PRESENTING AUTHOR

Lee Thomasson

Lee Thomasson[1], Danielle Dye[1], Emma Hadfield[1], Robert Steuart[2], Robert Trengove[2], Carl Mousley[1]

[1] Curtin School of Diagnostic and Therapeutic Sciences, Curtin Medical Research Institute; [2] Curtin Medical Research Institute

PRESENTATION

O-29

Tuesday 20

16:12

Riverside Ballroom

KEYWORDS

- Sec61 translocon
- Lipid metabolism
- Saccharomyces cerevisiae
- Very long chain fatty acids

Saccharomyces cerevisiae is a central model microbe and an industrial lipid producer, yet how the endoplasmic reticulum (ER), the principal site of lipid biosynthesis, governs the partitioning of fatty acids between storage and membrane pools remains incompletely understood. We addressed this using the Sec61 translocon-gating mutant sss1-7, profiling wildtype, sss1-7, and its SEC61N302L genetically encoded suppressor with a novel targeted reversed-phase LC-QqQ-MS/MS methodology also presented here, quantifying 17 lipid classes and resolving triacylglycerols (TAGs) to acyl-chain composition while requiring no specialised instrumentation.

We find that defective translocon gating established a distinct lipidomic state in which sss1-7 showed a ~3.8-fold TAG expansion, with disproportionate rerouting of very-long-chain (≥C22) fatty acids into TAGs (1.2% to 7.0% of TAG pool) and reciprocal exclusion of fully saturated TAG species (34% to 5% of TAG pool), underpinned by a structurally biased diacylglycerol precursor pool and mirrored by C26-ceramide sphingolipid enrichment. Additionally, membrane remodelling coupled directional monomethyl-phosphatidylethanolamine to phosphatidylcholine flux alongside selective lysophosphatidylcholine turnover. These changes were largely reversed by SEC61N302L and occurred with minimal transcriptional induction of lipid-associated genes, indicating post-transcriptional redistribution rather than biosynthetic upregulation.

Translocon gating therefore acts as a significant determinant of yeast lipid partitioning, directing very-long-chain fatty acids into TAGs. Beyond this biologically mechanistic and physiological interest, it also identifies the Sec61 translocon as a potentially relevant node for the industrial engineering of yeast strains rich in storage lipids constituted of these specific very-long-chain fatty acids, critical for the biomanufacturing of high value oleochemicals like lubricants, waxes, and pharmaceuticals.

ORAL PRESENTATION

O-30

Investigating the Diabetic Lipidome: A DBS-Based High-Resolution Mass Spectrometry Approach

PRESENTING AUTHOR

Jayden Lee Roberts

Jayden Lee Roberts[1], Luke Whiley[2], Nicola Gray[3], Melvin Gay[4], Elaine Holmes[3], Julien Wist[5], Jeremy K Nicholson[6], Nathan Lawler[3]

[1] University of Notre Dame Australia; [2] Curtin University; [3] Murdoch University; [4] Bruker Daltonics; [5] Universidad del Valle; [6] Imperial College London

PRESENTATION

O-30

Tuesday 20

16:24

Riverside Ballroom

KEYWORDS

microsampling

dried blood spots (DBS)

patient-centric sampling

HRMS

ion mobility

INTRODUCTION

Comprehensive lipidomic analyses have identified distinct lipid signatures in diabetics, indicating novel lipid markers may aid personalised approaches to monitoring disease. Dried blood spot(DBS) microsamples offer minimally-invasive and cost-effective methods for collecting/storing small sample volumes (typically <50µL). DBS also enable increased sampling frequency and simplified transport via mail, suitable for remote/resource-limited settings. However, lipidomic profiling of DBS for the diabetic context is limited, and further research is required before DBS can be routinely applied in longitudinal, decentralised, and remote settings.

METHODS

Participants (n=17 diabetic, n=17 non-diabetic controls) performed DBS self-collection using advanced microsamplers (10µL, CapitainerB®). Samples were analysed using an optimised untargeted 4D (timsTOF) method for semi-quantitative lipid analysis using UHPLC-HRMS. The 15-minute workflow enables analysis of 96 samples per day, covering 25 lipid subclasses spanning fatty acyls, glycerolipids, glycerophospholipids, sphingolipids, and sterols (n=432 unique lipids).

RESULTS

and Discussion: DBS lipid profiling revealed significant differences between diabetic individuals and non-diabetic controls ($R^2Y=0.77$, $Q^2=0.41$, $p=0.05$). The lipid species driving this difference were consistent with those in matched traditional venous plasma samples ($R^2Y=0.83$, $Q^2=0.49$, $p=0.05$) and capillary plasma samples ($R^2Y=0.72$, $Q^2=0.49$, $p=0.05$), highlighting self-collected DBS as a reliable alternative to venous phlebotomy. Associations between lipid species and diabetes-related measures (Problem Areas in Diabetes scale:PAID-20, and Patient Activation Measure:PAM-13), indicate lipidomics may provide insight into relationships between lipid profiles, psychosocial factors, and health-management behaviors in diabetes.

CONCLUSION

Taken together with our previously demonstrated DBS lipid stability, preliminary findings indicate feasibility for remote, longitudinal self-sampling with DBS, enabling personalised health monitoring for diabetes.

ORAL PRESENTATION

O-31

Towards Quantitative FTIR-Based Plasma Lipid Profiling: A Design of Experiments Approach to Optimising Lipid Extraction

PRESENTING AUTHOR

Satnam Kaur

Satnam Kaur[1], Shekhar Kumta[1]

[1] RMIT University, Melbourne, Victoria Australia, Melbourne University, Melbourne, Victoria Australia, Northern Health, Melbourne, Victoria Australia

PRESENTATION

O-31

Tuesday 20

16:36

Riverside Ballroom

KEYWORDS

- FTIR spectroscopy
- Lipid extraction
- Plasma lipid profiling
- Design of Experiments (DOE)
- Quantitative lipid analysis

Accurate quantification of plasma lipids is fundamental to the discovery and validation of biomarkers associated with cardiometabolic, inflammatory and other chronic diseases. Fourier-transform infrared (FTIR) spectroscopy has emerged as a rapid, reagent-free and cost-effective analytical technique for lipid profiling; however, its quantitative reliability is highly dependent on efficient sample preparation and lipid extraction. Optimising lipid extraction is therefore essential for developing robust FTIR-based methods suitable for clinical and translational applications.

This study employed a structured Design of Experiments (DOE) using simulated human plasma spiked with palmitic acid (C16:0), oleic acid (C18:1 cis) and elaidic acid (C18:1 trans) to systematically compare MTBE extraction and the Folch method. The experimental design incorporated single-lipid, binary and ternary mixtures across three concentration levels to investigate the effects of lipid composition, concentration and sample complexity on extraction efficiency and quantitative performance. Following extraction, samples were analysed by attenuated total reflectance FTIR (ATR-FTIR) spectroscopy, and quantitative performance was assessed using calibration-based quantification, lipid recovery, reproducibility and statistical analyses.

The study demonstrates the capability of ATR-FTIR to provide reliable quantitative analysis of lipid extracts while highlighting the influence of extraction methodology on quantitative performance. The systematic comparison establishes an evidence-based framework for assessing extraction methods, identifying critical parameters affecting quantitative accuracy and precision, and defining suitable conditions for analytical method validation. This work represents an important step towards the development of a rapid, reproducible and clinically translatable FTIR workflow for plasma lipid profiling, supporting future biomarker discovery and the development of FTIR-based diagnostic assays.

ORAL PRESENTATION

O-32

Lipidomic investigation of atherosclerotic plaque in S. pneumoniae inoculated ApoE-/- mice

PRESENTING AUTHOR

Kanishka Senthilvel

Kanishka Senthilvel[1], Silvia Lee[2], Dana Hicks[3], Shelley Waters[1], Benjamin Bartlett[1], Herbert Ludewick[4], Betty Yeung[5], Nicola Gray[6], Girish Dwivedi[7]

[1] APEX Heart Lab, Harry Perkins Institute of Medical Research, Perth, Western Australia, Australia, School of Medicine, The University of Western Australia, Perth, Western Australia, Australia; [2] APEX Heart Lab, Harry Perkins Institute of Medical Research, Perth, Western Australia, Australia, School of Medicine, The University of Western Australia, Perth, Western Australia, Australia, Department of Microbiology, Pathwest Laboratory Medicine, P; [3] Centre for Computational and Systems

PRESENTATION

O-32

Tuesday 20

16:48

Riverside Ballroom

KEYWORDS

Lipidomics

MALDI-MSI

BACKGROUND

Atherosclerosis is a chronic inflammatory disease identified by plaque formation. Prior studies demonstrate an increased risk of cardiovascular events post-pneumonia infection. Altered lipid processing is pivotal in plaque development and disease progression. In this study, we investigate the lipid composition of atherosclerotic plaques in the aortic root of ApoE-/- mice following upper respiratory tract (URT) colonisation with Streptococcus pneumoniae.

METHODS

ApoE-/- mice (n=14) were fed a high-fat diet for 8 weeks and intranasally inoculated with 107 CFU of S. pneumoniae strain P1121 (infected, n=7) or phosphate-buffered saline (control, n=7). After 21 days, mice were euthanised, and their aortic roots were collected and serially sectioned for histological analysis and matrix-assisted laser desorption/ionisation mass spectrometry imaging (MALDI-MSI).

RESULTS

Histological analysis characterised plaque morphology and quantified lipid content in the aortic root. MALDI-MSI and univariate ROC analysis on the root identified 3 metabolites unique to infected mice and 4 to control mice that were present in >50% of samples within the group (n ≥4). Lipid species associated with pro-inflammatory effects, such as Hexosylceramide (HexCer; d18:1/18:0) and Lysophosphatidic acid (LPA; 18:1), were observed in infected mice, while control mice showed lipid species, including Cholesteryl ester (CE; 18:2).

CONCLUSION

This pilot study established a dual-disease model that successfully identified distinct lipid features differentiating infected and control groups. The results demonstrate that bacterial colonisation of the URT may result in altered plaque composition, impacting cardiovascular risk. Future studies exploring different dosages and inoculation periods of bacteria would be critical to fully characterise the underlying pathophysiology.

ORAL PRESENTATION

O-33

Lipidomic profiling of cancer cell membranes reveals signatures for therapeutic targeting

PRESENTING AUTHOR

Sonia Henriques

Aurelie Benfield[1], Reuben Young[2], Grégoire Philippe[1], Rangika Perrera[1], Luis Pedro Coelho[1], Stephen Blanksby[1], Sonia Henriques[1]

[1] Queensland University of Technology; [2] University of Wollongong

PRESENTATION

O-33

Wednesday 21

09:30

Riverside Ballroom

KEYWORDS

- Lipid membrane
- Cancer Progression
- Outer leaflet characterisation
- Targeted therapy
- Membrane Biophysics

Cancer cell membranes shape tumour progression, drug response, and the activity of membrane-targeting therapeutics, yet quantitative information on their lipid composition and surface organisation remains limited. Here, we present a curated lipidomic dataset of cancer cell membranes and matched healthy counterparts designed to support physiologically relevant membrane modelling and therapeutic development. Using quantitative shotgun lipidomics, selective phospholipase digestion, and cell-surface charge measurements, we mapped total lipid composition, outer-leaflet glycerophospholipid distributions, and membrane electrostatic properties across multiple cancer models. The dataset reveals that whole-cell lipid class composition is relatively conserved among melanoma models but more heterogeneous across other cancer types. Compared with healthy cells, cancer membranes show reduced sphingomyelin proportions and a consistently more negative surface charge, with melanoma cells displaying the strongest electrostatic signature. Notably, surface charge did not uniformly correlate with externalised anionic phospholipids, indicating that cancer membrane physicochemical properties cannot be inferred from phospholipid asymmetry alone. By integrating lipid abundance, leaflet distribution, and membrane charge, this resource provides a benchmark for the lipidomics community and an enabling platform for cancer membrane research. It supports the construction of biologically relevant membrane models, improves interpretation of lipid-dependent membrane behaviour, and guides the rational design of therapeutics that exploit cancer-specific membrane features.

ORAL PRESENTATION

O-34

Assigning sn-position in glycerophospholipids by hyphenation of reversed-phase chromatography and collision-/ozone-induced dissociation mass spectrometry

PRESENTING AUTHOR

Qiyu Liu

Qiyu Liu[1], Benjamin Matthews[2], Tristan Wallis[2], Frederic Meunier[3], Reuben Young[4], Shane Ellis[4], Berwyck Poad[5], Sam Brydon[1], David Marshall[6], Stephen Blanksby[7]

[1] School of Chemistry and Physics, Queensland University of Technology, QLD; [2] Queensland Brain Institute, University of Queensland, QLD; [3] Queensland Brain Institute and School of Biomedical Sciences, University of Queensland, QLD; [4] School of Chemistry and Molecular Bioscience, University of Wollongong, NSW; [5] Thermo Fisher Scientific, Richlands, QLD; [6] Central Analytical Research Facility, Queensland University of Technology, QLD; [7] School of Chemistry and Physics and

PRESENTATION

O-34

Wednesday 21

09:45

Riverside Ballroom

KEYWORDS

Glycerophospholipids

Regioisomers

Brain lipids

CID/OzID

Glycerophospholipids (GPs) are the major component of cellular membranes, with their lipidome tightly regulated to maintain critical biophysical properties. The brain lipidome is especially complex, comprising diverse GPs across different membranes and cell types that support intricate processes such as signal transduction. Despite advances in lipidomics, much structural diversity remains unresolved due to limitations in distinguishing GP regioisomers arising from the positions of fatty acyl esterification on the glycerol backbone. These positional isomers, defined by the stereospecific numbering (sn) system, are poorly separated by liquid chromatography and are distinguished only by product ion ratios under collision-induced dissociation (CID), which are head-group dependent. This reliance often leads to structural assumptions based on canonical ideas, such as the presumption of polyunsaturated fatty acyl chains at the sn-2 position.

Here we report a streamlined strategy for sn-isomer resolution of GPs that combines reversed-phase LC with sequential collision- and ozone-induced dissociation (CID/OzID) mass spectrometry. This workflow leverages enhanced chromatographic separations alongside the structure-specific fragmentation patterns of sn-isomers to enable unambiguous identification and relative quantification across diverse GP classes and fatty acyl chain combinations. Applied to mouse brain lipid extracts, this methodology revealed striking class-dependent regioisomer presentations. In one such example, phosphatidylcholines (PC) and phosphatidylethanolamines (PE) present to be exclusively in the canonical 18:0/20:4 configuration. Conversely, the corresponding phosphatidylserine was found to present with both PS 18:0/20:4 and PS 20:4/18:0. These isomer-resolved data indicate a complex isomeric landscape within the brain lipidome and point to distinctive biosynthetic and acyl chain remodelling pathways between GP classes.

ORAL PRESENTATION

O-35

Spatial Lipidomics Reveals Lipid Signatures Associated with Plaque Instability and Rupture in Medial-Derived Carotid Atherosclerotic Plaques

PRESENTING AUTHOR

Hanane Belhoul-Fakir

Hanane Belhoul-Fakir[1], Jaclyn Swan[2], Susan Wu[1], Berin Boughton[2], Shirley Jansen[3], Juliana Hamzah[1]

[1] Targeted Drug Delivery, Imaging & Therapy, Harry Perkins Institute of Medical Research, QEII Medical Centre, Nedlands, Western Australia, 6009, Australia; [2] La Trobe Institute for Sustainable Agriculture and Food, La Trobe University, Bundoora, Victoria 3086, Australia.; [3] School of Population Health, Curtin University, Bentley, Perth, WA 6102, Australia

PRESENTATION

O-35

Wednesday 21

10:00

Riverside Ballroom

KEYWORDS

Atherosclrosis

Lipid species

Medial plaques

Stroke

Ferroptosis

BACKGROUND

Stroke is a leading cause of death and disability worldwide. In Australia, one in four strokes occurring in people under 65 and most strokes are ischemic. Carotid atherosclerosis is a major cause of ischemic stroke, with plaque rupture involving fibrous cap thinning or intraplaque hemorrhage. However, the mechanisms that determine whether plaques rupture through fibrous cap thinning or intraplaque hemorrhage remain unknown. In this study, we classified carotid plaques by the location of lipid-rich regions within the intimal or medial layer and hypothesised that these phenotypes differ in instability and rupture mechanisms.

AIM

To compare the spatial lipid profiles of media-originated carotid plaques and conventional intimal carotid plaques and identify lipid species associated with plaque instability.

METHOD

Fresh carotid endarterectomy plaques (n=9) were classified as media-originated (n=4) or intima-originated (n=5) and analysed by MALDI and MALDI-2-MSI. Lipids were annotated in SCiLS Lab and confirmed by LC-MS/MS on bulk tissue extracts. Differential abundance and pathway enrichment analyses were performed using MetaboScape (Bruker, v2025).

RESULT

Forty-four unique lipid species were detected in medial plaques in positive and 163 in negative ion modes, compared with 19 and 33, respectively, in intimal plaques. Ferroptosis-related lipids, including two hallmark ferroptosis driver species were identified exclusively in medial plaques.

CONCLUSION

Spatial lipidomics revealed a distinct lipid signature in medial-originated plaques enriched for ferroptosis-associated lipids, implicated in plaque instability. These findings provide new insights into the biological basis of plaque destabilisation and may improve risk stratification and targeted treatment of high-risk ischemic stroke patients.

ORAL PRESENTATION

O-36

A tale of two glycerols: Uncovering the structural diversity of phosphatidylglycerol congeners through multi-dimensional analysis

PRESENTING AUTHOR

Therese M. Fulloon

Therese M. Fulloon[1], Qiyu (Monica) Liu[1], Rangika Perera[2], Samuel C. Brydon[2], David L. Marshall[3], Kathleen M. Mullen[1], Stephen J. Blanksby[2]

[1] School of Chemistry and Physics, Queensland University of Technology, Brisbane, AUSTRALIA; [2] School of Chemistry and Physics & Central Analytical Research Facility, Queensland University of Technology, Brisbane AUSTRALIA; [3] Central Analytical Research Facility, Queensland University of Technology, Brisbane AUSTRALIA

PRESENTATION

O-36

Wednesday 21

10:15

Riverside Ballroom

KEYWORDS

multi-dimensional analysis

cyclic ion-mobility mass spectrometry

phosphatidylglycerols

micro-organisms

regio/stereo-isomers

Glycerophospholipids are defined by a glycerol-phosphate backbone. This category includes numerous lipid classes that integrate a diglycerol-phosphate moiety, further increasing the potential for class, regio- and stereo-isomerism. The number of acyl chains esterified to the four vacant hydroxyl groups define several distinct lipid classes: one acyl chain forms lysophosphatidylglycerol (LPG); two form either phosphatidylglycerol (PG) or bis(monoacylglycerol)phosphate (BMP), depending on the positions of esterification; three define hemi-bis(monoacylglycerol)phosphate (hemi-BMP); and four produce bis(diacylglycerol)phosphate (BDP). These structural congeners are biosynthetically related but retain distinct cellular functions. For example, BMP is involved in lysosomal lipid degradation whereas PG contributes to membrane structure and antiviral activity. Structural complexity is further increased by the presence of chiral centres within each glycerol, wherein the stereospecific numbering (i.e., sn-1/sn-2/sn-3) defines a range of possible regioisomers based on the position(s) of acyl chain esterification. The lipidome of a cell, tissue or organism may contain numerous lipid class, regio- or stereo-isomers that demand multi-dimensional analytical approaches to enable separation and structural characterisation. Current lipidomic workflows primarily rely on liquid chromatography (LC) coupled with mass spectrometry (MS) and collision-induced dissociation (CID) for lipid class separation and sum composition identification. Typically, these techniques do not resolve glycerol-derived regioisomers (sn) and stereoisomers (R/S) thus limiting structural assignment(s). Here, we present a multidimensional workflow integrating LC, CID and cyclic ion mobility-mass spectrometry (cIM-MS) that affords isomeric resolution and unambiguous structural identification of diglycerol-phosphate containing glycerophospholipids. Application of this technology reveals significant variation in lipid isomer populations between gram-positive and gram-negative bacteria.

ORAL PRESENTATION

O-37

RNA Capture and Retention in Lyotropic Lipid Mesophases

PRESENTING AUTHOR

Dev Chauhan

Dev Chauhan[1], Dr Soumya De[1], Dr Albert Fahrenbach[1], Dr Anna Wang[1]

[1] School of Chemistry, UNSW Sydney, Australia; Australian Centre for Astrobiology UNSW Sydney, Australia; UNSW RNA Institute, UNSW Sydney, Australia

PRESENTATION

O-37

Wednesday 21

11:15

Riverside Ballroom

KEYWORDS

- Self-assembly
- RNA delivery
- soft colloidal materials
- Stereochemistry

Lyotropic self-assembly of amphiphiles produces nanostructured mesophases whose internal connectivity governs solute transport and retention. Conventional lipid vesicles, however, poorly accommodate large macromolecules such as RNA because closed bilayers limit passive uptake. Here we evaluate glycerol monodecanoate (GMD), a simple monoglyceride amphiphile, as a minimal system to examine RNA transport in hydrated lipid assemblies.

GMD forms interconnected lipid mesophases rather than discrete vesicles, generating continuous aqueous pathways that allow RNA to spontaneously partition from solution. RNA uptake subsequently induces a structural reorganization of the assembly, promoting a transition from mesophase networks to vesicular structures that significantly enhances RNA retention. Transport is length-dependent, with mononucleotides and short oligonucleotides exhibiting distinct retention behavior consistent with confinement within the lipid–water network. Surprisingly, enantiomerically pure and racemic GMD produce chemically indistinguishable assemblies that nonetheless display different RNA transport behaviors, revealing that subtle differences in chiral packing can influence mesoscale diffusion within lipid nanostructures.

These results demonstrate that macromolecular cargo can reorganize lipid mesophases and trigger vesicle formation, highlighting glycerol monodecanoate as a minimal amphiphile for responsive lipid nanostructures relevant to RNA delivery, lipid nanoparticles, and other soft colloidal materials.

ORAL PRESENTATION

O-38

Targeted lipid nanoparticle delivery of plasmalogen precursors as a therapeutic strategy to restore plasmalogen levels and enhance cellular resistance to stress-mediated injury

PRESENTING AUTHOR

Pukjera Jena Rungreang

Pukjera Jena Rungreang[1], Amy W. Liang[2], Matilda van Buuren-Milne[2], Thy Duong[2], Natalie A. Mellett[2], Sudip Paul[2], Kevin Huynh[2], Mark Louis P. Vidallon[2], Thomas G. Meikle[2], Xiaowei Wang[2], Yow Keat Tham[2], Peter J. Meikle[2]

[1] Baker Department of Cardiometabolic Health, The University of Melbourne; [2] Baker Heart and Diabetes Institute, Melbourne, Australia

PRESENTATION

O-38

Wednesday 21

11:27

Riverside Ballroom

KEYWORDS

Lipid nanoparticle

lipid delivery

plasmalogen modulation

stress-related injury

lipidomics

INTRODUCTION

Plasmalogens are vinyl-ether-linked glycerophospholipids essential for membrane organisation and antioxidant defence. They are therapeutically attractive because their levels can be modulated by supplementing with their metabolic precursor, alkylglycerols (AKG). However, their slow incorporation into tissues has limited use under acute stress settings. Lipid nanoparticles (LNPs) have revolutionised nucleic acid delivery, yet there remain untapped opportunities to deliver bioactive lipids that target novel pathways. We proposed that LNPs could mediate AKG delivery to restore plasmalogen levels and enhance cellular resistance to stress-related injury.

METHODS

Characterisation of LNP physicochemical properties and stability was determined via dynamic light scattering. The immunomodulatory effects of AKG-LNPs were evaluated using RAW 264.7 mouse macrophages following a 24-hour pre-treatment and LPS exposure (100 ng/mL, 6-h), followed by qPCR analysis of inflammatory gene expression. Lipidomics via LC-MS was conducted to profile changes in lipid composition following treatment. Statistical significance ($p<0.05$) was determined using one-way and repeated-measures ANOVAs with multiple comparisons.

RESULTS

Exposure of AKG-LNP to fetal bovine and human serum did not significantly change particle size or polydispersity, indicating colloidal stability under biologically relevant conditions. Macrophages pre-treated with 25 and 50 µM AKG-LNPs, followed by LPS, exhibited a ~28 and 43% increase in plasmalogen levels, respectively, compared with LPS-treated cells ($p<0.05$). Moreover, these treatments were associated with reductions in LPS-induced Tnf-α expression to ~10- and ~11-fold, respectively ($p<0.05$), suggesting attenuation of LPS-induced inflammatory activation.

CONCLUSION

LNP delivery of AKG represents a rapid, novel therapeutic strategy that restores plasmalogen levels and provides cellular protection against stress-related injury.

ORAL PRESENTATION

O-39

Developing Potent and Selective Ceramide Synthase 6 Inhibitors for the Treatment of Metabolic Dysfunction in the Liver

PRESENTING AUTHOR

Patricia Schwarzkopf

Patricia Schwarzkopf[1], Helen Lim[1], Georgia Watt[1], Karen Denny[1], Anthony Don[2]

[1] School of Medical Sciences, Charles Perkins Centre, The University of Sydney, Camperdown, New South Wales, Australia; [2] School of Medical Sciences, Charles Perkins Centre, and Brain and Mind Centre, The University of Sydney, Camperdown, New South Wales, Australia

PRESENTATION

O-39

Wednesday 21

11:39

Riverside Ballroom

KEYWORDS

- Metabolic Dysfunction-Associated Fatty Liver Disease
- Ceramide
- Ceramide Synthase
- Small Molecule Inhibitors
- Drug Discovery

Metabolic dysfunction-associated fatty liver disease (MAFLD) is becoming increasingly prevalent, with only 1 TGA-approved pharmacotherapy. Its characteristic steatosis and insulin resistance phenotypes are driven by lipid excess, particularly saturated fats, and promote the onset of type II diabetes and liver fibrosis. A substantial body of evidence implicates the enzyme ceramide synthase 6 (CerS6) and its lipid product C16:0 ceramide in MAFLD progression, while CerS2-derived C22-C24 ceramides are protective. Importantly, there are no specific CerS6 inhibitors available. We employed acoustic ejection mass spectrometry to identify small molecule CerS6 inhibitors from a library of ~150,000 compounds, and counter screened the top 40 hits against the protective CerS2 enzyme. Compounds with low micromolar IC50 values for CerS6 and poor CerS2 inhibition were subjected to further evaluation. Compound CS6I1 emerged as a promising lead for inhibitor development, exhibiting IC50 values of 7.49μM and 10.54μM for CerS1 and 6 respectively, and no inhibitory effect on the other CerS enzymes. Untargeted quantification of >200 sphingolipid, phospholipid and neutral lipid species in CS6I1-treated HEK293E and Huh7 cells revealed highly selective reduction of C16:0 ceramide and its derivatives at sub-micromolar concentrations. Enzyme kinetic assays showed that CS6I1 competitively inhibits C16-Coenzyme A binding but displays non-competitive inhibition for dihydrosphingosine substrates of CerS6. This was supported by molecular docking and site-directed mutagenesis, indicating that Ser215 within the enzyme's active site confers selectivity of CS6I1 for CerS6 over other isoforms. Further evaluation of these compounds in vitro and in vivo should lead to a new class of therapeutics for MAFLD.

ORAL PRESENTATION

O-40

Inhibition of lysosomal beta glucocerebrosidase promotes de novo synthesis in a neuronal model

PRESENTING AUTHOR

Jennifer Saville

Jennifer Saville[1], Ashleigh Lake[2], Maria Fuller[3]

[1] Genetics and Molecular Pathology, SA Pathology at the Women's and Children's Hospital, School of Medicine, Adelaide University; [2] Genetics and Molecular Pathology, SA Pathology at the Women's and Children's Hospital, School of Biological Sciences, Adelaide University; [3] Genetics and Molecular Pathology, SA Pathology at the Women's and Children's Hospital, School of Medicine and School of Biological Sciences, Adelaide University

PRESENTATION

O-40

Wednesday 21

11:51

Riverside Ballroom

KEYWORDS

sphingolipids

neurons

sphingolipid metabolism

Sphingolipid metabolism is a tightly regulated, interconnected pathway with ceramide at its centre. Here we assessed the impact on sphingolipid metabolism when a key catabolic enzyme, beta-glucocerebrosidase (GCase), that produces ceramide is inhibited. Given sphingolipids are essential in the brain, acting as regulators of neuronal growth, development, and function, we explored this in the SH-SY5Y neuroblastoma line.

SH-SY5Y were treated with conduritol B-epoxide (CBE) to inhibit GCase activity. Cells were harvested at 1 and 10 days representing immature and mature neurons, respectively. De novo sphingolipid synthesis was assessed by measuring incorporation of 13C16-palmitate into sphingolipid species using liquid chromatography-electrospray ionisation-tandem mass spectrometry. Expression of enzymes involved in de novo sphingolipid synthesis was assessed by RT-qPCR and western blotting.

GCase inhibition resulted in the expected accumulation of glucosylceramide by day 1. Immature neurons showed decreases in ceramides which were then increased in mature neurons. Mature neurons also showed increases in dihexosylceramides and the simple gangliosides, particularly in shorter chain species. Incorporation of 13C16-palmitate was increased throughout the sphingolipid pathway, including a 1.2-fold increase in ceramide and a 3-fold increase in GM2 gangliosides. While transcript levels of key biosynthetic enzymes were unchanged, western blotting identified an approximately 30% reduction in protein expression at day 3, suggesting dynamic post-transcriptional or temporal regulation of this pathway.

Our results show that sphingolipid dyshomeostasis from GCase inhibition changes with neuron maturation. Labelling studies suggest an increase in de novo synthesis as a compensatory mechanism employed to offset for the loss of ceramide from turnover of GlcCer.

ORAL PRESENTATION

O-41

Acetaminophen-Mediated Alteration of Cardiolipin distribution in Rat Organ Tissue

PRESENTING AUTHOR

HAY-YAN WANG

HAY-YAN WANG[1], CHIH-LE LIU[1], AI-YUN TSAI[1], JI-HAN LI[1], SHIH-YU HUANG[1], MO-HSUAN HSU[1]

[1] Department of Biological Sciences, National Sun Yat-Sen University, Taiwan

PRESENTATION

O-41

Wednesday 21

12:03

Riverside Ballroom

KEYWORDS

acetaminophen

cardiolipins

MALDI-MSI

Acetaminophen (paracetamol; N-acetyl-para-aminophenol; APAP) is one of the most widely utilized non-steroid anti-inflammatory drugs worldwide. A small portion of the APAP metabolism is mediated by the oxidation of cytochrome P450s, generating a highly reactive and toxic product N-acetyl-p-benzoquinone imine which leads to multiple mitochondrial events in overdose (OD) condition. Cardiolipins (CLs), a mitochondrial inner membrane lipid holding critical functions in the mitochondrial electron transport chain are presumed affected by APAP metabolism. Its distribution patterns in liver and kidney under APAP exposure were explored by Matrix-assisted laser desorption/ionization (MALDI)-mass spectrometry imaging (MSI).

The mean MSI spectra of the in situ control rat livers CLs revealed with the 70-, 72- and 74-carbon CL groups (CL70, CL72, CL74 respectively), where CL72 group dominated the distribution. Oral high-dose acute (HDA) APAP feeding moderately enhanced and changed the distribution patterns of CL70 and CL72 groups, while mildly diminished the presentation of the CL74 group. Moderate dose, weeklong (MDW) oral APAP exposure decreased the CL presentations of all three CL groups. The mean MSI spectra of the control rat kidney showed species-specific CL distributions along the cortex (CTX), outer medulla (OM), and inner medulla (IM). The HDA exposure decreased the presentation of all three CL groups, some even lose their specific patterns. The MDW exposure further decreased the presentation of some CTX- and IM-dominant species, while reduced the overall OM CL signals to minimum. The physiological significance of APAP exposure and CL changes to the critical functions of organs and tissues was enumerated and postulated.

ORAL PRESENTATION

O-42

COmbiNtorial Chemistry for QUantitative Standards (CONCQUEST): Quantitative Phospholipid Standards via Randomized De Novo Synthesis

PRESENTING AUTHOR

Yanning Li

Yanning Li[1], Secilia Garza[1], J. Thomas Brenna[1]

[1] University of Texas at Austin

PRESENTATION

O-42

Wednesday 21

12:15

Riverside Ballroom

KEYWORDS

- Lipidomics
- Phosphatidylcholine
- Quantitative mass spectrometry
- Calibration standards
- Acyl randomization

BACKGROUND

and Objectives

Phospholipid quantitation by LC-MS remains a challenge, given a vast combinatorial space of species whose ionization efficiencies vary with acyl chain length and unsaturation. Current quantitation depends on analytical standards and constructing calibration curves, which are costly and for many species, nonexistent. Building on our prior demonstration of the CONCQUEST approach for triacylglycerols (Wang et al., RCM 2020), we now aim to randomize acyl chains on phospholipid backbones by Steglich esterification, producing standards for phospholipid classes from their fatty acid (FA) profiles.

METHODS

Free fatty acids saponified from Saccharomyces cerevisiae were coupled onto sn-glycero-3-phosphocholine by EDC/DMAP-catalyzed Steglich esterification. FA percentages, measured by GC-FID, predict each PC species' abundance via binomial statistics. d31-16:0 and d35-18:0 (2:1 molar ratio) served as sentinels confirming randomization. For each species, calculated abundance was compared to its ^{13}C-corrected $[M+Na]^+$ peak area. The response factor: calculated/observed abundance, normalized to 16:0/18:1 PC, expresses relative ionization efficiency. Native yeast PC was then quantified as raw peak area × response factor.

RESULTS

The synthesized PC was fully randomized: sentinel ratios PC(d31-16:0/18:1)/PC(d35-18:0/18:1) and PC(d31-16:0/18:0)/PC(d35-18:0/18:0) were 2.006 and 2.048, within −0.1% and +2% of the expected 2.008. Three independently extracted native yeast PC samples, quantified with response factors from three syntheses, agreed within 7% standard deviation.

CONCLUSION

s

Randomized de novo synthesis is a practical route to quantitative PC standards. Because each is built from the sample's own liberated fatty acids, it is inherently matched to the tissue and extends to any glycerolipid class amenable to acyl randomization.

CONTRIBUTED ABSTRACTS

Lightning Talks

Rapid Fire

Abstracts from our ECRs who have just 3 minutes to present their research interests and win the audience vote.

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RAPID FIRE

RF-01

Quantification and Localisation of New Brain Lipid Synthesis Using Deuterium Oxide and High Resolution Mass Spectrometry

PRESENTING AUTHOR

Catherine Zhang

Catherine Zhang[1], Jesse A. Michael[2], Jonathan D. Teo[1], Huitong Song[1], Mika T. Westerhausen[2], Shadrack M. Mutuku[2], Shane R. Ellis[2], Anthony S. Don[1]

[1] The University of Sydney; [2] The University of Wollongong

PRESENTATION

RF-01

ALSO POSTER 5

Sunday 18

17:30

Riverside Ballroom

KEYWORDS

- Deuterium
- Mass spectrometry imaging
- Lipid metabolism
- Remyelination

Myelin is the lipid-rich membrane that surrounds neuronal axons and is essential for neurological function in vertebrates. The development of therapeutics that stimulate myelin repair to treat demyelinating disorders such as multiple sclerosis is hampered by the inability to distinguish newly-synthesised from pre-existing myelin. We therefore developed a method to quantify and localise new myelin lipid synthesis in the mouse brain. Deuterium oxide was administered for two weeks in the drinking water of mice fed normal chow, chow containing the demyelinating toxin cuprizone, or during spontaneous remyelination following cuprizone withdrawal. Liquid chromatography-tandem mass spectrometry and matrix-assisted laser desorption/ionisation mass spectrometry imaging (MSI) were used to quantify and localise the newly synthesised, deuterated lipids in the brain. While most glycerophospholipids were constitutively deuterated, deuteration of myelin-enriched sulfatides, hexosylceramides, and phosphatidylethanolamine plasmalogens was only apparent during remyelination. Most deuterium atoms were found in the fatty acyl chains, indicative of de novo lipid synthesis. MSI showed selective localisation of deuterated hexosylceramides and phosphatidylethanolamine plasmalogen species in the corpus callosum, the white matter tract that is most heavily demyelinated in response to cuprizone. Additionally, MSI captured rapid myelin lipid synthesis in subregions of the corpus callosum where remyelination was not apparent by histological staining. In summary, the method described herein provides the means to quantify and spatially profile dynamic lipid synthesis across diverse biological contexts, including understanding myelin homeostasis and preclinical evaluation of remyelinating therapeutics.

RAPID FIRE

RF-02

Circulating fatty acid oxidation intermediates predict survival in ageing men: a prospective metabolomics study

PRESENTING AUTHOR

Manthan Sharma

Manthan Sharma[1], Jaya SIngh[1], Reika Masuda[1], Nicola Gray[2], Nathan G. Lawler[3], Bu B. Yeap[4], Luke Whiley[1], Elaine Holmes[1]

[1] Centre for Computational and Systems Medicine, Health Futures Institute, Murdoch University, Harry Perkins Institute, Murdoch, WA 6150, Australia; [2] Centre for Computational and Systems Medicine, Health Futures Institute, Murdoch University, Harry Perkins Institute, Murdoch, WA 6150, Australia, School of Medical, Molecular and Forensic Sciences, Murdoch University, Murdoch, WA 6150, Australia; [3] Centre for Computational and Systems Medicine, Health Futures

PRESENTATION

RF-02

ALSO POSTER 9

Sunday 18

17:34

Riverside Ballroom

KEYWORDS

fatty acids

aging

COX proportional hazard survival analysis model

Mitochondrial fatty acid β-oxidation is crucial for metabolic homeostasis in ageing. Acylcarnitines are important intermediates in fatty acid transport and oxidation, and their accumulation may indicate impaired mitochondrial metabolic flexibility. We examined the independent association of circulating acylcarnitines with mortality risk in a prospective cohort of older men.

We used a targeted metabolomics platform based on LC-MS/MS to analyse plasma samples from 1,135 older men, aged 79 to 97 years. Survival analyses were performed using multivariable Cox proportional hazards regression models, adjusted for age, body mass index, smoking status, diabetes, history of cardiovascular disease, and creatinine concentrations. Significant metabolites were identified after Benjamini-Hochberg correction for multiple testing. Hierarchical clustering was used to analyse relationships among survival-associated metabolites.

After multivariable adjustment and correction for multiple testing, several acylcarnitine species were significantly associated with all-cause mortality. The strongest associations were observed for dicarboxylic acylcarnitines, adipoyl-L-carnitine [CAR(6:0)-DC] and sebacoyl-L-carnitine [CAR(10:0)-DC], and hydroxylated species including 3-hydroxyhexanoyl-L-carnitine [CAR(6:0)-OH] and lauroyl-L-carnitine [CAR(12:0)-OH], consistent with perturbations in fatty acid oxidation pathways. Hazard ratios were 1.17 to 1.20 per standard deviation increase in metabolite abundance (all adjusted $P < 0.001$). Cluster analysis revealed that mortality-associated acylcarnitines were part of a distinct metabolic network including intermediates of short-, medium- and long-chain fatty acid oxidation.

Circulating acylcarnitines are independently associated with long-term mortality in older men, suggesting that perturbed fatty acid oxidation, a key metabolic feature of adverse aging trajectories. These findings demonstrate the utility of targeted clinical lipidomics and survival modelling for biomarker discovery of biological ageing in population cohorts.

RAPID FIRE

RF-03

Investigating Microglial Lipid Metabolism in Response to Amyloid-Beta: Implications for Alzheimer’s Disease

PRESENTING AUTHOR

Ayesha Templeman

Ayesha Templeman[1], Luke Gray Whiley[1], Melissa Eccles[2], Giuseppe Verdile[1], David Groth[1]

[1] School of Diagnostic and Therapeutic Sciences, Curtin, Curtin University, Bentley, Western Australia 6102, Australia; [2] School of Medical and Health Sciences, Edith Cowan University, Joondalup, WA 6027, Australia

PRESENTATION

RF-03

ALSO POSTER 12

Sunday 18

17:38

Riverside Ballroom

KEYWORDS

Lipidomics

Microglia

Alzheimer's Disease

Amyloid-beta

Alzheimer’s Disease (AD) is a multifactorial neurodegenerative disorder in which lipid dyshomeostasis has emerged as a key contributor to disease onset and progression. Alterations in the lipid composition of the brain are closely linked to hallmark AD pathologies, such as amyloid-beta (Aβ) accumulation, tau pathology and neuroinflammation. Microglia are central to this process, as they regulate lipid uptake, metabolism and trafficking. However, in AD, microglial lipid handling becomes dysregulated leading to impaired phagocytosis and a pro-inflammatory phenotype that exacerbates neuronal damage. As such, examining the relationship between the lipidome and functional outcomes in microglial cells will elucidate how microglial dysregulation drives AD pathogenesis. This project aims to investigate how Aβ oligomer exposure reshapes the microglial lipidome and alters downstream cellular functions, comparatively to classic innate immune stimuli. By integrating comprehensive lipidomic profiling with functional assays, this study will examine how lipid metabolic pathways influences lipid droplet dynamics, inflammatory responses and phagocytotic capacity in microglial cells. The significance of this work lies in advancing mechanistic understanding of how specific lipid pathways regulate microglial function, as many lipid species remain unidentified. Insights into lipid perturbations will help identify which lipid metabolic processes are most influential in shaping microglial behaviour. This may also reveal novel targets for therapeutic intervention, providing the foundation for more precise strategies aimed at modulating the microglial lipidome in AD.

RAPID FIRE

RF-04

The lipid landscape behind harmful extracellular vesicle release in vascular aging

PRESENTING AUTHOR

Elizaveta (Eliza) James

Elizaveta (Eliza) James[1], Jennifer Gamble[2], Georges Grau[3], Yanfei (Jacob) Qi[1]

[1] Heart Research Institute; [2] Centenary Institute; [3] University of Sydney

PRESENTATION

RF-04

ALSO POSTER 20

Sunday 18

17:42

Riverside Ballroom

KEYWORDS

- Senesence
- Vascular dysfunction
- Healthy Aging
- Lipidomics

Endothelial cells are among the earliest cell types to undergo senescence during organismal ageing. Senescent endothelial cells secrete pro-inflammatory factors collectively termed the senescence-associated secretory phenotype (SASP), propagating vascular inflammation and dysfunction. Large extracellular vesicles (lEVs) are increasingly recognised as mediators of intercellular communication and an important component of the SASP. However, the mechanisms regulating senescent lEV (seno-lEV) release and their pathogenic effects remain poorly understood.

Cholesterol is a major constituent of the plasma membrane (PM), the primary site of lEV biogenesis. Beyond its structural role, cholesterol regulates membrane organisation, vesicle budding, and exocytosis, all of which influence lEV secretion. This study investigates whether PM cholesterol is a viable target for limiting harmful seno-lEV release.

We established a robust isolation and characterisation pipeline for seno-lEVs derived from senescent primary human endothelial cells using nanoparticle tracking analysis, spectral flow cytometry, transmission electron microscopy, and Western blotting. We found that lEV release is significantly increased in senescent endothelial cells and that seno-lEVs induce pro-inflammatory responses in young recipient endothelial cells, supporting their pathogenic role. We are investigating the inflammatory signalling pathways activated by seno-lEVs.

Up to 90% of free cholesterol resides within the endothelial PM. Lipidomic analysis revealed elevated PM cholesterol in senescent endothelial cells. Our laboratory identified a unique lipid transporter that specifically delivers cholesterol to the PM. Using lentiviral knockdown, we selectively restrict PM cholesterol accumulation and are currently testing whether lowering PM cholesterol limits seno-lEV release and attenuates its pathological effects to identify novel therapeutic targets for vascular ageing.

RAPID FIRE

RF-05

Lipid remodelling defines divergent cancer immunotherapy outcomes

PRESENTING AUTHOR

Nicola Principe

Nicola Principe[1], Amber-Lee Phung[1], Dana Hicks[2], Kriya Velayutham[1], Cher van der Heijden[1], Melvin Chin[3], Prerak Trivedi[4], Nicola Gray[2], Jonathan Chee[1]

[1] Institute for Respiratory Health, The University of Western Australia, Nedlands, WA, Australia; [2] Centre for Computation and Systems Medicine and Australian National Phenome Centre, Health Futures Institute, Murdoch University, Perth, WA, Australia; [3] Sir Charles Gardiner Hospital, Institute for Respiratory Health, The University of Western Australia, Nedlands, WA, Australia; [4] St Vincents Institute, Melbourne, VIC, Australia

PRESENTATION

RF-05

ALSO POSTER 33

Sunday 18

17:46

Riverside Ballroom

KEYWORDS

cholesterol

lung cancer

mesothelioma

immunotherapy

BACKGROUND

Immune checkpoint blockade (ICB) produces durable cancer regression in a minority of patients, but majority of treated patients are non-responsive to ICB. Tumour lipids contribute to a ICB non-responsive tumour immune microenvironment (TiME), including supporting regulatory T cells (Tregs) proliferation and suppressive function. We compared tumour lipid and Treg profiles between ICB responders and non-responders and investigated whether targeting lipid metabolism could alter the TiME and improve ICB response.

METHODS

We used an established mouse model in which ICB generates responders and non-responders. We assessed TiME at baseline and seven days after ICB. Tumours underwent targeted lipidomics following IPA90 extraction with analysis by LC-MS/MS using a SCIEX QTRAP 7500+ mass spectrometer. Tumour infiltrating Tregs were analysed by RNA sequencing and flow cytometry. We treated mice with cholesterol-lowering therapy alone or in combination with ICB, and assessed changes in Treg phenotype and therapeutic response.

RESULTS

Lipid profiles and Treg frequencies were similar between responders and non-responders at baseline. Responding tumours following ICB had increased cholesteryl esters, LPC/PC and reduced SM/Cer. These changes coincided with reduced intratumoral Tregs, and downregulation of cholesterol homeostasis genes in Tregs. A cholesterol lowering drug reduced intratumoural Tregs, their expression of lipid uptake receptor CD36, and increased responders when combined with ICB.

CONCLUSION

These findings identify complex remodelling of lipid profiles in the TiME post ICB that could contribute to ICB response. Future work will investigate the cell-specific lipid mechanisms across T cell subsets in relation to ICB and in combination with cholesterol-lowering drugs.

RAPID FIRE

RF-06

Integrative Modelling of Lipidomics and Lipoproteomics: A Comparative Approach for BMI Prediction

PRESENTING AUTHOR

Novia Minaee

Novia Minaee[1], Alanah St-James[1], Luke Whiley[2], Nicola Gray[3], Reika Masuda[4], Bu B Yeap[5], Samantha Lodge[1], Kevin Wong[6], Jeremy K Nicholson[7], Julien Wist[8], Elaine Holmes[9]

[1] Centre for Computational and Systems Medicine, Health Futures Institute, Murdoch University, Perth, Australia; [2] Centre for Computational and Systems Medicine, Murdoch University, Perth, Australia; [3] Medical, Molecular and Forensic Sciences, Murdoch University, Perth, Australia; [4] Australian National Phenome Centre, Murdoch University, Perth, Australia; [5] Medical School, University of Western Australia, Perth, Australia; [6] School of Information Technology, College of Science, Technology,

PRESENTATION

RF-06

ALSO POSTER 13

Sunday 18

17:50

Riverside Ballroom

KEYWORDS

lipid

lipoprotein

feature selection

multi-omics integration

Body mass index (BMI) is a widely used surrogate of adiposity but does not fully capture the biological complexity of metabolic dysfunction. Lipidomics and lipoproteomics provide complementary molecular information that may better reflect cardiometabolic status. This study used BMI prediction as a clinically relevant benchmark to compare the predictive performance and biological interpretability of feature selection methods for integrated lipidomic and lipoproteomic data.

Serum samples from the Busselton Healthy Ageing Study (BHAS; N = 1,806), comprising 112 lipoprotein parameters measured by ^{1}H NMR spectroscopy and 895 lipid species quantified by mass spectrometry, were analysed. Eight feature selection methods (Lasso, Ridge, Elastic Net, Random Forest, Support Vector Machine, OPLS-R, HPLS-R and SELECT) were evaluated across repeated train-test splits. Model performance was assessed using RMSE, R^2Y, Q^2Y, feature stability and biological interpretability.

Lasso achieved the strongest predictive performance (RMSE = 0.749, R^2Y = 0.455, Q^2Y = 0.419). A consensus framework combining top-ranked features across methods with correlation-based redundancy reduction produced comparable validation performance (RMSE = 0.799, R^2Y = 0.381, Q^2Y = 0.362) while identifying a more diverse and biologically informative biomarker panel, including sphingomyelins, phosphatidylcholines and specific triacylglycerides. These findings demonstrate that consensus-based feature selection can improve biological interpretability without compromising predictive performance, providing a practical framework for integrating high-dimensional lipidomic and lipoproteomic data.

RAPID FIRE

RF-07

A Systematic Evaluation of Extraction and 4D-MS Parameters for Comprehensive Human Blood Lipidomics

PRESENTING AUTHOR

Nurul Saadah Ahmad

Nurul Saadah Ahmad[1], Rafidah Ahmad[2], Rosni Ibrahim[3], Toh Leong Tan[4], Syarul Nataqain Baharum[2], Zamberi Sekawi[3]

[1] Department of Medical Microbiology, Faculty of Medicine and Health Sciences, Universiti Putra Malaysia, Serdang, Selangor, Malaysia., Department of Emergency Medicine, Faculty of Medicine, Universiti Kebangsaan Malaysia, Kuala Lumpur, Malaysia.; [2] Metabolomics Research Laboratory, Institute of Systems Biology (INBIOSIS), Universiti Kebangsaan Malaysia, Bangi 43600, Selangor, Malaysia.; [3] Department of Medical Microbiology, Faculty of Medicine and Health Sciences,

PRESENTATION

RF-07

ALSO POSTER 22

Sunday 18

17:54

Riverside Ballroom

KEYWORDS

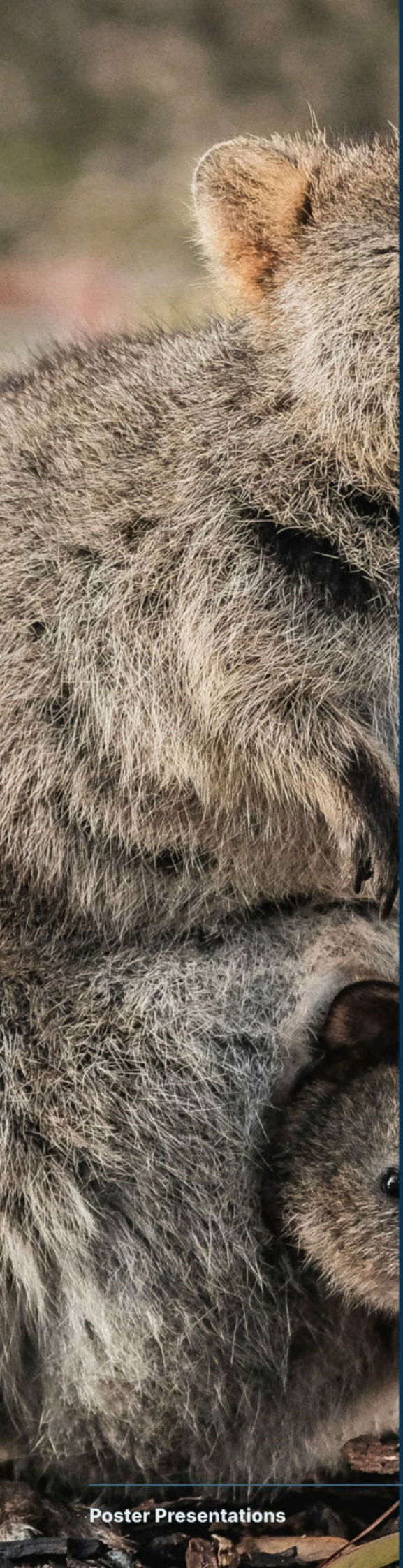
lipid extraction

human serum

human plasma

ion mobility

The rapidly evolving field of lipidomics requires robust and optimized analytical workflow to ensure comprehensive lipid coverage and reproducibility. However, methodological variability in sample preparation and analytical conditions remains a major challenge in untargeted lipidomic studies. This study systematically evaluates key parameters to establish an optimal 4D LC-IM-QTOF-MS workflow for maximizing lipid yield from human serum and plasma. We performed a multi-factorial evaluation of four lipid extraction methods (Folch, Isopropanol [IPA], Butanol-Methanol [BUME] and Methyl tert-butyl ether [MTBE]), two sample types (human serum vs. plasma), column types (C8 vs. C18), and ionization modes (positive vs. negative). Analyses were conducted on a 4D liquid chromatography ion mobility-quadrupole time-of-flight (4D LC-IM-QTOF-MS) platform. Performance was assessed based on the total number of detected lipid features, peak area intensity and lipid class annotation. Among the four extraction methods, isopropanol (IPA) extraction yielded the highest total lipid peak area and number of detected lipid species. Chromatographic separation using C8 column showed better retention and separation across diverse range of lipid classes, particularly for non-polar lipids as compared to C18. Positive ionization mode significantly detected a broader spectrum of phospholipids and sphingolipids than negative mode. Under these optimized conditions, serum and plasma exhibited largely overlapping lipid profiles with distinct differences detected between the two matrices. Therefore, a workflow integrating IPA extraction, C8 column and positive ion mode provides the most comprehensive lipid coverage for human serum and plasma, offering a standardized framework for 4D-MS-based untargeted lipidomics, enhancing data reproducibility in human lipidomics research.



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Poster Presentations

Poster abstracts are presented in poster-number order for quick reference during the meeting.

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POSTER PRESENTATION

P-01

High-throughput regioisomer resolved analysis of phosphoinositides

PRESENTING AUTHOR

Kevin Huynh

Alexandra Faulkner[1], Matilda van Buuren-Milne[1], Thy Duong[1], Nat Mellett[1], Sudip Paul[1], Peter Meikle[1], Yow Keat Tham[1], Kevin Huynh[1]

[1] Baker Heart and Diabetes Institute

POSTER

Poster 1

Monday 19

12:45

Silver Room

KEYWORDS

phosphoinositide

phosphatidylinositol phosphate

high-throughput

Phosphoinositides (PIPs) are low-abundance signalling lipids whose biological functions depend on both the number and position of phosphate groups on the inositol headgroup. Their analysis by liquid chromatography-mass spectrometry (LC-MS) remains difficult because of poor stability, non-specific surface adsorption and the need for hazardous derivatisation procedures. We developed a simplified, derivatisation-free LC-MS workflow for high-throughput regioisomer-resolved measurement of phosphatidylinositol monophosphate, diphosphate and triphosphate species.

An optimised method to isolate PIPs were extracted using a monophasic butanol:methanol:water procedure containing EDTA, medronic acid and ammonium formate. phosphate-dependent surface losses. Chromatographic separation was performed on a C8 column using solvents comprising of water, acetonitrile and isopropanol with ammonium bicarbonate and medronic acid in the mobile phases. Targeted analysis was conducted on an Agilent 6495C triple quadrupole and Agilent Revident Q-TOF.

The workflow achieved femtomole-on-column sensitivity without methylation or another chemical derivatisation. Furthermore, it provides coverage of many complex anionic lipids beyond phosphoinositides. Two acquisition formats were established: a five-minute method for high-throughput phosphoinositide profiling and a 15-minute method providing headgroup regiosiomer resolution. We apply this method across diverse cell and tissue types, highlighting substantial re-arrangement of phosphoinositides isomers upon immune activation, and heterogeneity across different tissue types.

This method provides a practical platform for scalable phosphoinositide analysis using standard LC–MS instrumentation. It reduces sample handling, avoids highly reactive derivatisation reagents and enables both rapid screening and regioisomer-resolved characterisation of phosphoinositide signalling across biological samples. The approach is readily adaptable to tissues, cultured cells and translational cohort studies.

POSTER PRESENTATION

P-02

Development of a targeted lipid plasma method with a propanol free mobile phase

PRESENTING AUTHOR

Lee Thomasson

Robert Trengove[1], Lee Thomasson[2], Carl Mousley[2], Danielle Dye[2], Emma Hadfield[2], Robert Steuart[1], Pat Metharom[1], Robert Plumb[3]

[1] Curtin Medical Research Institute; [2] Curtin School of Diagnostic and Therapeutic Sciences; [3] Waters Corporation

POSTER

Poster 2

Tuesday 20

12:45

Silver Room

KEYWORDS

plasma lipidomics

reversed phase

phenyl hexyl

Lipid analysis methodologies typically use either untargeted reversed phase liquid chromatography, targeted HILIC approaches, or a shotgun lipidomics approach. Reversed phase use of a C18 stationary phase requires the use of an isopropanol based mobile phase for total lipid analysis. A targeted HILIC method, using a HILIC or Amide column, can elute all classes of lipids without an isopropanol based mobile phase. However, the HILIC separation is a polarity based separation and the lipid coverage can be limited by the number of lipids to be analysed within each “polarity” group, even with latest generation triple quadrupole systems.

We have developed a targeted reversed phase LC-MS/MS method with a phenyl hexyl stationary phase and an acetonitrile based mobile phase with a triple quadrupole mass spectrometer. The use of acetonitrile in the mobile phase facilitates polar metabolite analysis and lipid analysis with the same instrument/mobile phase system and negates the use of isopropanol (inferior purity compared to the water & acetonitrile) and the need to rinse the system to remove isopropanol post analysis.

The method currently semi-quantitates over 800 targeted lipids in 12 minutes and has been developed in parallel with a targeted HILIC approach.

POSTER PRESENTATION

P-03

A Comprehensive Targeted Lipidomics Workflow for Characterisation of Multiple Biological Matrices

PRESENTING AUTHOR

Dana Hicks

Dana Hicks[1], Monique Ryan[2], Harrison Szemray[1], Lakshini Herat[3], Markus Schlaich[4], Bu Yeap[5], Girish Dwivedi[6], Julien Wist[7], Luke Whiley[8], Nicola Gray[9]

[1] Centre for Computational and Systems Medicine, Murdoch University, Perth, Western Australia; [2] Australian National Phenome Centre, Murdoch University, Perth, Western Australia; [3] Dobney Hypertension Centre, Medical School - Royal Perth Hospital Unit, The University of Western Australia, Perth, Western Australia; [4] Dobney Hypertension Centre, Medical School - Royal Perth Hospital Unit, The University of Western Australia, Perth, Western Australia, RPH Research Foundation, Perth,

POSTER

Poster 3

Monday 19

12:45

Silver Room

KEYWORDS

Targeted lipidomics

INTRODUCTION

Lipidomics is increasingly used to investigate biological pathways underlying health and disease. While untargeted approaches support discovery, targeted LC-MS/MS provides reproducible semi-quantitative measurements for large-scale biological and clinical studies. However, many targeted methods provide limited lipidome coverage, restricting pathway-level interpretation and mechanistic insight. Here, we describe an expanded targeted LC-MS/MS method for lipid profiling across diverse biological matrices.

METHOD

Lipid profiling across 29 subclasses was performed using a 16 min reversed-phase method targeting 2,076 lipid species and 87 stable isotope-labelled internal standards. Adipose, heart, kidney, liver, pancreas, and plasma samples were analysed using an ExionLC™-SCIEX Triple Quad™ 7500 system. Peak integration and quality control (QC) filtering were performed using in-house pipelines, retaining lipid species with relative standard deviations (RSDs) <30%.

RESULTS

Lipid species meeting the pooled-QC criterion (RSD <30%) ranged from 1,419 in plasma and 1,472 in adipose to 1,839 in pancreas, with heart (1,752), liver (1,758), and kidney (1,816) also demonstrating high analytical reproducibility. Median pooled-QC RSDs ranged from 11.8% to 14.2% across tissues and from 8.3% to 10.4% in plasma. Reproducibility differed by lipid subclass and biological matrix, with cardiolipins and several glycerophospholipid subclasses showing the greatest variability.

CONCLUSION

This expanded targeted lipidomics workflow provides broad lipid subclass coverage while maintaining robust analytical performance across diverse biological matrices. The increased coverage supports deeper interrogation of lipid pathways associated with mitochondrial function, fatty acid metabolism, lipid signalling, sphingolipid biology, and plasmalogen/ether lipid metabolism in mechanistic and clinical studies.

POSTER PRESENTATION

P-04

Application of Zeno SWATH DIA in multi-tissue metabolomic profiling

PRESENTING AUTHOR

Thusitha Rupasinghe

Qing Zhao[1], Guo Shou Teo[1], Chenyuan Huang[2], Jiong-Wei Wang[2], Thusitha Rupasinghe[3], Paul Baker[4], Markus R. Wenk[5], Hyungwon Choi[1]

[1] Singapore Lipidomics Incubator (SLING), Life Sciences Institute, National University of Singapore, Singapore; [2] Yong Loo Lin School of Medicine, National University of Singapore, Singapore; [3] SCIEX, Australia; , 4^SCIEX, US; [5] College of Health and Life Science, Hamad Bin Khalifa University, Qatar

POSTER

Poster 4

Tuesday 20

12:45

Silver Room

KEYWORDS

Untargeted

ZenoTOF 7600

Multi tissue

IDA

SWATCH

Untargeted metabolomics has emerged as a method of choice in discovery-driven and hypothesis-generating research. Using a HRMS instrument, untargeted analysis can be performed in either data-dependent acquisition (DDA, as information-dependent acquisition, or IDA) or data-independent acquisition (DIA, or SWATH-MS). Although IDA remains the most widely used approach in the metabolomics field, conventional IDA workflows suffer from inherent limitations due to their stochastic nature in MS/MS spectra acquisition, biased towards abundant precursor ions. In contrast, SWATH-MS acquisition offers a truly unbiased MS/MS spectral collection, enhancing the MS2-level detection of low-abundance metabolites that are otherwise missed in IDA. In this study, we describe a scalable quantitative metabolomics workflow based on sequential application of IDA and SWATH-MS acquisition using the ZenoTOF 7600 system. Mice were subjected to left anterior descending artery occlusion followed by reperfusion. In total, twelve tissues were collected at four time points post-surgery, including the heart, liver, lung, spleen, kidney, intestine, colon, brain, white adipose tissue (WAT), brown adipose tissue (BAT), skeletal muscle, and plasma. Tissue and plasma samples were extracted using a biphasic methyl tert-butyl ether (MTBE)/methanol (MeOH)/H_2O system to separate organic and aqueous fractions. Metabolic changes in the infarct region of the left ventricle are the primary focus of the myocardial infarction mouse model. For downstream analysis, differentially changed metabolites were identified based on the highest-quality quantifiers. To further validate our biological findings, a list of significantly altered metabolites was examined by mining all high-quality MS1- and MS2-level quantifiers from the Zeno SWATH DIA data and up to three MS2-level quantifiers that passed our filtering pipeline were incorporated into the visualization, alongside their corresponding MS1-level features on the left. A high degree of concordance was observed between MS1- and MS2-level peak features, demonstrating the capability of Zeno SWATH to provide MS2-level quantification with higher confidence.

The organic fraction was separated using a reverse phase (RP) ZORBAX RRHD Eclipse Plus C18 column (2.1 x 50 mm, 1.8 µm). The aqueous fraction was separated using a SeQuant ZIC-cHILIC PEEK column (100 x 2.1, 3um,100Å). Lipids and polar metabolites were analyzed using RP and HILIC methods, respectively, with ZenoTOF 7600. IDA with Zeno trap generated rich metabolite spectra from multiple tissues. After preliminary filtering of IDA data, customized spectral libraries with 2047 IDs (HILIC-positive), 676 IDs (HILIC-negative), and 2541 IDs (RP-positive) were generated for SWATH-MS data extraction. Zeno SWATH DIA enabled deep profiling of tissue samples, yielding quantification of approximately 900-1000 metabolites with high confidence after applying rigorous quality control criteria. This profile comprised over 600 lipids and approximately 300 non-lipid metabolites. In addition, metabolite coverage in individual tissue samples was significantly enhanced by matching against the customized spectral library pooled from all 12 tissue types in the IDA experiment.

POSTER PRESENTATION

P-06

MStargetR: A reproducible, auditable, and containerised R workflow for end-to-end targeted mass spectrometry data processing

PRESENTING AUTHOR

Harrison Szemray

Harrison Szemray[1], Vimalnath Nambiar[1], Dana Hicks[1], Samantha Lodge[2], Julien Wist[3], Nathan Lawler[4], Luke Whiley[5]

[1] Centre for Computational and Systems Medicine Murdoch University, School of Medical Molecular and Forensic Science Murdoch University; [2] Centre for Computational and Systems Medicine Murdoch University; [3] Centre for Computational and Systems Medicine Murdoch University, Chemistry Department, Universidad del Valle; [4] Centre for Computational and Systems Medicine Murdoch University, School of Allied Health Murdoch University; [5] Centre for Computational and Systems

POSTER

Poster 6

Tuesday 20

12:45

Silver Room

KEYWORDS

Lipidomics

Mass Spectrometry

Reproducibility of Results

Software

Targeted LC-MS/MS metabolic phenotyping pre-processing relies on a disjointed toolchain of vendor software, manual peak integration, and in-house quality control (QC) scripts, introducing variability between users and laboratories that complicates cross-study comparisons. Existing tools address only part of the problem: vendor platforms allow detailed inspection but lock data into proprietary formats; open-source MRM processors automate peak picking but force inflexible integration decisions at export; and Skyline offers vendor-neutral chromatogram review but is not a scriptable, R-native batch workflow with an integrated QC layer. As a result, no unified open-source workflow currently combines automation, auditable peak review, and reproducibility in a single deployable toolkit.

MStargetR is an open-source R package integrating Skyline and msConvert into a fully scriptable, R-native workflow, automating vendor file conversion, peak integration, and quality control from a single R session. Containerisation via version-pinned Docker images ensures reproducible execution across systems, with Apptainer support for HPC environments where Docker is restricted. A Shiny-based interface supports code-free access. Critically, each analytical plate produces a fully populated Skyline project file (.sky) that can be reopened for chromatogram inspection and manual boundary correction, with results reimported without rerunning upstream steps, preserving human review without breaking reproducibility.

Applied to a 128-sample targeted lipidomics dataset (1,161 lipid MRM targets), MStargetR recovered 949 features (81.7%) at RSD <30% across long-term reference QCs. Throughput testing up to 4200 samples averaged 4.1 seconds per sample, demonstrating robust analytical performance and predictable scaling on a standard 8-core workstation without specialist computational infrastructure.

POSTER PRESENTATION

P-07

Bridging Lipid Annotation and Chemometric Analysis Through a Cross-Platform Nontargeted Lipidomics Workflow

PRESENTING AUTHOR

Chris Fouracre

Mark Sartain[1], Emma E. Rennie[1], Chris Fouracre[1], Sarah M. Stow[1], Sandra Gallego[1], Jeremy Koelmel[1]

[1] Agilent Technologies

POSTER

Poster 7

Monday 19

12:45

Silver Room

Agilent Technologies

Agilent

As adoption of non-targeted lipidomics expands, software tools must provide high-confidence, high-coverage annotations within a user-friendly framework. MassHunter Explorer offers a workflow-driven interface for feature extraction, statistical analysis, and MS/MS-based compound identification, while LipidMatch Suite provides deep lipid annotation capabilities and tools for manual review and curation. We demonstrate an integrated workflow that exchanges information between these software platforms and enables comprehensive lipid characterization of a brain lipid extract. This approach yielded over 800 lipid annotations and highlights the importance of chemometrics, streamlined curation, and integrated visualization tools for robust lipidomics interpretation.

POSTER PRESENTATION

P-08

Development of a metabolomic screening algorithm for the early detection of genetic disorders using newborn dried bloodspots: The NewbornsInSA Study

PRESENTING AUTHOR

Tomas Rozek

Tomas Rozek[1]

[1] Department of Genetics and Molecular Pathology, SA Pathology, Adelaide, SA, 5000 Australia

POSTER

Poster 8

Tuesday 20

12:45

Silver Room

KEYWORDS

Lipidomics

mass spectrometry

New Born Screening

Blood Spots

NewbornsInSA

BACKGROUND

The NewbornsInSA research study explores a novel multi-omic model of newborn bloodspot screening (NBS) that combines semi-targeted metabolomic profiling, genomic sequencing, and clinical factors to identify newborns at risk of early onset, treatable genetic conditions. Here we describe the development and validation of a metabolomic screening algorithm aimed at identifying newborns that may be at risk for one of these conditions.

METHODS

and Results: Untargeted metabolomics was performed on 2,648 retrospective dried bloodspots from term newborns with no known genetic disorder and 180 cases with genetically confirmed diagnoses. These samples formed a discovery cohort and a validation cohort. Untargeted metabolomics and lipidomics were performed using high resolution mass spectrometers operated in both positive and negative ionisation modes using different Ultra-Performance Liquid Chromatography separations. The discovery cohort (n = 2,193) was used for the development of a novel screening algorithm. Supervised machine learning statistics analysis were applied to identify discriminatory features and performance assessed using the confirmed genetic diagnoses in the validation cohort (n = 635). Features have been selected for semi-targeted metabolomics and the performance of the algorithm is being evaluated. Preliminary analyses found that our metabolomic screening algorithm could identify conditions screened in traditional NBS programs, such as Phenylketonuria and X-linked adrenoleukodystrophy, and shows promise for conditions not yet included.

CONCLUSION

The semi-targeted algorithm successfully identified a range of conditions in our discovery and validation cohorts, and its performance will be further evaluated in a larger retrospective cohort, and >1,000 prospectively enrolled newborns who underwent genomic NBS.

POSTER PRESENTATION

P-10

Enabling Precision and Translational Lipidomics Through Bespoke Deuterated Lipids from ANSTO's National Deuteration Facility

PRESENTING AUTHOR

Karyn Wilde

Karyn Wilde[1], Tamim Darwish[1]

[1] National Deuteration Facility- Australian Nuclear Science and Technology Organisation

POSTER

Poster 10

Tuesday 20

12:45

Silver Room

ANSTO

National Deuration Facility

KEYWORDS

- Deuterium Labelling
- Deuterated Lipids
- Labelled Lipids

Quantitative lipidomics increasingly underpins diagnostic development, regulatory science, and next-generation therapeutics. Yet accurate mass-spectrometric quantification remains challenged by matrix effects, ionisation variability, and instrument drift. Stable isotope internal standards, particularly deuterated lipids with defined isotopic incorporation, remain the gold standard for achieving analytical robustness. Recognising a global shortfall in the availability of structurally precise deuterated lipids, the National Deuteration Facility (NDF), an open-access national capability co-funded by NCRIS, has established a specialised capability for the bespoke synthesis of high purity, deuterium labelled lipid molecules. Building on longstanding expertise in phospholipid deuteration, NDF now produces a diverse suite of labelled sterols, phospholipids, glycerides, and unsaturated lipid chains. These molecules support advanced characterisation techniques, while enabling high resolution MS quantitation studies. Importantly, NDF has demonstrated the ability to synthesise deuterated analogues of many lipids central to mRNA lipid nanoparticle (LNP) technologies. As LNPs become foundational to vaccines, gene therapies, and precision lipidomics workflows, access to labelled ionisable lipids, helper lipids, and PEG-lipids is increasingly critical for mechanistic studies, stability assessments, and regulatory grade quantification. NDF's bespoke lipid standards are now being deployed across multiple industry and clinical research partnerships in Australia and overseas, supporting the development of emerging diagnostic assays and translational lipidomics platforms. These collaborations highlight the essential role of precise isotopic labelling in enabling reproducible, quantitative lipidomics suitable for clinical validation. This poster presentation will outline some recent advances in deuterated-lipid synthesis and showcase some applications, demonstrating how national-scale isotope labelling capability is accelerating the future of precision and translational lipidomics. Information will also be provided on the NDF access modes available for research and industry users.

POSTER PRESENTATION

P-11

Clinical Translation of Targeted Lipidomics: Validation of BREASTEST plus v2 for Early Breast Cancer Detection

PRESENTING AUTHOR

Mark David

Mark David[1], Kalpesh Shah[1], David Peake[1], Kathy Koskiris[1], Lisa Ren[1], Fatemeh Vafaee[2], Tim Huang[2], David Speakman[3]

[1] BCAL Diagnostics Ltd, Sydney, NSW 2000, Australia; [2] OmniOmics.ai, Pagewood, NSW 2035, Australia; [3] The Peter MacCallum Cancer Centre, Melbourne, VIC

POSTER

Poster 11

Monday 19

12:45

Silver Room

KEYWORDS

- Lipidomics
- Clinical Translation
- Breast Cancer
- Validation (Analytical
- Algorithm and Clinical)

Breast cancer diagnosis continues to rely predominantly on imaging and tissue biopsy, highlighting the need for complementary blood-based biomarkers that can improve diagnostic pathways. BREASTEST plus™ - version 2 is the next generation of a targeted lipidomic assay developed using liquid chromatography-tandem mass spectrometry (LC-MS/MS) and machine learning to support early breast cancer detection. Building on the first commercially available lipidomic oncology assay (BREASTEST plus™), the updated platform expands the clinical utility of the test through an increase in the biomarker panel from six to sixteen lipids and incorporates the clinical variables age and body mass index (BMI) within a regularised logistic regression classification model. The algorithm also incorporates an indeterminate classification to improve confidence around borderline results.

The assay employs a targeted LC-MS/MS workflow on a SCIEX 6500+ platform using polarity-switching electrospray ionisation, stable isotope-labelled internal standards and quantitative multiple reaction monitoring (MRM), with the workflow developed to support future high-throughput clinical implementation. The assay is currently undergoing comprehensive analytical and pre-analytical validation in accordance with clinical laboratory requirements to support implementation in an accredited diagnostic laboratory.

The classification algorithm was developed using approximately 500 prospectively collected plasma samples, with independent clinical validation currently being performed in a separate cohort of approximately 300 Australian women. In parallel, technology transfer activities are underway to establish the assay within an international reference laboratory, providing the foundation for future inter-laboratory validation and broader clinical implementation.

Final analytical validation and clinical performance data will be presented.

POSTER PRESENTATION

P-14

Lipidomic analysis reveals a distinct PUFA-rich profile in endometriosis

PRESENTING AUTHOR

Sarah W. Walsh

Sarah W. Walsh[1], Fiona Wood[2], Lucy W. Barrett[2], Andrew W. Stevenson[2], Jayden L. Roberts[3], Monique J Ryan[4], Luke Whiley[5], Cecilia M. Prêle[6], Nicola Gray[7]

[1] School of Medical, Molecular and Forensic Sciences (Murdoch University), Burn Injury Research Unit (The University of Western Australia), Fiona Wood Foundation,; [2] Burn Injury Research Unit (The University of Western Australia), Fiona Wood Foundation; [3] Australian National Phenome Centre (Murdoch University), Centre for Computational and Systems Medicine (Murdoch University), The University of Notre Dame; [4] Australian National Phenome Centre (Murdoch University), Centre for

POSTER

Poster 14

Tuesday 20

12:45

Silver Room

KEYWORDS

- Endometriosis
- Lipidomics
- Polyunsaturated fatty acids
- Triacylglycerols
- Free fatty acids

BACKGROUND

Endometriosis is a chronic, heterogeneous disorder with chronic inflammatory features that affects approximately 1 in 9 women of reproductive age and remains frequently underdiagnosed owing to the lack of robust non-invasive biomarkers. Identifying disease-specific metabolic signatures may facilitate earlier diagnosis and improve patient care.

AIM

To characterise the plasma lipidomic signature associated with surgically confirmed endometriosis compared to age-matched women without the disease.

METHODS

Plasma lipid species were quantified by targeted LC-MS/MS in women with clinically confirmed endometriosis (n = 44) available through the ENDORIGINS biobank (Perth, Western Australia) and age-matched controls (n = 23). Cases were stratified as minimal (superficial peritoneal lesions; n = 15) or extensive (ovarian endometrioma and/or deep infiltrating endometriosis; n = 29). Group differences were investigated using OPLS-DA, then followed by subsequent two-sided Mann-Whitney U and Kruskal-Wallis tests with Benjamini-Hochberg false discovery rate correction (FDR < 0.05) to determine differentially abundant lipids.

RESULTS

Plasma lipid profiles differed significantly between endometriosis and controls, with 138 lipids showing differential abundance, including 52 elevated in endometriosis. Endometriosis was characterised by increased free fatty acids including FA(18:2) and FA(18:3), alongside polyunsaturated fatty acid-rich (PUFA) diacylglycerols (DG(16:0_18:2)) and long-chain, highly unsaturated triacylglycerols enriched in FA18:2 and FA18:3 (TG(54:7_FA18:2)). In contrast, controls showed higher monoacylglycerols (MG(16:1), MG(18:1)), shorter-chain, less polyunsaturated di- and triacylglycerols (TG(40:0_46:0)), and phospholipids including PI(16:0_14:0) and PE(18:1_18:2).

CONCLUSION

Endometriosis is associated with a distinct systemic PUFA-rich plasma lipid profile. Further studies are required to validate these findings and their potential as disease biomarkers.

POSTER PRESENTATION

P-15

A unified workflow for small molecule analysis for exposomics and molecular epidemiology

PRESENTING AUTHOR

Steve Wilson

Erica Forsberg[1], Aiko Barsch[2], Steve Wilson[6], Cristian De Gobba[3], Tuan Hai Pham[4], Guido Dallman[4], Alice Limonciel[4], Matthew R. Lewis[5]

[1] Bruker Scientific LLC, San Jose, CA, United States; [2] Bruker Daltonics GmbH & Co. KG, Bremen, Germany; [3] Bruker Nordic Denmark, Filial of Bruker Nordic AB, Copenhagen, Denmark; [4] Biocrates life sciences gmbh, Innsbruck, Austria; [5] Bruker UK Ltd., Coventry, United Kingdom; [6] Bruker Australia

POSTER

Poster 15

Monday 19

12:45

Silver Room

Bruker

A hybrid LC-MS workflow was developed to combine targeted quantification and untargeted ion mobility-enabled metabolomics in a single platform. Targeted analysis uses validated calibration models and stable isotope-labeled internal standards for absolute quantification, while untargeted analysis uses trapped ion mobility spectrometry (TIMS) with data-dependent acquisition for feature discovery and structural annotation. Two TIMS acquisition modes were evaluated: DDA-PASEF and DDA with Mobility Range Enhancement (DDA-MoRE), which broadens the accessible mobility region for high-mobility, lower-mass species. This parallel strategy maximizes information yield from each sample injection.

Eleven human plasma samples and NIST SRM 1950 reference plasma were prepared using dual derivatization (PITC and 3NPH) and analyzed in triplicate. Data were processed with dedicated software for absolute quantification and untargeted feature extraction, annotation, and in-silico derivatization. Quantitative precision was high, with median CV of 6.5% in spiked plasma and 3.4% in NIST SRM 1950. Accuracy of representative metabolites against certified reference values showed generally <15% deviation.

Untargeted analysis enabled confident annotation of 474 derivatized targets across both chemistries, complemented by features identified exclusively through predicted derivatization patterns. Principal component analysis of a proof-of-concept donor comparison revealed distinct separation of one lipemic sample. Rule-based lipid annotation showed that lysophosphatidylcholines, absent from the targeted panel but present in the untargeted dataset, drove this differentiation. Additional small molecules

annotated through in-silico prediction further expanded the accessible chemical space. All spectra were archived in a digital metabolome for future discovery.

Ongoing work will evaluate priority-directed DDA (prio-MoRE and prio-PASEF), which preferentially selects targeted compounds for fragmentation before allocating remaining duty cycle to untargeted precursor sampling. This is expected to improve MS/MS coverage and spectral quality for targeted analytes while preserving broad metabolome discovery. By uniting rigorous quantification with expanded metabolite annotation, this workflow provides a scalable framework for large-cohort studies in exposomics and biomarker discovery.

POSTER PRESENTATION

P-16

Lipidome-wide analyses provide novel insights into the relationship between lipid metabolism, type 2 diabetes and prediabetes risk.

PRESENTING AUTHOR

Habtamu Bedimo Beyene

Habtamu Bedimo Beyene[1], Tingting Wang[1], Michelle Cinel[1], Natalie Mellett[1], Thy Duong[1], Matilda van Buuren-Milne[1], Alexandra Faulkner[1], Jingqin Wu[1], Gavriel Olshansky[1], Jonathan Shaw[1], Dianna Magliano[1], Melissa Southy[2], Roger Milne[2], Allison Hodge[2], Corey Giles[1], Kevin Huynh[1], Peter Meikle[1]

[1] Baker Heart and Diabetes Institute; [2] Monash University

POSTER

Poster 16

Tuesday 20

12:45

Silver Room

KEYWORDS

- Lipidome-wide associations
- T2D
- impaired fasting flucose and impaire glucose tolerance
- large cohorts
- mass spectrometry

Type 2 diabetes (T2D) is characterised by elevated fasting glucose, insulin resistance, and impaired glucose tolerance. We report lipid associations with T2D, fasting blood glucose (FBG), and 2-hour post-load glucose (2h-PLG).

Using the Australian Diabetes, Obesity and Lifestyle Study (AusDiab; n=10,339) and Melbourne Collaborative Cohort Study (MCCS; n=2,996), we assessed associations of 743 lipid species across 38 classes with T2D, FBG, and 2h-PLG, independently of clinical measures.

In AusDiab, 556 lipid species were associated with prevalent T2D and 365 with 5-year incident T2D at FDR<0.05. Lipids containing odd- and branched-chain fatty acids, including 17:0 and 16:0;Me, 18:2-containing lipids, and sphingolipids with sphingadiene bases were inversely associated with T2D, whereas deoxyceramides and dihydroceramides were positively associated. Most associations replicated in MCCS.

Deoxyceramides, sphingomyelins with sphingadiene bases, and lipids containing odd- and branched-chain fatty acids showed similar associations with FBG and 2h-PLG. However, long-chain acylcarnitines were positively associated with 2h-PLG, whereas 18:2-containing lipids were inversely associated. These species showed weaker or opposite associations with FBG, indicating distinct lipidomic profiles of fasting and post-load glucose regulation. Acylcarnitines were also positively associated with 2h-PLG but inversely associated with FBG. For example, AC 16:1 was positively associated with 2h-PLG (β=0.25, 95% CI 0.23–0.27) but inversely associated with FBG (β= −0.31, 95% CI −0.34 to −0.27).

These findings identify lipidomic signatures of T2D and distinct metabolic profiles underlying impaired fasting glucose and impaired glucose tolerance, supporting lipidomics for characterising heterogeneous prediabetes phenotypes and potentially improving biological understanding of the mechanisms contributing to divergent prediabetes subtypes.

POSTER PRESENTATION

P-17

Targeted lipidomics identifies markers for skin collection in a minimally invasive micro-biopsy punch

PRESENTING AUTHOR

Kailyn O'Connor

Kailyn O'Connor[1], Stuart Kushon[1], Andrew Gooley[1]

[1] Trajan Scientific and Medical

POSTER

Poster 17

Monday 19

12:45

Silver Room

KEYWORDS

Micro-biopsy

Skin

Ceramide

Blood

Harpera

Traditional skin sampling typically involves a painful and invasive 3-5 mm punch biopsy, which can result in bleeding, infection and scarring. With increasing sensitivity of molecular techniques, micro-biopsy offers a promising minimally invasive alternative for tissue collection and characterisation. This study evaluated whether lipidomic analysis could characterise tissue collections from Trajan's Harpera micro-biopsy punch, as well as distinguish any skin-derived material from blood contamination. Several ceramides (Cer) were selected as markers for skin tissue collection, given their abundance in the stratum corneum, while phosphatidylcholine (PC) and sphingomyelin (SM) lipids were investigated as indicators of blood contamination, due to their abundance in red blood cells (3). Micro-biopsy needles used to collect samples from both the forearm and palm of the hand were photographed under a microscope and qualitatively classed as having no, low or high blood content before tissue extraction in butanol: methanol and analysis via LC-MS/MS. Regardless of the blood content recorded, all micro-biopsy devices showed evidence of tissue collection, with higher ceramide levels relative to extraction blanks. The consistent ceramide presence in all skin samples could be referenced as a ratio against the blood-associated PC and SM lipids to flag samples for blood contamination. This research presents a means to evaluate micro-biopsy skin collections through lipidomics workflows, providing a basis for further optimisation of both the device design and sampling methodology. Further investigations may aid in developing a pipeline for robust characterisation of healthy and diseased skin tissue.

POSTER PRESENTATION

P-18

High-throughput Single-cell Multi-OMICS Using Standard-flow LC and a Multi-reflecting TOF MS

PRESENTING AUTHOR

Marten F. Snel

Marten F. Snel[1], Qian Ying Xu[2], Scarlet Ferrinho[2], Lee A. Gethings[3], Robert S. Plumb[4], Adriana Tirado[4], David Heywood[3], Olivier Cexus[5], Paul Townsend[6], E.N. Clare Mills[2]

[1] Waters Australia, Rydalmere, NSW, Australia; [2] University of Surrey, Guildford, UK; [3] Waters Corporation, Wilmslow, UK; [4] Waters Corporation, Milford, MA, USA; [5] Gustave Roussy, Paris, France; [6] Liverpool John Moores University, Liverpool, UK

POSTER

Poster 18

Tuesday 20

12:45

Silver Room

Waters™

Waters

KEYWORDS

single-cell

multi-omics

mass spectrometry

Single-cell lipidomics and metabolomics are increasingly important for uncovering cell-to-cell biochemical heterogeneity often masked in bulk analyses. However, extremely low analyte abundance in individual cells necessitates highly sensitive and robust analytical workflows. HT29 and CaCO2 gastrointestinal and cancer research cell lines were used to demonstrate a unified LC-MS strategy for high-throughput discovery profiling.

Single cells were isolated using the IsoPick microfluidic platform and deposited directly into LC vials. Lipids were extracted with ice-cold isopropanol containing deuterated internal standards, while polar metabolites were extracted with 0.1% formic acid using a validated method. Extracts were analysed on a U(H)PLC system equipped with a 2.1×100 mm phenyl-hexyl column operated at 0.4 mL/min and 70 °C, coupled to a multi-reflecting TOF mass spectrometer running DIA and DDA modes with enhanced duty cycle (EDC).

Processed datasets (LipoStar for lipids, MARS for metabolites) produced more than 200 lipid identifications and over 150 polar metabolites per single cell (n=5), spanning amino acids, organic acids and nucleotides. High-resolution MS1 and MS2 data, boosted by EDC for ~10× higher MS2 signal, improved confidence in structural elucidation. The phenyl-hexyl chemistry eliminated isopropanol from the mobile phase, reducing background signal and enhancing signal-to-noise.

This streamlined LC-MS configuration enables comprehensive single-cell lipidomics and metabolomics without hardware changes, providing an efficient and scalable platform for multi-OMIC discovery studies.

POSTER PRESENTATION

P-19

Lipidomic Characterisation of Human Monocytes, Macrophages and Monocyte-Derived Microglia-Like Cells

PRESENTING AUTHOR

Matilda van Buuren-Milne

Matilda van Buuren-Milne[1], Alexandra George[2], Thy Duong[1], Nat Mellet[1], Peter Meikle[2], Kevin Huynh[2]

[1] Metabolomics Laboratory, Baker Heart and Diabetes Institute; [2] Metabolomics Laboratory, Baker Heart and Diabetes Institute, Baker Department of Cardiometabolic Health, The University of Melbourne, Australia, Baker Department of Cardiovascular Research, Translation and Implementation, La Trobe University

POSTER

Poster 19

Monday 19

12:45

Silver Room

KEYWORDS

- Targeted lipidomics
- Neurodegenerative disease
- Microglia
- Monocytes
- Macrophages

Neurodegenerative disorders cause gradual deterioration of neurons which primarily affect the brain and nervous system. Inflammation and immune activation occur in the early stages of disease and activated microglia and macrophages have become central to disease pathology. Lipids play essential roles in the functioning and identification of these cells. Human monocyte-derived microglia-like cells (MDMi) offer an in vitro approach that is both relatable to human pathology and time efficient. Using a targeted lipidomic method, this project focused on characterising the lipid composition of human monocytes, monocyte derived macrophages and MDMi.

Monocytes were isolated from whole blood using negative magnetic selection (n = 3). Monocytes were differentiated into macrophages (M-CSF 50ng/ml) and MDMi (GM-CSF 10ng/ml & IL-34 100ng/ml) over 14 days in triplicates. Lipids from extracted cells were screened using a targeted lipidomics platform (830 species & 52 classes).

MDMi had substantially higher dihydrosphingolipids, including sphingomyelins (SM 42:0;O2, 3.46-fold higher, p=6.55 x 10-8) and ceramides (Cer 18:0;O2/24:0;O, 7.32-fold, p=6.14 x 10-8). Interestingly, sphingodiene containing sphingolipids were lower (SM 18:2;O2/24:1, 0.73-fold, p=2.50 x 10-7). Many glycerophospholipids also differed, including ether lipids (PC O-34:2, 2.53-fold higher, p=3.55 x 10-6) and phosphatidylserines (PS 40:3, 0.45-fold lower, p =2.9 x 10-5).

These results provide novel insights into the lipid remodeling during monocyte-to-microglia-like differentiation and indicate many lipid features distinguish MDMi cells from matched macrophages. This model provides a potential platform for investigating microglia biology in neurodegenerative diseases.

POSTER PRESENTATION

P-21

Glass-Like Well Plates for High-Throughput Lipidomics: Limiting Analyte Loss and Reducing Extractables

PRESENTING AUTHOR

Kailyn O'Connor

Kailyn O'Connor[1], Lidia Konkol[1], Andrew Gooley[1]

[1] Trajan Scientific and Medical

POSTER

Poster 21

Monday 19

12:45

Silver Room

TRAJAN

Trajan

Careful selection of sample contact surfaces is critical for reducing background contaminants and maximising analyte recovery in sensitive analytical workflows. For hydrophobic analytes and strong organic solvents, inert glass surfaces are preferred, as extended contact with polypropylene (PP) can result in non-specific adsorption and increased extractables (1, 2). However, the use of glass vials or inserts introduces handling and alignment challenges in automated systems, which favour the well plate (WP) format. To determine whether the Pure-Pass™ glass-like coated (GLC) WP can overcome both analytical and automation challenges, lipids were extracted from blood plasma for two independent protocols and analysed by GC- and LC-MS. In both cases, Pure-Pass™ GLC WPs reduced background contaminants compared to the PP WP by at least 20% and maintained the same performance as both glass vials and glass inserts in terms of lipid recoveries. A separate study, conducted in partnership with Helmholtz Munich, also demonstrated that Pure-Pass™ GLC WPs are interchangeable with glass, showing excellent correlation across 19 lipid classes (R2 = 0.99). This surface compatibility could be attributed in part to the high purity Pure-Pass™ PP base, which was shown to significantly reduce extractables in strong organic solvents compared to standard commercially available PP. Pure-Pass™ GLC WPs demonstrate high analyte recoveries and low extractables, showing glass-like performance can be achieved without compromising throughput in an increasingly automated field.

POSTER PRESENTATION

P-23

Characterising Coffee Lipids from Green Bean to Brew: A Lipidomic Approach

PRESENTING AUTHOR

Jermaine Marie Ann Fabella

Jermaine Marie Ann Fabella[1], Tobias Kretzschmar[1], Parth Patel[1], Lei Liu[1]

[1] Southern Cross University

POSTER

Poster 23

Monday 19

12:45

Silver Room

KEYWORDS

Green coffee beans

roasted coffee beans

brewed coffee

lipid composition

lipidomic

Coffee is a globally traded commodity valued for its economic importance, distinct flavour, aroma and potential health-related benefits. Lipids are important contributors to coffee quality and sensory characteristics, yet their composition can change substantially throughout processing, roasting and brewing. Despite growing interest in coffee lipidomics, compositional studies have predominantly focused on green coffee beans. As a result, the transformation of lipid classes throughout the bean-to-cup process remains incompletely characterised. Recent advances in lipidomics provide an opportunity to investigate the coffee lipidome with greater coverage. However, key lipid classes, particularly diterpene esters, are inconsistently detected. The study evaluates key analytical challenges in coffee lipidomics and identifies approaches to improve the detection and characterisation of coffee lipids across processing stages, from green beans to brewed coffee.

POSTER PRESENTATION

P-24

Targeting lipid and cholesterol metabolism in endometrial cancer

PRESENTING AUTHOR

Claudia E. Tiplady

Claudia E. Tiplady[1], Hui-Ming Lin[2], Sarah Hancock[3], Ellen Olzomer[1], Lisa G. Horvath[4], Rhonda Farrell[4], Peter Meikle[5], Frances L. Byrne[1]

[1] School of Biotechnology & Biomolecular Sciences, Faculty of Science, University of New South Wales, Sydney, Australia; [2] Advanced Prostate Cancer Group, Garvan Institute; [3] Victor Chang Institute, Sydney, Australia; [4] Chris O'Brien Lifehouse, Sydney, NSW, Australia; [5] Baker Heart and Diabetes Institute, Melbourne, VIC, Australia

POSTER

Poster 24

Tuesday 20

12:45

Silver Room

KEYWORDS

- Endometrial Cancer
- Lipidomics
- Cholesterol
- Lipids

Endometrial cancer (EC) arises from epithelial cells of the uterus and is the most common gynaecological malignancy in Australia. Nearly 2,000 Australian women are diagnosed each year, and the incidence of this malignancy is rising worldwide. Obesity is a major risk factor for EC, contributing up to 50% of cases. Standard of care treatments for women with advanced EC include carboplatin and paclitaxel but these only extend progression-free survival by ~13 months. The rise in cases and lack of more effective therapies highlights the urgent need to discover new drug targets and therapies to treat EC.

Previous research by the Byrne laboratory showed lipid synthesis is elevated in obesity-related EC. In this study, we expanded on these findings by conducting bulk RNA-Seq and lipidomics on endometrial tissues from obese and lean postmenopausal women, with and without endometrial cancer. Lipidomics was also performed human endometrial cell lines. Results showed cholesterol metabolism genes, molecular pathways and metabolites, specifically cholesterol esters, are enriched in endometrial tumours from both lean and obese women. We then investigated the cytotoxicity of cholesterol synthesis inhibitors in human EC cells and normal endometrial cells. Results show that Avasimibe, an ACAT (acetyl-CoA acetyltransferase 1) inhibitor, has greater toxicity to EC cells compared to healthy epithelial endometrial cells and has additive cytotoxic effects when combined with carboplatin. Therefore, inhibitors of cholesterol metabolism, such as ACAT inhibitors, could represent new therapeutics to treat EC.

POSTER PRESENTATION

P-25

Serum lipidomics reveals class-specific lipid remodelling and altered acyl chain composition in Parkinson's disease

PRESENTING AUTHOR

Jack Price

Jack Price[1], Mari Muldmaa[2], Katrin Sikk[2], Pille Taba[3], Luke Whiley[4], Sulev Koks[5]

[1] Perron Institute for Neurological and Translational Science, Perth, WA, Australia; [2] Department of Neurology, North Estonia Medical Centre, Tallinn, Institute of Clinical Medicine, Faculty of Medicine, University of Tartu, Tartu, Estonia; [3] Institute of Clinical Medicine, Faculty of Medicine, University of Tartu, Clinic of Neurology, Tartu University Hospital, Tartu, Estonia; [4] Curtin Medical Research Institute, School of Diagnostic and Therapeutic Sciences, Dementia Centre of Excellence, enAble

POSTER

Poster 25

Monday 19

12:45

Silver Room

KEYWORDS

- Glycerolipids
- Glycerophospholipids
- Arachidonic acid
- Parkinson's Disease
- Levodopa-induced dyskinesia

BACKGROUND

Lipid metabolism is increasingly implicated in Parkinson's Disease (PD) pathology, yet the specific lipid classes and acyl chain compositions involved, and their relationship with PD clinical symptoms, remain poorly characterised. Objectives: We aimed to identify serum lipid signatures that differentiate people with PD (PwPD) from healthy controls and to evaluate associations with levodopa-induced dyskinesia (LID).

METHODS

Serum collected from 58 PwPD and 58 age- and sex-matched non-PD controls was analysed using targeted liquid chromatography - mass spectrometry (LC-MS). Group differences were assessed using univariate and multivariate approaches. Logistic regression was used to identify lipids associated with LID among PwPD.

RESULTS

Significantly lower levels of glycerolipids and glycerophospholipids were observed in PwPD compared to controls, alongside higher levels of lysophosphatidylethanolamines (LPE) and lysophosphatidylcholines (LPC). Analysis of individual lipid species revealed a lipid remodelling of the acyl chain composition of glycerolipids and glycerophospholipids, with decreased esterified arachidonic, adrenic, docosapentaenoic, and docosahexaenoic acids. Non-esterified arachidonic acid ($p = 5.80e-6$) and eicosapentaenoic acid ($p = 3.65e-10$) were elevated in PwPD. In secondary analyses, no lipids discriminated between PwPD with and without LID; however, these models were constrained by modest subgroup sizes and clinical heterogeneity between groups, limiting conclusions regarding lipid-based discrimination of LID status.

CONCLUSION

s: These findings demonstrate disease-associated remodelling of acyl chain composition across multiple lipid classes in PD, particularly selective depletion of esterified polyunsaturated fatty acids from glycerolipids and glycerophospholipids. Replication in independent cohorts is warranted to establish whether these pathways represent viable therapeutic targets.

POSTER PRESENTATION

P-26

Total Lipid Extract (TLE) from Bovine Liver (Avanti) on Orbitrap Tribrid Apex Small Molecule MS

PRESENTING AUTHOR

Jose Barreto

Jose Barreto[1]

[1] Thermo Fisher Scientific, Australia

POSTER

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Tuesday 20

12:45

Silver Room

Thermo Fisher Scientific

Purpose: In this study, Thermo Scientific Orbitrap Tribrid Apex Small Molecule Mass Spectrometer, is accessed for use in structural lipidomics using multiple fragmentation techniques including utraviolet photodissociation (UVPD).

METHODS

Preliminary analysis was done using Total Lipid Extract (TLE) from Bovine Liver (Avanti) on Orbitrap Tribrid Apex Small Molecule MS. This was followed up with analysis of Lipid Extract from the flower and parts of the plant Agapanthus.

RESULT

Resolution of isomers and annotation of unknown lipid species is shown

POSTER PRESENTATION

P-27

Ion mobility separation of lipids using a MALDI XS source with a Waters Cyclic IMS P20

PRESENTING AUTHOR

Marten Snel

Marten Snel[1], Mark Towers[2], Emmanuelle Claude[2], Joanne Ballantyne[2], Patricia Prabutzki[3], Eva Cuypers[3]

[1] Waters Australia, Rydalmere, NSW, Australia; [2] Waters Corporation, Wilmslow, UK; [3] M4I, Maastricht University, Maastricht, NL

POSTER

Poster 27

Monday 19

12:45

Silver Room

KEYWORDS

- Mass Spectrometry
- Ion Mobility
- Imaging
- Isomers

Confident identification of lipids in mass spectrometry imaging (MSI) is challenging due to the diversity of lipid species and the prevalence of isomeric and isobaric forms within narrow m/z ranges. As a result, MSI peaks are often assigned to a single lipid even when multiple structures could be present. These isoforms can vary greatly in biological function, and enzymes frequently show strong selectivity, making accurate discrimination essential for linking spatial lipid patterns to biochemical pathways.

Ion mobility offers an effective means to resolve isomeric species. Here, we demonstrate isomer separation using a Waters™ Cyclic™ IMS P20 instrument fitted with a MALDI XS source. With a focus on the separation of Lysophosphatidylcholine (LPC) sn-1 18:0/0:0 and LPC sn-2 0:0/18:0, we found isomer separation starting to occur at 20 passes resulting in cycle time of 348.58 ms.

For single cell MALDI imaging experiments, MDA-MD-231 cells cultured under normoxic and anoxic conditions were coated with 4-(dimethylamino) cinnamic acid (DMACA) via sublimation using 2 ml of 5 mg/ml DMACA in acetone/methanol (1:1). The samples were imaged at pixel sizes of 20 µm. MALDI XS laser settings were successfully defined for the analysis of MDA cells using a continuous raster imaging mode. MDA cells can be seen as highlighted by the m/z 760.54 peak and m/z 782.56 peak of PC 34:1. Some co-localization of extracted LPC was observed for the cells, but further work will need to be performed to assess other MALDI matrices to reduce background interference.

POSTER PRESENTATION

P-28

Sex-Specific Lipid Signatures Associated with Paracetamol Exposure

PRESENTING AUTHOR

Shaimaa Khandaker

Shaimaa Khandaker[1], Nicola Gray[1], Samantha Lodge[1], Luke Whiley[1,2]

[1] Centre for Computational and Systems Medicine and Australian National Phenome Centre, Health Futures Institute, Murdoch University, Perth, Western Australia 6150, Australia; [2] Curtin Medical Research Institute, Curtin University, Perth, Western Australia 6102, Australia

POSTER

Poster 28

Tuesday 20
12:45
Silver Room

KEYWORDS

- acetaminophen
- lipidomics
- sulfated lipids
- non-targeted
- liquid chromatography-mass spectrometry

INTRODUCTION

Paracetamol (acetaminophen, APAP) is widely used, yet its metabolic effects remain poorly characterised. Prior evidence suggests APAP may compete with endogenous sulfation pathways, potentially altering sulfated hormone metabolism. We investigated sex stratified, age adjusted serum lipid signatures associated with APAP exposure in participants from the Airwave Health Monitoring Study, focusing on sulfated lipids (sulfatides).

METHODS

We analysed open access non-targeted serum lipidomics data generated by the UK National Phenome Centre using liquid chromatography-mass spectrometry (LC MS). Data from 2,966 participants (39.4% female; median age 42) were acquired using hydrophilic interaction liquid chromatography (HILIC, positive mode) and reversed phase chromatography (positive and negative modes), yielding 348 annotated lipid features. Serum data were pre-processed and APAP exposure was defined by low (<16.0) and high (>17.50) log2 intensity thresholds, excluding 99 serum samples between these ranges (16-17.50), retaining 2,867 final participants. Sex stratified comparisons were performed, with all analyses adjusted for age. Mann-Whitney U tests were applied with Benjamini-Hochberg correction across all features or, for sulfatides, within class.

RESULTS

Sex stratified analyses identified 42 APAP associated lipids in males, including higher triglycerides, diglycerides, ceramides and lower ether linked phospholipids and sulfatides, while none reached significance in females. Sulfatides exhibited sex divergent responses: eleven species tended to decrease in exposed males but increase in exposed females, with five showing significant sex-specific interactions (FDR $q < 0.05$).

CONCLUSION

APAP exposure was associated with male specific lipid perturbations and opposing sulfatide responses between sexes, highlighting potential sex dependent metabolic consequences.

POSTER PRESENTATION

P-29

Profiling Short-Chain Acylcarnitines in Human Peripheral Blood Mononuclear Cells

PRESENTING AUTHOR

Thy Duong

Thy Duong[1], Kevin Huynh[2], Peter Meikle[2]

[1] Baker Heart and Diabetes Institute; [2] Baker Heart and Diabetes Institute, Baker Department of Cardiometabolic Health, Melbourne University, Baker Department of Cardiovascular Research Translation and Implementation, La Trobe, Faculty of Medicine, Nursing and Health Sciences, Monash

POSTER

Poster 29

Monday 19

12:45

Silver Room

KEYWORDS

short chain acylcarnitine

LC-MS/MS

PBMCs

lipidomics

Peripheral blood mononuclear cells (PBMCs) have been used increasingly as a non-invasive way to investigate immune responses in aging. Short-chain acylcarnitines (SCACs; 2–5 carbons) are vital metabolites for mitochondrial health and neurotransmitter function. While plasma long-chain acylcarnitines are known to be altered with age, short- and medium-chain (6–14 carbons) acylcarnitines in PBMCs remain unexplored.

To address this, we utilised existing PBMC lipidomics extracts from an aging study to evaluate their detectability and association with age. Metabolites were separated on an Agilent Poroshell 120 Aq-C18 column using 10 mM ammonium formate in water (solvent A) and 90% acetonitrile/10% water (solvent B). We targeted 2–14 carbon acylcarnitines on an Agilent 6495C QqQ mass spectrometer in positive ion mode.

Our analysis successfully identified SCACs in human PBMCs. We performed a linear regression to evaluate the relationship between SCACs levels and age. After normalising to free carnitine and adjusting for sex and BMI, AC 6:0 increased by 28.7% per decade, AC 2:0 increased by 21.3%, and AC 6:0 [b] increased by 23.3% ($p = 0.0065$, 0.019, and 0.044 respectively).

This pilot study provides a valuable description of SCACs in human PBMCs from an ageing cohort. Our approach demonstrates the feasibility of utilising the existing lipidomics extracts to perform additional metabolic profiling and analysis. These initial results provide the validation necessary for the profiling of larger studies, and provide a deeper understanding of SCACs, their role in PBMCs, and their link to diseases of ageing.

POSTER PRESENTATION

P-30

Optimization and Clinical Validation of a Cell-Free Cholesterol Efflux Capacity Assay Using Immobilized Liposomes-bound Magnetic Beads in Statin-Treated Patients

PRESENTING AUTHOR

Wanida Laiwattanapaisal

Wisanu Niamsi[1], Natchaya Vanwong[2], Ryunosuke Ohkawa[3], Wanida Laiwattanapaisal[2]

[1] Graduate Program in Clinical Biochemistry and Molecular Medicine, Department of Clinical Chemistry, Faculty of Allied Health Sciences, Chulalongkorn University, Bangkok 10330, Thailand; [2] Department of Clinical Chemistry, Faculty of Allied Health Sciences, Chulalongkorn University, Patumwan, Bangkok 10330, Thailand; [3] Department of Clinical Bioanalysis and Molecular Biology, Graduate School of Medical and Dental Sciences, Institute of Science Tokyo, Tokyo, Japan

POSTER

Poster 30

Tuesday 20

12:45

Silver Room

KEYWORDS

cholesterol efflux capacity

cell-free CEC

immobilized liposome-bound magnetic beads

cardiovascular disease

HDL functionality

Coronary artery disease (CAD) risk assessment traditionally relies on standard lipid profiles to determine the need for lipid-lowering therapy. As increasing HDL-C level has not shown clear CAD benefits, clinical interest has shifted toward HDL functionality, particularly cholesterol efflux capacity (CEC), an emerging predictor of cardiovascular events. In this research, a cell-free immobilized liposome-bound magnetic beads (ILMs) assay has been proposed for CEC. The protocol has been optimized to improve homogeneity by introducing a sonication step before freeze-thaw cycles to reduce vesicle size of the liposome and serve as cholesterol donors. The assay requires only a 1-hour incubation time, which drastically increases its throughput. We assessed its clinical potential in 41 subjects (21 CVD, 20 non-CVD) who were actively managed on lipid-lowering therapy, enabling the direct assessment of residual risk detection. The results demonstrated that the ILM assay effectively differentiated the patient groups. Normalized CEC was significantly reduced in the CVD cohort compared to non-CVD controls ($p = 0.023$). Bivariate analysis confirmed positive correlates with HDL-C ($r = 0.329$, $p = 0.036$), supporting its association with HDL functionality. Integrating ILM CEC values with standard clinical metrics (age, sex, and HDL-C) into a multivariable risk model could support diagnostic accuracy, achieving an AUC of 0.845 ($p < 0.001$). In conclusion, the ILM assay shows promise as a rapid and automatable platform for HDL functionality assessment, with potential to identify residual cardiovascular risk beyond standard lipid profiles. Further validation in larger cohorts is warranted to establish its clinical utility in CAD risk prediction.

POSTER PRESENTATION

P-31

Root-shoot lipidome interactions in Medicago truncatula after drought stress

PRESENTING AUTHOR

Alastair Ross

Alastair Ross[1], Jesus Echaide Aquino[2], Asyraf Amnan[3], Mariana Pires Barboza[2], Christina Reumiller[1], Erin Lee[1], Evelyne Maes[1], Paul Dijkwel[3], Artemio Mendoza-Mendoza[2], John Hampton[2]

[1] New Zealand Institute of Bioeconomy Science; [2] Lincoln University; [3] Massey University

POSTER

Poster 31

Monday 19
12:45
Silver Room

KEYWORDS

Pasture

Drought-stress

Medicago

Lipidomics

Sphingolipids

Pasture plants underpin an important part of global food supply, and pasture systems are under threat from climate change. Lipid changes may play a role in drought response and adaptation. Using Medicago truncatula, a model pasture legume, we analysed how root and shoot lipidomes change with drought stress. M truncatula seeds were planted and after emergence kept well-watered for 14 days, after which one group had water withheld for 16 days (drought stress) and one group kept well-watered (n=9 replicates/group, each replicate pooled from 7 plants). At the end of drought stress plants were harvested, root and shoots were separated, processed and extracted using water:methanol:chloroform (1/1/1). The non-polar phase was analysed using untargeted lipidomics in positive mode and with data independent acquisition. Data were processed using MSDIAL and features with MS1 or MS2 identity matches screened using an in-house lipid retention time prediction algorithm. 224 and 738 lipid features were putatively identified in shoots and roots respectively. Data were analysed using PCA, MUVR and O2PLS. Lipids differed between drought-stressed compared to well-watered plants in both roots and shoots based on cross-validated classification analysis. Well-watered M truncatula was most strongly characterised by higher concentrations of shoot membrane lipids and chloroplast associated lipids, while drought stressed plants were associated with high concentrations of root lipid classes including oxidised triacylglycerol and hexosylceramides. Root-shoot integration identified a limited number of coordinated drought responsive lipid classes, predominantly involving ceramide phosphate, hexosylceramide and acylhexosylceramide species (Pearson r=0.81-0.86), suggesting

POSTER PRESENTATION

P-32

Acetonitrile-Derived Acetamidinium Adduct Formation Reduces Signal Intensity in LC–MS Lipidomics

PRESENTING AUTHOR

Nat Mellett

Nat Mellett[1], Thy Duong[1], Juliane David[1], Peter Meikle[2], Kevin Huynh[3]

[1] Baker Heart and Diabetes Institute; [2] Baker Heart and Diabetes Institute, Baker Department of Cardiovascular Research, Translation and Implementation, La Trobe University; [3] Baker Heart and Diabetes Institute, Baker Department of Cardiometabolic Health, The University of Melbourne

POSTER

Poster 32

Tuesday 20
12:45
Silver Room

KEYWORDS

LCMS

adducts

signal loss

The use of adducts in liquid chromatography mass spectrometry (LC–MS) lipid analysis is essential for measuring lipid species that do not readily form protonated [M+H]+ ions. Instead, alternative adducts, such as ammonium, enable detection of neutral lipid classes including triglycerides and cholesteryl esters. We recently identified the accumulation of acetamidine in solvent under certain storage conditions, resulting in unwanted adduct formation reducing signal sensitivity across a large range of lipid species and classes. Overtime, this led to increased analytical variability, signal loss, and ultimately retention time drift. In targeted lipidomics, acetamidinium adduct formation can introduce analytical variability and potentially confound lipid identification. For example, PC 34:1 [M+acetamidinium]+ occurs at a similar precursor m/z to PC O-40:7 [M+H]+ (m/z 818.6382 and 818.6058, respectively), and both generate the characteristic phosphocholine product ion at m/z 184.0733, although the species are chromatographically resolved. As acetamidine accumulates, signal intensity decreases for many lipid species, particularly those detected as ammonium adducts, while the apparent PC O-40:7 signal increases, providing a practical marker of acetamidine contamination. Acetamidine formation accelerated at elevated temperatures, whereas replacing acetonitrile with methanol prevented detectable contaminant formation. However, such solvent substitution may not be compatible with all applications and would require method revalidation. Here, we identify acetamidine formation as a previously underappreciated source of LC–MS variability that can alter lipid signal intensities and, in some cases, lead to misidentification of lipid species.

POSTER PRESENTATION

P-34

Enhancing the biodistribution of artificial lipid droplets as a novel lipid particle delivery system in vivo

PRESENTING AUTHOR

Ebony Monson

Ebony Monson[1], Zahra Telikani[1], Irumi Amarasinghe[1], Shuai Nie[2], Marcus Hofer[3], Kazuhide Okuda[1], Karla Helbig[1]

[1] La Trobe Institute for Molecular Sciences; [2] Bio21 Molecular Science & Biotechnology Institute, The University of Melbourne; [3] The University of Sydney

POSTER

Poster 34

Tuesday 20

12:45

Silver Room

KEYWORDS

- Lipid droplets
- immunity
- particle delivery
- lipidomics

We have shown that lipid droplets (LDs) are rapidly upregulated early during viral infection, driving antiviral cytokine production and restricting viral replication. Recently, we demonstrated that LDs can also be secreted and taken up by neighbouring cells, conferring protection during infection. Harnessing this process presents an exciting opportunity to develop LD-based antiviral therapies. This study aimed to optimise an artificial LD (aLD) system for in vivo use. We first characterised the lipidome of virally induced LDs to identify lipid changes associated with infection and determine key lipids to include in an aLD system. Antiviral LDs showed a significant shift in their lipidomic composition, including (1) a global increase in long and very-long polyunsaturated fatty acids such as docosahexaenoic acid and arachidonic acid, and (2) a modest but significant enrichment of phosphatidylinositol and phosphatidylethanolamine in the LD membrane. Recapitulating the native antiviral LD lipidome, we developed an aLD incorporating a complex phospholipid membrane, and triglyceride core. These aLDs showed uptake across 12 cell types in vitro, with enhanced uptake in brain cell types; astrocytes and neurons. aLDs demonstrated a broad biodistribution in vivo following intravenous injection in mice, and cardinal vein injection of zebrafish. aLDs localised to lungs, heart, liver and most excitingly to the brain and facial region within 6 hours and persisted for over 48 hours. As a small lipid particle system, aLDs with complex membranes demonstrate superior organ delivery compared to extracellular vesicles and lipid nanoparticles, offering a promising platform for antiviral cargo delivery to the brain.

POSTER PRESENTATION

P-35

Pre-Analytical Matrix Considerations in Sepsis Lipidomics: A Systematic Review Comparing Serum and Plasma Profiling in Bacterial Infections

PRESENTING AUTHOR

Nurul Saadah Ahmad

Nurul Saadah Ahmad[1], Rosni Ibrahim[1], Toh Leong Tan[2], Muhammad Faizan Abdul Shukor[3], Syarul Nataqain Baharum[4], Zamberi Sekawi[1]

[1] Department of Medical Microbiology, Faculty of Medicine and Health Sciences, Universiti Putra Malaysia; [2] Department of Emergency Medicine, Faculty of Medicine, Universiti Kebangsaan Malaysia; [3] Research Centre for Chemical Defence, Universiti Pertahanan Nasional Malaysia; [4] Metabolomics Research Laboratory, Institute of Systems Biology (INBIOSIS), Universiti Kebangsaan Malaysia

POSTER

Poster 35

Monday 19

12:45

Silver Room

KEYWORDS

serum

plasma

sepsis

lipidomics

systematic review

Blood-based lipidomics offers a promising approach for identifying sepsis biomarkers; however, differences in lipid species profiles between serum and plasma pose challenges for data interpretation and clinical translation. This systematic review (PROSPERO: CRD420251086177) evaluated pre-analytical matrix considerations by synthesizing lipidomic evidence across five databases (Scopus, PubMed, Ovid MEDLINE, Web of Science, Google Scholar). Nine studies (2016–2025) employing targeted or untargeted lipidomics in adult bacterial sepsis cohorts were analyzed to systematically compare serum and plasma profiles, methodological workflows and clinical associations. Synthesized results revealed reproducible candidate biomarker trajectories across both matrices, characterized by consistent elevations of triacylglycerols (TAG) and ceramides, marked depletion of lysophosphatidylcholines (LPC) and bidirectional shifts in phosphatidylcholines (PC) and sphingomyelins (SM). Crucially, pre-analytical mechanics drove distinct matrix-specific performance: the natural clotting process during serum preparation triggers platelet activation and bioactive lipid release, yielding broader lipid species coverage and stronger prognostic performance. Conversely, halting coagulation during plasma collection preserves an in vivo circulatory metabolic snapshot, offering higher discriminatory accuracy for acute diagnosis. However, substantial methodological heterogeneity was identified in sample collection, storage, extraction protocols and mass spectrometry platforms. Ultimately, both matrices provide valuable clinical insights and observed performance trade-offs reflect primary study designs and pre-analytical sample handling rather than intrinsic matrix superiority. Standardizing pre-analytical and analytical lipidomic workflows is essential to mitigate methodological heterogeneity, ensure inter-study reproducibility and successfully translate sepsis lipid biomarkers into precision clinical practice.

POSTER PRESENTATION

P-36

Circular Bioprocessing: A study of spent media and biomass recycling with Umbelopsis isabellina

PRESENTING AUTHOR

Jasmeet Kaur Khanijou

Jasmeet Kaur Khanijou, Kah Yin Lee, Christian Hermansen, Yoganathan Kanagasundaram, Sreenivasa Reddy Puniredd

[1] Singapore Institute of Food and Biotechnology Innovation

POSTER

Poster 36

Tuesday 20
12:45
Silver Room

KEYWORDS

Fatty Acids

Precision Fermentation Sustainability

Precision fermentation provides a sustainable alternative for producing food ingredients and biomolecules. However, the fermentation process generates spent media and residual biomass which are usually discarded even though they may have compounds with the potential for reuse or recovery. In this work, a circular bioprocess was developed, where spent media recycling and spent biomass valorisation was incorporated in Umbelopsis isabellina fermentations for lipids such as fatty acids. With varying proportions of recycled spent media tested over multiple cycles, 50% spent media reuse together with targeted nutrient supplementation supported the fermentation process. The spent biomass was freeze dried and subjected to analysis using Liquid Chromatography Quadrupole Time-of-Flight Mass Spectrometry (LC-QTOF-MS) after dichloromethane-methanol (1:1, v:v) extraction. Fatty acids such as palmitic acid, hexadecenoic acid, heptadecoic acid, heptadecenoic acid, stearic acid, oleic acid, linoleic acid, linolenic acid, arachidic acid, gadoleic acid, and docosanoic acid were identified and quantified. Highest amounts of fatty acids were recovered from day 2 of round 1 of fermentation followed by day 3 of round 2 of fermentations. In a separate experiment, spent biomass was subjected to solvent-free supercritical CO2 extraction prior to lipidomics profiling on the LC-QTOF-MS. The fatty acids extracted before and after supercritical CO2 extraction were quantified whereby further optimization to the supercritical CO2 extraction conditions would be required to improve fatty acid extraction yield. Overall, spent media recycling coupled with solvent-free supercritical CO2 extraction has the potential to improve fermentation process sustainability and provide a viable alternative to producing food ingredients such as fatty acids.

POSTER PRESENTATION

WITHDRAWN

Lipidomic Remodeling of Erythrocyte Membranes During Oxidative Stress in Chronic Kidney Disease: Bridging Eryptosis and Precision Lipidomics for Early Biomarker Discovery

PRESENTING AUTHOR

Hafsa Hussnain

Hafsa Hussnain[1]

[1] University of Agriculture Faisalabad Pakistan

POSTER

KEYWORDS

Chronic Kidney Disease

Lipidomics

Eryprosis

Oxidative stress

Erythrocyte membrane Biomarkers

Chronic kidney disease (CKD) is a progressive multisystem disorder affecting more than 10% of the global population and remains a major contributor to cardiovascular morbidity and premature mortality. Despite advances in clinical management, current biomarkers such as serum creatinine and estimated glomerular filtration rate primarily reflect established renal dysfunction rather than the early molecular alterations driving disease progression. Emerging evidence suggests that oxidative stress-induced eryptosis contributes to erythrocyte dysfunction in CKD; however, the accompanying lipidomic remodeling of erythrocyte membranes has received limited attention.

Here, a hypothesis-driven translational framework integrates oxidative stress biology, eryptosis, and erythrocyte membrane lipidomics to identify novel biomarkers for early CKD progression. We propose that oxidative injury promotes remodeling of membrane phospholipids, including phosphatidylcholine, phosphatidylethanolamine, sphingomyelin, lysophosphatidylcholine, and oxidized phospholipids, resulting in membrane destabilization, phosphatidylserine externalization, and enhanced eryptosis. These molecular alterations are hypothesized to precede measurable declines in conventional renal function, thereby offering a minimally invasive opportunity for earlier disease detection and improved risk stratification. Integrating high-resolution lipidomic profiling with established oxidative stress and eryptosis biomarkers may provide a mechanistically informed platform for precision nephrology. Such an approach could facilitate earlier diagnosis, identify patients at increased cardiovascular risk, and support the development of targeted antioxidant and membrane-protective therapies.

This perspective highlights erythrocyte membrane lipidomics: an underexplored, clinically relevant research direction at the interface of redox biology and precision lipidomics. Future translational studies combining lipidomics, flow cytometry, and clinical phenotyping may establish erythrocyte membrane lipid signatures as novel biomarkers for the early detection and monitoring of CKD progression.

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Thank you

We look forward to welcoming the global lipid research community to Perth in October 2026.

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